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Theoretical analysis of diffusion flames and development of a Large Eddy Simulation model for turbulent hydrogen diffusion flames

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Analyse théorique des flammes de diffusion et développement d'un modèle de simulation aux grandes échelles pour les flammes de diffusion turbulentes à l'hydrogène

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Abstract — This thesis focuses on the modeling of diffusion flames in hydrogen-air flames. These diffusion flames can be found in many present and future chamber concepts aiming to burn hydrogen. Three key fundamental issues which are often ignored in many recent approaches are discussed here:

- While flame strain is recognized by the whole community as the main parameter controlling the flame structure of diffusion fronts, the exact definition of flame strain is ambiguous and is revisited here for both stagnation point laminar flames and for turbulent DNS and LES flames. A single consistent definition for flame strain is derived and compared to all others.

- Many arguments used in the literature for diffusion flames are based on a classical theory which is derived assuming that density is constant. In hydrogen-air diffusion flames, this assumption is totally wrong: even if the flame is not burning, density changes from 1 to 15 when going from pure hydrogen to pure air. The constant density classical theory for one-dimensional stagnation point flames is revisited using analytical derivations and extended to account for variable density and shown to be more precise than the constant density approach.

- Finally the thesis discusses spatial resolution issues and flame/turbulence interaction models. It shows that no satisfactory approach exists today for diffusion flamelets in turbulent flows, especially at high pressures where these thicknesses can be much less than all other characteristic lengths. This under resolution is a critical issue which must be treated before discussing flame/turbulence interaction. This is done by avoiding numerical artefacts such as artificial viscosity, by introducing a new model called DFTC (Diffusion Front Thickness Control) which thickens diffusion fronts enough to resolve them on the mesh, as done in the well-known TFLES model for premixed flames.

Mots clés : Diffusion, simulation numérique, résolution, modélisation, combustion de l'hydrogène, flamme-turbulence interaction.

Résumé — Cette thèse porte sur la modélisation des flammes de diffusion dans les mélanges hydrogène-air.

Ces flammes de diffusion sont présentes dans de nombreux concepts de chambres de combustion actuels et futurs destinés à la combustion de l'hydrogène. Trois problèmes fondamentaux, souvent négligés dans les approches récentes, sont abordés ici :

• Bien que l'étirement de la flamme soit reconnu par la communauté scientifique comme le principal paramètre contrôlant la structure des fronts de diffusion, sa définition exacte reste ambiguë et est réexaminée ici, tant pour les flammes laminaires au point de stagnation que pour les flammes turbulentes simulées par DNS et LES. Une définition unique et cohérente de l'étirement de la flamme est établie et comparée à toutes les autres.

• De nombreux arguments utilisés dans la littérature concernant les flammes de diffusion reposent sur une théorie classique qui suppose une densité constante. Dans les mélanges hydrogène-air, cette hypothèse est totalement erronée: même en l'absence de combustion, la densité varie de 1 à 15 lors du passage de l'hydrogène pur à l'air pur. La théorie classique à densité constante des flammes unidimensionnelles au point de stagnation est réexaminée à l'aide de dérivations analytiques et étendue pour prendre en compte la densité variable. Elle se révèle plus précise que l'approche à densité constante.

• Enfin, la thèse aborde les problèmes de résolution spatiale et les modèles d'interaction flamme/turbulence. Elle montre qu'il n'existe actuellement aucune approche satisfaisante pour les flammes de diffusion dans les écoulements turbulents, en particulier à hautes pressions où ces épaisseurs peuvent être bien inférieures à toutes les autres longueurs caractéristiques. Cette sous-résolution est un problème critique qui doit être traité avant d'aborder l'interaction flamme/turbulence. Pour ce faire, on évite les artefacts numériques tels que la viscosité artificielle, en introduisant un nouveau modèle appelé DFTC (Diffusion Front Thickness Control) qui épaissit suffisamment les fronts de diffusion pour les résoudre sur le maillage, comme dans le modèle TFLES bien connu pour les flammes prémélangées.

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Nomenclature

Acronyms

AMR	Automatic Mesh Refinement
AVBP	Compressible reactive CFD solver developed at CERFACS
BC	Boundary Condition(s)
CAN2AV	Tool used to convert a CANTERA solution into an AVBP initialization
CANTERA	open-source suite of tools for problems involving chemical kinetics, thermodynamics, and transport processes
CD	Constant Density model
CFD	Computational Fluid Dynamics
CMC	Conditional Moment Closure
DFTC	Diffusion Front Thickness Control
DGAC	Direction Générale de l'Aviation Civile
DNS	Direct Numerical Simulation
F-TACLES	Filtered Tabulated Chemistry for LES
FRI	Flame Resolution Index
FTI	Flame/Turbulence Interaction
FTIC	Flame/Turbulence Interaction Consumption criterion
FVI	Flame/Vortex Interaction
HPC	High Performance Computing
LAM	Modification of the laminar diffusivity term
LC	Laminar Consumption criterion
LES	Large Eddy Simulation
LQ	Laminar Quenching criterion
LTMAX	Maximum-temperature conservation criterion
ONERA	Office national d'études et de recherches aérospatiales

PDF	Probability Density Function	
PHYDROGEN	Research project dedicated to hydrogen combustion applications	
RANS	Reynolds-Averaged Navier-Stokes	
REAC	Modification of the reaction term	
RES	Resolution criterion	
RMS	Root Mean Square	
SMR	Steam Methane Reforming	
SPLF	Strained Stagnation Point Laminar Flame	
TFLES	Thickened Flame LES	
TNF	Turbulent Nonpremixed Flames workshop	
TUR	Modification of the turbulent diffusivity term	
VD	Variable Density model	

Greek Symbols

α	Scaling factor applied to turbulent diffusivity in DFTC	–
$\alpha_{\text{sup}}, \alpha_{\text{inf}}$	Coefficients in the analytical molecular-weight model	*model-dependent*
β	Fractal exponent in the flame-surface formulation	–
$\beta_{\text{sup}}, \beta_{\text{inf}}$	Coefficients in the analytical molecular-weight model	*model-dependent*
χ	Scalar dissipation rate	s^{-1}
$\Delta h^0_{f,k}$	Standard formation enthalpy of species k	$\text{J} \cdot \text{kg}^{-1}$
Δt	Time step	s
ΔT^0	Fuel/oxidizer inlet-temperature difference	K
Δ	Mesh/filter/Cell size	m
δ_c	Inner cutoff scale in the fractal flame model	m
δ_f	Diffusion-flame thickness	m
δ_{ij}	Kronecker delta	–
$\dot{\Omega}$	Reaction/heat-release	*domain-dependent*
$\dot{\omega}$	Local production/consumption rate	$\text{kg} \cdot \text{m}^{-3} \cdot \text{s}^{-1}$
$\dot{\Omega}_F$	Integrated fuel-consumption rate per unit flame area	$\text{kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$
$\dot{\Omega}_T$	Heat release integrated rate over the computational domain	*domain-dependent*
$\dot{\omega}_T$	Local heat-release rate	$\text{W} \cdot \text{m}^{-3}$

ϵ	Turbulent kinetic-energy dissipation rate	$\mathrm{m}^2 \cdot \mathrm{s}^{-3}$
$\eta_{\mathrm{sup}}, \eta_{\mathrm{inf}}$	Coefficients in the analytical heat-capacity model	*model-dependent*
$\gamma_{\mathrm{sup}}, \gamma_{\mathrm{inf}}$	Coefficients in the analytical temperature model	*model-dependent*
κ	Flame stretch rate	s^{-1}
κ_{curv}	Curvature contribution to flame stretch	s^{-1}
Λ	Pressure eigenvalue used in CANTERA	*model-dependent*
λ	Thermal conductivity	$\mathrm{W} \cdot \mathrm{m}^{-1} \cdot \mathrm{K}^{-1}$
$\lambda_{\mathrm{sup}}, \lambda_{\mathrm{inf}}$	Coefficients in the analytical heat-capacity model	*model-dependent*
μ	Dynamic viscosity	$\mathrm{Pa} \cdot \mathrm{s}$
ν	Kinematic viscosity	$\mathrm{m}^2 \cdot \mathrm{s}^{-1}$
ν'	Stoichiometric coefficient	–
ϕ	Generic scalar quantity	–
$\Phi_{i,j}$	Wilke-law mixture coefficient	–
ψ	Generic flame-front scalar / integration variable, depending on context	–
ρ	Density	$\mathrm{kg} \cdot \mathrm{m}^{-3}$
Σ	Flame-surface density	m^{-1}
σ_{ij}	Total stress tensor	Pa
τ	DFTC relaxation time	s
τ_η	Kolmogorov timescale	s
τ_t	Turbulent integral timescale, l_t/u'	s
τ_{ij}	Viscous stress tensor	Pa
Θ	Flame wrinkling factor, $\Theta = L_f/L_y$	–
ξ	Random/integration variable in PDF formulations	–
Ξ_Δ	Subgrid flame-wrinkling factor	–
ζ	Reduced coordinate incorporating strain and diffusion	–

Roman Symbols

$\mathscr{A}$	Strain model	s^{-1}
A	Flame-surface area	m^2
a	Local strain rate	s^{-1}
a_Q	Quenching strain rate	s^{-1}

b	Exponent in the diffusivity law $D \propto T^b$	–
C_p	Mixture heat capacity at constant pressure	$\mathrm{J \cdot kg^{-1} \cdot K^{-1}}$
D_k	Molecular diffusivity of species k	$\mathrm{m^2 \cdot s^{-1}}$
D_t	Turbulent diffusivity	$\mathrm{m^2 \cdot s^{-1}}$
D_{th}	Thermal diffusivity	$\mathrm{m^2 \cdot s^{-1}}$
D_{jk}	Binary diffusion coefficient between species j and k	$\mathrm{m^2 \cdot s^{-1}}$
E	Total non-chemical energy	$\mathrm{J \cdot kg^{-1}}$
F	Local thickening factor	–
$f(z)$	Thickening distribution function	–
F_r	Reaction-rate scaling factor	–
$f_{k,j}$	Volume force acting on species k in direction j	$\mathrm{m \cdot s^{-2}}$
H	Distance between the two burners/inlets	m
h	Auxiliary variable $h = \partial_\zeta z$ in the VD model	–
h_k	Specific enthalpy of species k	$\mathrm{J \cdot kg^{-1}}$
j	Diffusive mass flux	$\mathrm{kg \cdot m^{-2} \cdot s^{-1}}$
K	Dimensionless DFTC relaxation parameter	–
K_0	VD-model constant $W_0^{-2} T_0^{2-b}$	*model-dependent*
L_f	Flame length in two-dimensional simulations	m
l_t	Turbulent integral length scale	m
L_y	Domain length in the y direction	m
Le	Lewis number	–
N	Number of chemical species	–
n_i, n_j	Components of the unit normal vector to the flame front	–
P	Pressure	Pa
p	Empirical temperature-correction factor in Chapter 3	–
Q	Heat-release parameter in the analytical temperature model	*model-dependent*
q_i	Heat flux component	$\mathrm{W \cdot m^{-2}}$
R	Universal gas constant	$\mathrm{J \cdot mol^{-1} \cdot K^{-1}}$
r	Radial coordinate	m
R_1, R_2	Principal radii of curvature of the flame front	m

r_v	Inlet velocity ratio $v_{\text{oxy}}/v_{\text{fuel}}$	–
r_{hyp}	Ratio used to assess the diffusivity power law	–
s_d	Flame displacement speed	$\text{m} \cdot \text{s}^{-1}$
T	Temperature	K
t	Time	s
$T_{\text{sup}}, T_{\text{inf}}$	Analytical temperature branches	K
u'	RMS turbulent velocity fluctuation	$\text{m} \cdot \text{s}^{-1}$
u, v, w	Cartesian velocity components	$\text{m} \cdot \text{s}^{-1}$
V	Scaled radial velocity in CANTERA, $V = v/r$	s^{-1}
V_i^c	Correction diffusion velocity	$\text{m} \cdot \text{s}^{-1}$
$V_{k,i}$	Diffusion velocity component of species k	$\text{m} \cdot \text{s}^{-1}$
W	Mixture molecular weight	$\text{kg} \cdot \text{mol}^{-1}$
X_k	Mole fraction of species k	–
Y	Mass-weighted spatial coordinate used in the VD model	m
Y_k	Mass fraction of species k	–
z	Mixture fraction	–

Subscripts/Superscripts

$(\cdot)'$	Fluctuating quantity
$(\cdot)^0$	Inlet value or standard-state quantity depending on context
$(\cdot)^*$	Modified, thickened, or intermediate quantity depending on context
$(\cdot)^t$	True/physical quantity, as in FRI^t
$(\cdot)^{\text{res}}$	Resolved quantity, as in FRI^{res}
$\dot{(\cdot)}$	Rate/source term
η	Kolmogorov scale/timescale, as in τ_η
$\overline{(\cdot)}$	Filtered/averaged quantity depending on context
$\vec{(\cdot)}$	Vector quantity
$\widetilde{(\cdot)}$	LES-filtered quantity
F,fuel	Fuel
H_2, O_2, H_2O, N_2	Quantity associated with the corresponding species
i, j	Spatial/component indices

k	Chemical-species index
$n, n+1$	Current and next controller iteration
Q	Quenching
0	stagnation value
artif	Artificial/numerical quantity
curv	Curvature contribution
fit	Fitted quantity
in	Inlet value
inf	Branch corresponding to $z < z_{\text{st}}$
lam	Laminar quantity
max	Maximum value
min	Minimum value
oxy	Oxidizer-side quantity
res	Resolved quantity
st	Evaluated at stoichiometry
sup	Branch corresponding to $z > z_{\text{st}}$
target	Target value
th	Thermal quantity
tot	Total quantity
turb	Turbulent quantity

Part I

Introduction and objectives

Chapter 1

Introduction

In realistic combustion chambers, pressures can rise up to 10-50 bar [60, 95], because high pressure increases efficiency [8, 47]. This choice is optimal for industry but it represents a real challenge for simulations. Combustion involves several physical phenomena influencing each other at the same time (reactive chemistry, heat transfer, fluid mechanics, etc ...) [55]. Several scales need to be captured within a simulation: from microscopic scales where flame fronts live and interact with the surrounding turbulence [55] up to macroscopic scales since a gas turbine can be several meters long [92]. Facing high pressure, this disparity grows because pressure causes a decrease in length scale (reduction of flame front thickness, reduction of smallest turbulent scales too) [83, 111]. With that in mind, if a simulation must capture all events that happen during combustion, in gas turbines or in aeroengines for example, it becomes a nightmare in terms of computational cost. Ultimately, DNS is not an option here (most DNS papers are dedicated to academic studies [90, 97]) and LES represents the right balance between physical fidelity and computational cost.
It is possible to loosely summarize LES as a simulation that splits the problem into two parts: on one side, the resolved scales, where physics can be captured and is computed using a mesh that is coarser than DNS (typically of the order of 100 micrometers) but finer than RANS; on the other side, the unresolved scales, where physics occurs at a sub-grid level and therefore requires the presence of a model [103]. The frontier between these gets worse under high pressure. Even the resolution of the supposedly "resolved" flame front can represent an issue (as proved in this manuscript). If hydrogen is added in the equation, it tends to be worse: hydrogen, among many great features, has the particularity to create thinner flame fronts than other classical carbon-based fuels such as kerosene or methane (which also tends to decrease at high pressure) [88]. In addition, the intrinsic properties of hydrogen (very fast chemistry) promote the formation of diffusion flames. This type of flame is known, among other things, to be thinner than premixed flames [87] because they are strongly influenced by strain [23]. Strain decreases flame thickness and can be roughly linked to the velocity which can be quite high in industrial combustion chambers. Gathering all these mechanisms represents a challenge which has generated hundreds of papers over the last decades [85].

1.1 Hydrogen for gas turbines as a new fuel

Hydrogen is one of the most abundant elements in the universe and has been used industrially for more than a century [113]. Despite the growing attention it receives today as an energy vector, its traditional applications have primarily been associated with the chemical and petrochemical industries [49]. Most of the hydrogen currently produced worldwide is consumed in oil refining, ammonia synthesis, methanol production, and various industrial chemical processes [15, 35]. In these applications hydrogen acts as a feedstock rather than as an energy source. Consequently, hydrogen is not a new industrial molecule; what is new is the growing interest in its use as a fuel capable of supporting the transition toward low-carbon energy systems [20]. Environmental and economic ambitions of France and the rest of the world follow this trend during the past years [31, 72, 73, 75, 99].

The renewed interest in hydrogen originates from the need to reduce greenhouse-gas emissions while maintaining reliable and flexible energy production. Unlike fossil fuels, hydrogen contains no carbon atoms and therefore does not generate carbon dioxide during combustion, only water. This characteristic has positioned hydrogen as a potential energy carrier for sectors that are difficult to decarbonize through direct electrification alone, including heavy industry, long-distance transportation, and power generation [15, 35]. Hydrogen is therefore increasingly considered as a strategic component of future energy systems and an important pillar of net-zero emission strategies [109].

However, hydrogen should not be regarded as a primary energy source. Similar to electricity, hydrogen must first be produced from another energy source before it can be utilized [15]. Its environmental impact therefore depends not only on its final use but also on the production pathway employed. Several production routes currently exist, leading to the well-known hydrogen color classification summarized in Table 1.1 [2, 15, 35].

Although this color classification provides a convenient technological framework, it does not fully characterize the environmental performance of hydrogen production, which ultimately depends on life-cycle emissions and the carbon intensity of the energy used during production [15].

The environmental benefits of hydrogen are therefore closely linked to its production. Conventional fossil-based hydrogen production remains associated with significant carbon emissions, whereas renewable and low-carbon production pathways can substantially reduce the global warming potential of the hydrogen supply chain [15]. Nevertheless, large-scale deployment of green hydrogen requires considerable quantities of electricity, water resources, and dedicated infrastructure [35]. As a result, hydrogen should not be viewed as a universal solution to decarbonization but rather as a complementary energy vector whose deployment must be targeted toward applications where alternative low-carbon solutions are difficult to implement [35].

Beyond environmental considerations, hydrogen presents several technical advantages as a fuel. Its gravimetric lower heating value is approximately three times larger than that of conventional hydrocarbon fuels, making it an attractive energy carrier on a mass basis [101]. Furthermore, hydrogen can be produced from a wide variety of primary energy sources and enables long-term storage of renewable electricity through power-to-hydrogen pathways [15]. These characteristics have motivated significant research efforts toward hydrogen utilization in combustion systems, including industrial burners, internal combustion engines, and gas turbines [93, 101].

Table 1.1: Classification of the main types of hydrogen according to their production pathway. [2, 15, 35]

Type	**Production pathway**	**Energy source**	**Carbon emissions**
Grey hydrogen	Steam methane reforming (SMR)	Natural gas	High
Blue hydrogen	SMR with CO_2 capture and storage	Natural gas + capture and storage	Medium
Green hydrogen	Water electrolysis	Renewable electricity	Very low if powered by renewables
Yellow or pink hydrogen	Water electrolysis	Nuclear electricity	Very low
Black hydrogen	Coal gasification	Bituminous coal	Very high
Brown hydrogen	Lignite gasification	Lignite	Very high
Turquoise hydrogen	Methane pyrolysis	Natural gas + heat	Medium
White hydrogen	Natural extraction	Geological hydrogen	depends on extraction, purification, and transport

However, hydrogen burners are more confidential than its carbon-based competitors and require a tremendous amount of work from the science community to be able to fully control this fuel's properties [50, 51]. Hydrogen possesses a very low density, high diffusivity, wide flammability limits, low ignition energy, and significantly higher laminar flame speeds than most hydrocarbon fuels [62, 109]. These properties strongly influence flame stabilization, ignition behavior, combustion efficiency, and safety. In particular, hydrogen-air mixtures are more susceptible to flashback, flame acceleration, and under certain conditions, detonation than methane-air mixtures [109]. Consequently, replacing methane or kerosene with hydrogen cannot be achieved by a simple fuel substitution, as combustion systems designed for hydrocarbon fuels may experience substantial changes in performance and operating limits [62].

In fact, hydrogen combustion introduces several challenges for gas-turbine combustors. Owing to its high reactivity and flame speed, hydrogen increases the risks of flashback and combustion instabilities, while the elevated flame temperatures that may occur under certain operating conditions can lead to increased NOx [109]. Furthermore, hydrogen flames exhibit strong interactions between chemical kinetics, turbulent mixing, molecular diffusion, and heat release, making their behavior significantly different from that of conventional natural-gas flames which requires rethinking designs [62]. For example, calculations have shown that changes in injection properties (timing and duration for ex-

ample) can play a major role in efficiency [107]. This example is one of the many reasons that explain the need for research on injectors dedicated to hydrogen and the creation of the project PHYDROGEN of which this thesis is a part (details in the next subsection).

These challenges motivate the need for advanced predictive tools capable of describing hydrogen combustion under realistic operating conditions. Experimental investigations provide valuable information but are often limited by measurement accessibility, cost, and safety constraints. Consequently, numerical simulations have become an essential component of hydrogen-combustion research. Recent studies emphasize the growing role of computational fluid dynamics (CFD), including Reynolds-averaged simulations (RANS), large-eddy simulations (LES), and direct numerical simulations (DNS) [85], for investigating hydrogen ignition, flame stabilization, turbulent combustion, and safety-related phenomena [109]. Such approaches enable the analysis of coupled transport, turbulence, and chemical-reaction processes that can hardly be isolated experimentally.

As hydrogen progressively emerges as a major energy carrier for future low-carbon systems, the ability to accurately predict its combustion behavior becomes increasingly important. The development of reliable CFD methodologies for hydrogen flames is therefore not only a scientific challenge but also a technological necessity for the design of safe, efficient, and low-emission combustion systems. In this context, gas turbines constitute a particularly relevant application, where the strong coupling between turbulence, mixing, and chemical kinetics requires advanced modeling strategies. The present work contributes to this effort by focusing on the numerical simulation of turbulent hydrogen combustion and the development of modeling approaches suitable for high-fidelity CFD calculations of non-premixed flames.

1.2 PHYDROGEN project

Following the need for new injectors dedicated to hydrogen, a collaboration between DGAC (direction général de l'aviation civil), ONERA (Office national d'études et de recherches aérospatiales) and CERFACS (Centre européen de recherche et de formation avancée en calcul scientifique) has been created in 2022 under the name PHYDROGEN. The aim of this project is to develop a comprehensive set of analysis tools for the study and optimization of a hydrogen injector design for gas turbines. Combining experimental investigations and numerical simulations, the project is organized into three main research directions:

- Experimental workbench dedicated to the study of hydrogen injectors and combustion chambers at high pressure (at ONERA).
- Optimization strategy to determine a set of key parameters that control the hydrogen injection through a parametric study with simulations (at ONERA).
- High fidelity academic simulations and development of models dedicated to hydrogen (at CERFACS).

This thesis is conducted within the framework of the latter activity and has been cofunded by PHYDROGEN and CERFACS. In this framework, several objectives have been set. High pressure and high fidelity 3D LES have been conducted using the combustion

model TFLES, described in the next section, in the PhD of J. Bertsch [3]. In the present work, the focus is on the models themselves, especially for diffusion flames.

1.3 Modeling multi-regime flames for hydrogen non-premixed flames

As mentioned above, the presence of hydrogen favors the formation of diffusion flames in combustion chambers. At atmospheric pressure the issue can be managed through optimal LES meshing strategy [24, 36], yet at high pressure it is not the same situation. Physical length scales tend to fall below sub-grid scales, which hinders the fidelity of the simulation (as it is demonstrated in this manuscript).
At this point, we can separate two issues:

- Flame/Turbulence Interaction (FTI) - is the flame/interaction model correct ?
- Resolution (RES) - is the flame structure produced through this model resolved on the flame grid ? There is no guarantee that a given FTI model will lead to a front which can be resolved on an arbitrary LES grid.

The second constraint (spatial resolution) has been often ignored for a long time but it becomes critical at high pressures. No FTI model can work if the 'resolved' front is resolved on two points. This thesis focuses on resolution issues even if it is difficult to separate from the first one (FTI). We begin here by discussing FTI models.

Existing LES models for non-premixed flames can be grouped into three broad categories according to the physical quantity used to describe combustion: flame-surface-based models, statistical models, and turbulent-mixing models [33, 104]. Although these approaches originate from different theoretical frameworks, they all aim to reconstruct the unresolved reaction rates resulting from the interaction between turbulence and chemistry. They all use the resolved fields to build the subgrid quantities. A summary is presented in the following subsections.

1.3.1 Flame-Surface Approaches

Flame-surface formulations originate from premixed-combustion modeling and describe combustion through the geometry and evolution of the flame front [104]. Their central assumption is that the flame structure can be represented by a thin reaction layer whose propagation is influenced by turbulence-induced wrinkling. The amount of flame surface contained within a control volume is commonly quantified through the flame surface density (FSD),

$$\Sigma = |\nabla \psi|, \tag{1.1}$$

where ψ denotes a scalar identifying the flame front, such as a progress variable or mixture-fraction isosurface. In LES, only a filtered flame surface density is resolved, and the unresolved contribution must be modeled. This leads naturally to the definition of a subgrid wrinkling factor,

$$\Xi_\Delta = \frac{\Sigma_{\text{tot}}}{\Sigma_{\text{res}}}, \tag{1.2}$$

which represents the ratio between the total flame surface and the resolved flame surface.

One important family of methods relies on filtered one-dimensional flame solutions. In these approaches, detailed laminar flame calculations are filtered on a grid which has the same size as the LES grid ($\sim 100\mu m$) and tabulated in advance, allowing the effects of chemistry to be incorporated into LES without solving a complete chemical mechanism during runtime. The filtered flame structure provides closure relationships for unresolved quantities, reducing computational cost while retaining detailed chemical information.

A representative example is the F-TACLES (Filtered Tabulated Chemistry for LES) methodology [34], in which filtered laminar flame solutions are used to reconstruct reaction rates and thermochemical variables. In this framework, the filtered progress-variable equation can be written as

$$\frac{\partial \bar{\rho}\tilde{c}}{\partial t} + \nabla \cdot (\bar{\rho}\tilde{\mathbf{u}}\tilde{c}) = \nabla \cdot (\rho D \nabla c) + \overline{\dot{\omega}_c}, \tag{1.3}$$

where the filtered reaction rate $\overline{\dot{\omega}_c}$ is obtained from pre-computed filtered flamelet solutions. Originally developed for premixed flames, the method has subsequently been extended to partially premixed and diffusion-flame configurations [21]. By construction, F-TACLES satisfies the RES criterion because flames are directly projected on the available LES mesh. One limitation of such approaches is that the filtering operation removes part of the flame-surface information. Consequently, additional modeling is required to account for unresolved flame wrinkling and subgrid flame-surface growth.

Another widely used flame-surface model is the Thickened Flame Model (TFLES) [17]. The basic idea is to artificially increase the flame thickness by modifying transport and reaction terms so that the flame can be resolved on the LES grid. The molecular diffusivity is increased by a thickening factor F,

$$D^* = FD, \tag{1.4}$$

while the reaction rate is reduced accordingly,

$$\dot{\omega}^* = \frac{\dot{\omega}}{F}. \tag{1.5}$$

This procedure preserves the laminar flame speed while increasing the flame thickness approximately as

$$\delta_L^* \approx F\delta_L. \tag{1.6}$$

which allows the resolved flame to be solved on the available LES grid by choosing F[1]. Since the thickening procedure reduces the flame surface area that would naturally be generated by turbulence, a compensating efficiency function or wrinkling factor must be introduced to recover the correct global reaction rate [13]. The filtered reaction rate therefore becomes

[1]Very large values of F will be needed for thin flames solved on coarse grids but from a theoretical point of view, this approach remains consistent and guarantees that the premixed flame will be resolved

$$\dot{\omega}_{LES} = \Xi_\Delta \frac{\dot{\omega}}{F}. \tag{1.7}$$

The success of TFLES depends strongly on the quality of the flame-wrinkling model used to represent unresolved flame-surface distortions in Ξ_Δ.

The concept of flame wrinkling is often linked to fractal descriptions of turbulent flames [38, 106]. In this framework, the flame surface is considered self-similar over a range of scales bounded by an outer cutoff related to the LES filter width Δ and an inner cutoff δ_c associated with the smallest physically relevant flame structures. The increase in flame surface area caused by turbulence can then be estimated through the fractal scaling law

$$\Xi_\Delta = \left(\frac{\Delta}{\delta_c}\right)^\beta, \tag{1.8}$$

where

$$\beta = B - 2, \tag{1.9}$$

and B denotes the fractal dimension of the flame surface. Although this methodology has been extensively investigated for premixed combustion, its application to non-premixed flames remains less established and requires additional assumptions regarding the nature of diffusion-flame surfaces [39].

1.3.2 Statistical Approaches

A second class of LES models is based on statistical descriptions of subgrid-scale fluctuations. Rather than explicitly tracking flame geometry, these methods represent unresolved combustion through probability density functions (PDFs).

The central idea is that thermochemical quantities such as temperature, species concentrations, or reaction rates can be expressed as statistical averages over the unresolved scales [37]. The filtered value of a scalar ϕ is obtained through

$$\widetilde{\phi} = \int \phi(\xi) P(\xi)\, d\xi, \tag{1.10}$$

where $P(\xi)$ represents the probability density function associated with the unresolved fluctuations.

The simplest implementation and a common choice for diffusion flames is the presumed PDF approach [19, 79]. In this methodology, the shape of the subgrid probability distribution is assumed *a priori*, typically using a beta distribution for mixture fraction,

$$P(Z) = \frac{Z^{a-1}(1-Z)^{b-1}}{B(a,b)}, \tag{1.11}$$

where a and b are determined from the resolved mean and variance of the mixture fraction z. This procedure provides a computationally efficient way of incorporating subgrid mixing effects into combustion calculations.

A more rigorous alternative is the transported PDF method [46]. Instead of prescribing the probability distribution, an additional transport equation is solved for the filtered density function itself,

$$\frac{\partial P}{\partial t} + \nabla \cdot (\mathbf{u}P) = \mathcal{M} + \mathcal{C}, \tag{1.12}$$

where $\mathcal{M}$ and $\mathcal{C}$ denote mixing and chemical source terms. This allows the evolution of unresolved fluctuations to be represented more accurately and avoids assumptions regarding the shape of the distribution. However, the method is significantly more expensive computationally [86].

Conditional Moment Closure (CMC) can also be interpreted within the statistical framework [56]. In CMC, chemical quantities are conditioned on a conserved scalar, typically the mixture fraction Z, so that

$$Q = Q(Z, t). \tag{1.13}$$

By solving transport equations for conditional averages, the method reduces the complexity of turbulence–chemistry interactions while preserving important information regarding local mixing states. CMC has proven particularly useful for capturing extinction and reignition phenomena in diffusion flames [16, 54].
CMC models, like PDF models, do not consider RES issue. It assumes that the method leads to fronts which will be resolved on the LES mesh, a doubtful proposition at high pressure which is not guaranteed by construction.

1.3.3 Mixing-Based Approaches

The third family of LES models focuses directly on turbulent mixing processes. These methods are based on the observation that, in non-premixed combustion, reaction rates are often controlled by the rate at which fuel and oxidizer are brought together by turbulent transport.

One of the most common approaches is the Eddy Dissipation Concept [96]. The Eddy Dissipation Concept assumes that chemical reactions occur within fine turbulent structures embedded in the flow. The exchange of mass and energy between these structures and the surrounding fluid determines the overall combustion rate. The reaction rate is generally expressed as

$$\dot{\omega} \propto \frac{\rho\, \gamma^*}{\tau^*}, \tag{1.14}$$

where γ^* is the fine-structure volume fraction and τ^* is the characteristic turbulent time scale. The model provides a practical compromise between detailed chemistry and computational efficiency and has been applied successfully to a variety of turbulent reacting flows. It has also been adapted on non-premixed flame too [65]

Another influential methodology is the Linear Eddy Model [53]. Rather than representing turbulent mixing through averaged quantities, Linear Eddy Model attempts to mimic the action of turbulent eddies on scalar fields using a sequence of stochastic rearrangement events. Molecular diffusion and chemical reaction remain governed by the one-dimensional transport equation.

$$\frac{\partial \phi}{\partial t} = D\frac{\partial^2 \phi}{\partial x^2} + \dot{\omega}_\phi, \tag{1.15}$$

while turbulent motions are represented through stochastic eddy-mapping operations. The model is particularly attractive because it captures small-scale mixing processes more explicitly than conventional closure approaches [9, 69]. Like CMC and PDF models, all these approaches do not consider the issue of spatial resolution RES. Only FTI is viewed as an objective.

1.3.4 Resolution issues

The previous sections show that resolution is not a criterion which is always taken into account to build subgrid FTI models. For example, presumed PDF expressions dedicated to non-premixed flames need the mixture fraction z and the scalar dissipation rate χ [80, 87]. Therefore if the physical length scales drastically drop, which is the case under high pressure, it might result in a critical situation where even resolved parameters are not well captured: the resolution becomes perilous and FTI models will ultimately fail. For example, the scalar dissipation rate $\chi = 2D(\nabla z)^2$ which is an input in many FTI models for diffusion flames becomes $\chi = 2D\left[(1-0)/\Delta\right]^2$ when the z profile is resolved only on two points separated by the mesh size Δ. This evaluation is totally controlled by the under resolution and does not depend on the flow anymore, an unacceptable solution. It is clear that the question of resolution must be addressed before questioning which FTI models need to be applied.
To study RES on FTI problems, this thesis uses two numerical tools: CANTERA for strained 1D flames and AVBP for multi-dimensional cases. The next two sections provide a short description of CANTERA and AVBP, the two numerical tools, one open-source and the other developed at CERFACS, that will be used throughout this manuscript.

1.4 CANTERA

CANTERA is used in this work for evaluating thermodynamic properties, transport properties, chemical source terms, and one-dimensional reacting-flow solutions [42]. The purpose of this section is twofold. First, it gives a concise description of CANTERA and of physical problems for which it is intended. Second, it records the governing equations and modeling hypotheses associated with the one-dimensional counterflow flame configuration implemented in CANTERA. This documentation is included so that the numerical results obtained from a CANTERA counterflow calculation can be interpreted as the solution of a clearly defined set of conservation equations, rather than a black-box computational output.

CANTERA automates the evaluation of chemical kinetic rates, thermodynamic state properties, and transport coefficients, so that detailed thermo-chemical models can be incorporated into reacting-flow calculations without reimplementing the underlying property and source-term evaluations for each application such as F-TACLES or TFLES [42]. The software can be accessed through several programming interfaces, including Python, Matlab, C/C++, and Fortran, and is used in a broad range of applications including

combustion, detonations, electrochemical energy conversion, fuel cells, batteries, plasmas, and thin-film deposition [42].
For one-dimensional flame calculations, CANTERA models flames stabilized in an axisymmetric stagnation flow and computes the solution along the stagnation streamline, corresponding to $r = 0$ [43]. A similarity solution is used to reduce the three-dimensional governing equations to a one-dimensional form along the axial coordinate z [43]. In the counterflow configuration, this reduction is particularly useful because the flame is represented through the axial variation of velocity, temperature, species composition, and diffusive fluxes between opposed streams. The governing equations implemented by CANTERA for steady axisymmetric stagnation flow follow the formulation given by Kee et al. [43, 52].
The principal geometrical hypothesis is axisymmetry. The axial coordinate is denoted by z, the radial coordinate by r, the axial velocity by u, and the radial velocity by v. CANTERA introduces the scaled radial velocity

$$V = \frac{v}{r} \tag{1.16}$$

The solution is evaluated along the stagnation streamline, where tangential velocity is assumed to be zero [43]. Under these assumptions, the steady stagnation-flow problem is written as a set of ordinary differential equations in z. The model is applicable both to ideal fluids and to non-ideal fluids described using real-gas equations of state [43]. In a standard ideal-gas counterflow flame calculation, the thermodynamic state is therefore closed using the selected gas-phase equation of state and the specified chemical mechanism.

The continuity equation solved by CANTERA for the one-dimensional stagnation flow is

$$\frac{\partial \rho u}{\partial z} + 2\rho V = 0, \tag{1.17}$$

where ρ is the density. The factor $2\rho V$ arises from the axisymmetric radial contribution and derived from Section 7.2 of Kee et al. [52]. The radial momentum equation is

$$\rho u \frac{\partial V}{\partial z} + \rho V^2 = -\Lambda + \frac{\partial}{\partial z}\left(\mu \frac{\partial V}{\partial z}\right), \tag{1.18}$$

where μ is the dynamic viscosity and $\Lambda = \frac{1}{z}\frac{dP}{dz}$ is the pressure eigenvalue, which is independent of z [43]. This equation expresses the balance between axial convection of the scaled radial velocity, radial strain, the pressure-eigenvalue term, and viscous diffusion in the similarity coordinate.

The energy equation implemented by CANTERA is

$$\rho C_p u \frac{\partial T}{\partial z} = \frac{\partial}{\partial z}\left(\lambda \frac{\partial T}{\partial z}\right) - \sum_k j_k \frac{\partial h_k}{\partial z} - \sum_k h_k W_k \dot{\omega}_k, \tag{1.19}$$

where C_p is the mixture heat capacity at constant pressure, T is the temperature, λ is the thermal conductivity, j_k is the diffusive mass flux of species k, h_k is the species enthalpy, W_k is the species molecular weight, and $\dot{\omega}_k$ is the molar production rate of species k [43].

The species equations are written, for each species k, as

$$\rho u \frac{\partial Y_k}{\partial z} = -\frac{\partial j_k}{\partial z} + W_k \dot{\omega}_k, \tag{1.20}$$

where Y_k is the mass fraction of species k [43]. This equation balances axial convection, divergence of diffusive mass flux, and chemical production or consumption. The local composition in a counterflow flame is therefore determined not only by chemistry, but also by the transport model selected for j_k.

CANTERA computes the species diffusive mass fluxes j_k using either a mixture-averaged or a multicomponent formulation [43]. In the mixture-averaged formulation, an intermediate flux is first computed as

$$j_k^* = -\rho \frac{W_k}{\overline{W}} D'_{km} \frac{\partial X_k}{\partial z}, \tag{1.21}$$

where $\overline{W}$ is the mean molecular weight, D'_{km} is the mixture-averaged diffusion coefficient for species k, and X_k is the mole fraction of species k. A correction is then applied:

$$j_k = j_k^* - Y_k \sum_i j_i^*. \tag{1.22}$$

This correction ensures that the sum of the species mass fluxes is zero, a condition that is not automatically guaranteed by the uncorrected mixture-averaged formulation [43]. This constraint is necessary because diffusive transport redistributes species mass without creating net mass flux in the mixture frame. Consequently, a counterflow simulation must be documented not only by its chemical mechanism and boundary conditions, but also by whether mixture-averaged or multicomponent transport is used and whether Soret diffusion is activated, which is not the case in this thesis.

Boundary conditions must be defined to close the one-dimensional problem. At an inlet boundary, CANTERA supplies the temperature T_0, species mass fractions $Y_{k,0}$, scaled radial velocity V_0, and mass flow rate $\dot{m}_0$ [43]. At the inlet point $z = z_{\text{in}}$, the imposed conditions include

$$T(z_{\text{in}}) = T_0, \tag{1.23}$$

$$V(z_{\text{in}}) = V_0, \tag{1.24}$$

$$\dot{m}_0 Y_{k,\text{in}} - j_k(z_{\text{in}}) - \rho(z_{\text{in}}) u(z_{\text{in}}) Y_k(z_{\text{in}}) = 0. \tag{1.25}$$

When the mass flow rate is specified, CANTERA also imposes

$$\rho(z_{\text{in}}) u(z_{\text{in}}) = \dot{m}_0. \tag{1.26}$$

For a counterflow diffusion flame, the flame structure is obtained from the inlet compositions, inlet temperatures, transport model, pressure, and imposed mass fluxes.
In summary, a one-dimensional counterflow flame simulation in CANTERA is the solution of a steady, axisymmetric stagnation-flow problem reduced to one dimension by a similarity formulation. The dependent variables include the axial velocity, scaled radial velocity, temperature, species mass fractions, and, where applicable, pressure eigenvalue and auxiliary control variables. The governing equations couple continuity, radial momentum, energy, species transport, molecular diffusion, and finite-rate chemistry. Therefore, a CANTERA counterflow calculation must specify the chemical mechanism, thermodynamic model, transport model, Soret setting, boundary conditions, pressure, inlet compositions, inlet temperatures, mass fluxes, and any continuation or flame-control strategy used in the computation [42, 43].

1.5 AVBP

AVBP is a compressible Computational Fluid Dynamics solver developed at CERFACS for the simulation of multi-species turbulent reacting and non-reacting flows [11]. AVBP is closely connected to applications in aeronautical combustion chambers and gas turbines with complex flows.
From an implementation perspective, AVBP is written in Fortran 90, designed for High Performance Computing platforms and can run on CPU-based nodes or GPU-based nodes. AVBP is a massively parallel, fully compressible, reactive turbulent-flow solver that solves the multi-species reactive Navier-Stokes equations in conservative form on both structured and unstructured grids, using a cell-vertex finite-volume method [94].
The governing equations solved by AVBP are the Navier-Stokes equations [87]. The equations below use ρ for density, u_i for the i-th velocity component, x_i for the spatial coordinate, p for pressure, Y_k for the mass fraction of species k, $V_{k,i}$ for the i-th component of the diffusion velocity of species k, and $\dot{\omega}_k$ for the species reaction rate.

The conservation equation for momentum is written in conservative form as

$$\frac{\partial \rho u_j}{\partial t} + \frac{\partial \rho u_i u_j}{\partial x_i} = -\frac{\partial p}{\partial x_j} + \frac{\partial \tau_{ij}}{\partial x_i} + \rho \sum_{k=1}^{N} Y_k f_{k,j}, \tag{1.27}$$

where $f_{k,j}$ denotes the volume force acting on species k in direction j. The viscous stress tensor is given by

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right), \tag{1.28}$$

with μ the dynamic viscosity and δ_{ij} the Kronecker delta [87].

The species conservation equation is written for each species as

$$\frac{\partial \rho Y_k}{\partial t} + \frac{\partial}{\partial x_i} \left[\rho \left(u_i + V_{k,i} \right) Y_k \right] = \dot{\omega}_k, \qquad k = 1, \ldots, N. \tag{1.29}$$

For a reacting multi-species calculation, this equation is therefore one of the locations where the combustion model and transport model enter the resolved conservation problem. The detailed chemical mechanism in this thesis is San Diego mechanism [100].
The total non-chemical energy E, defined in the source as the sum of sensible and kinetic energies, satisfies

$$\frac{\partial \rho E}{\partial t} + \frac{\partial}{\partial x_i} \left(p u_i + \rho u_i E \right) = \dot{\omega}_T - \frac{\partial q_i}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\sigma_{ij} u_i \right), \tag{1.30}$$

where $\dot{\omega}_T$ is the total heat release due to combustion and $\sigma_{ij} = \tau_{ij} - p\delta_{ij}$ is the stress tensor [87]. The heat-release and heat-flux terms are defined as

$$\dot{\omega}_T = -\sum_{k=1}^{N} \Delta h^0_{f,k} \dot{\omega}_k, \qquad q_i = -\lambda \frac{\partial T}{\partial x_i} + \rho \sum_{k=1}^{N} h_k Y_k V_{k,i}, \tag{1.31}$$

where $\Delta h^0_{f,k}$ is the formation enthalpy of species k, λ is thermal conductivity, T is temperature, and h_k is the enthalpy of species k.
The diffusion treatment is a central question as in CANTERA. AVBP uses a simplified

form based on the Hirschfelder-Curtiss approximation [48]. The approximation is first expressed as

$$V_k X_k = -D_k \nabla X_k, \qquad D_k = \frac{1 - Y_k}{\sum_{j \neq k} X_j / D_{jk}}, \tag{1.32}$$

where X_k is the mole fraction of species k and D_{jk} is a binary diffusion coefficient. Note that Prandtl and Schmidt numbers are set as constant and are different for each species [48].
Because the Hirschfelder-Curtiss approximation does not conserve global mass, AVBP introduces a correction velocity V^c. The corrected diffusion velocity is written

$$V_k = -D_k \frac{\nabla X_k}{X_k} + V^c. \tag{1.33}$$

With this correction, the species equation is rewritten as

$$\frac{\partial \rho Y_k}{\partial t} + \frac{\partial}{\partial x_i} \left[\rho \left(u_i + V_i^c \right) Y_k \right] = \frac{\partial}{\partial x_i} \left(\rho D_k \frac{W_k}{W} \frac{\partial X_k}{\partial x_i} \right) + \dot{\omega}_k, \tag{1.34}$$

where W_k is the molecular weight of species k, W is the mixture molecular weight, and the mole and mass fractions are related by $X_k = W Y_k / W_k$ [48].
The viscosity model contains an additional assumption that is especially relevant for hydrogen-air mixtures. A conventional AVBP model for transport coefficients uses a power law. However, for hydrogen-air mixtures, this simple power-law treatment is not suited because hydrogen viscosity is much lower than air viscosity. Therefore AVBP uses a Wilke law where the mixture viscosity is written [110].

$$\mu = \sum_i \frac{X_{\mathrm{i}} \mu_{\mathrm{i}}}{X_{\mathrm{i}} + \sum_{j \neq i} X_j \Phi_{i,j}} \tag{1.35}$$

with

$$\Phi_{i,j} = \frac{\left[1 + \left(\frac{\mu_{\mathrm{i}}}{\mu_{\mathrm{j}}} \right)^{1/2} \left(\frac{W_{\mathrm{j}}}{W_{\mathrm{i}}} \right)^{1/4} \right]^2}{\sqrt{8} \left(1 + \frac{W_{\mathrm{i}}}{W_{\mathrm{j}}} \right)^{1/2}}, \tag{1.36}$$

The governing equations are solved using the Lax-Wendroff scheme [59] which is an explicit finite volume centered scheme and is second order accurate in space and time.
In this thesis, time advancement is constrained through a CFL number of 0.7, chosen by the user and kept below the critical CFL threshold of the convection scheme.

1.6 Objectives of the thesis

Theory, CANTERA and AVBP are used here to construct a combustion model for diffusion flames starting from the flamelet assumption. This hypothesis states that under reasonable turbulent conditions (moderate Reynolds number) the turbulent flame front can be seen as a collection of laminar stagnation point flames. Thus, a full comprehension of this type of flame/configuration must be achieved to create a foundation of knowledge that will help build a model. It is primordial to identify the key parameters that characterize strained flames. Moreover, it is well-known that a diffusion flame, as opposed to a

premixed one, is subjected to high physical variations as a function of the strain which is not a trivial variable to define. In fact, there seems to be no global consensus on which definition the community should use for strain and several different expressions can be found in the literature. Since strain is the foundation that dictates the global behavior of non-premixed flames, it should be well defined. To tackle that, in chapter 2 a full study about strain definitions that ensure proper evaluations in configurations such as Strained stagnation Point Laminar Flame (SPLF) is carried out, which is a typical configuration for non-premixed flames under flamelet assumption. We will see that the intrinsic properties of hydrogen such as low density induce high variations of density through the flame front (in a Hydrogen-air SPLF context) which invalidates some models or definitions of strain. In fact, the present study is conducted on methane and hydrogen to ensure robustness of our definition, which reveals that high variations of density along the flame front play a non-negligible role on strain. This information leads us to reconsider the classical 1D model for non-premixed SPLF present in the literature.
Chapter 3 recalls a historical 1D model, assuming constant density, that aims to estimate the mixture fraction z, which unlocks the access to other key parameters such as the local flame thickness δ_f, the scalar dissipation rate χ or the total fuel consumption $\dot{\Omega}_F$. In the latter, one of the major hypotheses is that the density is considered as constant (which is mostly true for methane-air flames). Yet, the previous study on strain highlights the influence of such a variation, which is present for hydrogen-air SPLF.
A new 1D model for SPLF is then built under the main consideration of variable density, simply named variable density model (VD) as opposed to constant density model (CD) from the literature. In this context key parameters of diffusion flames have been studied over a wide range of operating points (variation of pressure, oxidizer/fuel temperature) as a function of strain. These two chapters establish a strong foundation and practical tools for understanding how to face resolution issues mentioned above.
This resolution issue must be fully apprehended if we want to build a model dedicated to it, the literature has already highlighted the fact that resolution can truly spoil the results even on non-premixed configurations [7]. To do so, chapter 4 starts with the introduction of a metric to capture the resolution in a non-premixed flame front, called Flame Resolution Index (FRI). FRI tells us, during the simulation, how many points are used in the flame. This criterion will be scrutinized in depth to reach the following ultimate goal: be able to rigorously separate resolved flame fronts from unresolved ones. Under the scope of this purpose, several test cases are performed at steady and unsteady conditions which are the SPLF configuration and the 2D interaction between a non-premixed flame and a couple of vortices (Flame/Vortex Interaction FVI), respectively. With the analysis of the results, a characterization of the question "in what way an unresolved situation impacts local and global behaviors of the flame ?" is addressed.
After gathering all results and unlocking answers to resolution questions, chapter 5 presents a model dedicated to ensure a proper resolution in the whole flame front called Diffusion Front Thickness Control (DFTC). Starting from the results of chapter 4, the model is based on a dynamic monitoring of the resolution index (FRI) during the simulation. DFTC follows the same underlying philosophy as TFLES [17, 61]: in other words we will thicken the flame. Thickening a non-premixed flame has already been done in the past [22, 25], yet it is the first time that the dynamic evolution of the thickening factor is solely designed to ensure proper spatial resolution in the flame front. Additionally, the require-

ments that the model must satisfy have been defined and a set of test cases has been used to challenge DFTC against these requirements. SPLF is the steady reference case and a 2D Flame/Turbulent interaction (FTI) is added as the unsteady reference to ensure the validity of the model. This chapter also highlights the range of action, in quantitative terms, available for adjusting the parameters in order to meet the different requirements. Finally, since TFLES and DFTC share the same spirit, a comparison is done between thickening premixed flames (TFLES) and thickening diffusion flames (DFTC) to underline the potential of DFTC showing that thickening diffusion flames can be as reasonable as thickening premixed flames.

Part II

Laminar stagnation point flames: Theory

To study turbulent flames and build LES models for simulation, a basic approach is to use flamelet ideas where the flames present in a turbulent flame brush are assumed to share many properties with laminar, stagnation point flames. This idea is the basis of the flamelet method and provides not only guidelines to guess how turbulent diffusion flames will react to strain, stretch, unsteady effects but also to incorporate these mechanisms into LES models.
In this part, two basic notions related to laminar, stagnation point, diffusion flames are revisited because they will be needed in Part III where LES models are developed. These two issues are:

- The exact definition of strain in stagnation point, diffusion, laminar flames and in turbulent flames. Since strain (or stretch which is equal to strain in the planar flames used for flamelet prototypes in all models) is the quantity used by all methods to predict the structure of diffusion flames and since diffusion flames are fully controlled by local strain, using coherent, exact definitions of strain in a stagnation point, diffusion flame is required to be able to compare for example a laminar flame computed with CANTERA and a local turbulent flamelet isolated in a 3D LES field. Although the definition of strain [10, 87] is unequivocal, its application in a strained diffusion flame is not and multiple definitions can be found for strain [14, 42, 76, 87]. The main reason for this is that strain cannot be known a priori before the full fields of velocity and density are known so that many authors have searched for a priori definitions of strain using only boundary conditions such as the inlet velocities of hydrogen and air and the distance between injectors in laminar flames. This question is discussed in detail in Chapter 2 where various definitions of strain are presented and compared in laminar, stagnation point flames computed with CANTERA or AVBP.

- The extension of classical constant density approaches to stagnation point, diffusion, laminar flames to cases where density changes are accounted for, a property which may be required for hydrogen air flames, where density changes are very strong. This theory is presented in details in Chapter 3. The main issue here is to recognize that LES models use many relations directly coming from constant density approaches and that this may lead to uncertainties in the final LES models. For example, local scalar dissipation and local strain rate in turbulent flows are often related using the constant density solution [87] ($\chi = a/\pi f(z)$), a simplification which may be detrimental in an LES approach. Similarly, quenching limits of flamelets are often taken from constant density (CD) theories and directly included in LES models. Chapter 3 describes the derivation of a full model assuming infinitely fast chemistry like the standard model but relaxing the constant density approximation. Relaxing this CD assumption is critical in H_2-Air flames not only because dilatation due to heat release but also because H_2 and air, even for cold flow, have density variations of 1 to almost 15 suggesting that density changes cannot be neglected. Constant density and variable density solutions are compared to numerical solutions to determine whether going to variable density formulae in LES models is required or not.

Chapter 2

Evaluations of strain in laminar diffusion flames

This chapter recalls the classical theory presented in textbooks to express strain (or stretch) in stagnation point flames and discusses why this field needs to be revisited to build LES models. Our first objective is to compare flame structures observed in DNS or LES, using AVBP, with the same structures observed in axisymmetric stagnation flames, using CANTERA. This is usually done in flamelet models by saying that the AVBP flamelet and the CANTERA one are the same if they have the same strain (for the CANTERA flame, this means having the same stretch since these flames are plane and have no curvature term) [1]. This is a reasonable approach, used by most of the community, but it relies on one assumption: strain can be measured in DNS/LES of AVBP and in the axisymmetric flame of CANTERA in the same way.
An examination of the literature shows that this is actually a weak point and that authors use inconsistent definitions which are not the same. This chapter tries to clarify this issue. Figure 2.1 illustrates the practical difficulty. In almost all analysis of DNS/LES results, a usual exercise is to compare flamelet structures extracted from DNS or LES with laminar stagnation point flame extracted from CANTERA. This exercise allows validation of models, to predict when a flamelet is quenched or not etc... However, the exercise is valuable only if the two flamelets (AVBP on one side, CANTERA on the other side) are compared for the same stretch. And stretch definitions are not the same in AVBP and CANTERA. In AVBP, in a turbulent flame, stretch requires necessarily the use of Candel & Poinsot definition [10] $\kappa = \partial_{x_j} u_i \left(\delta_{ij} - n_i n_j\right)$. In CANTERA, multiple definitions are used, sometimes based on local velocity gradients and sometimes on the inlet jet velocities and the distance between the burners H [14, 42, 76, 87]. We will show below that this is a source of errors and propose a single definition of stretch to use in AVBP and CANTERA: $\kappa = 1/\rho_{st}\ \partial_y \left(\rho v\right)|_{z=z_{st}}$ where v is the normal velocity to the front flame and $z = z_{st}$ the stoichiometric line.

[1] In this part, we will use "strain" to describe "stretch" in many places, since for diffusion flames, propagation speeds are zero and these two terms, strain and stretch, are equal

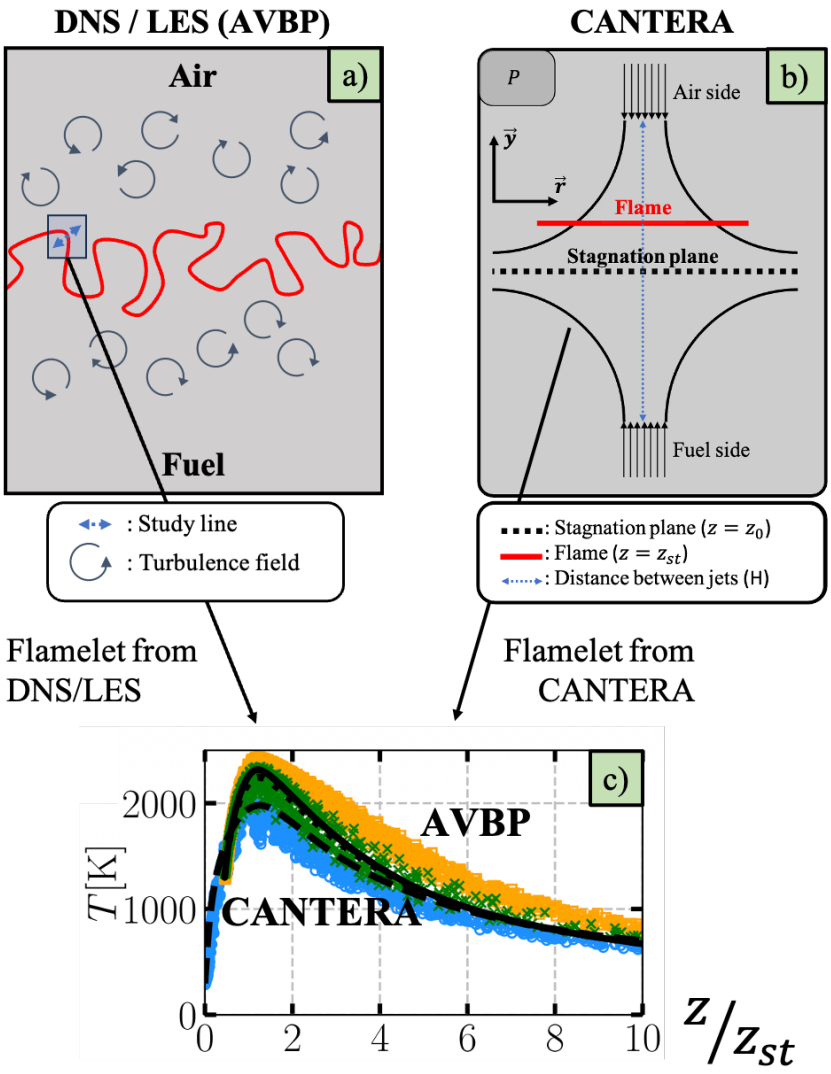


Figure 2.1: a) DNS/LES on AVBP where a diffusion front is wrinkled under the effect of turbulence. b) CANTERA stagnation point diffusion flame. c) Scatter plot displaying temperature versus z/z_{st} for CANTERA strained flames (lines) vs AVBP (points) in the HYLON burner [105]. The similarity between the data from a) and b) show the validity of the flamelet assumption.

2.1 Strain in stagnation diffusion flames: exact definition and approximate models

Strain (or stretch) is a key variable to describe a diffusion flame. Most variables of interest in flames can be evaluated as a function of it. For example, total heat release $\dot{\Omega}_T$ or the thickness δ_f of a diffusion flame both depend strongly on strain. Additionally, the scalar dissipation rate χ is proportional to strain [87] [2] and this variable is the foundation of several flame turbulence interaction (FTI) models [80, 84, 87].
Strain should be determined precisely so that flames can be compared. To do so, this section proposes to review definitions and models found in the literature and add definitions that will allow to capture strain accurately in configurations considered in this thesis. The purpose of this section is to identify consistent definitions of strain that can be easily evaluated and compared throughout any simulation.

2.1.1 Strain definitions

Strain is named by the letter a in this thesis. For the sake of clarity, we distinguish our definitions of a from other evaluations found in the literature by indicating it with capital letters. For example, $\mathscr{A}_1$ stands for the first model while a_1 characterizes the first true definition.
We begin by deriving "true" definition of strain and stretch: starting from the basics, any flame can be subject to stretch, named κ, which represents the alteration of the flame front under the effect of a surrounding non-uniform velocity field. Stretch is the temporal derivative of the flame surface, named A, divided by the flame surface itself [10, 68]:

$$\kappa = \frac{d_t A}{A} \tag{2.1}$$

This definition is exact but difficult to use. It is possible to express stretch as a function of the two mechanisms which create stretch:

1. the flow velocity gradient (strain)
2. the flame curvature and displacement speed.

Starting from Eq. (2.1), a derivation of the variation of A shows that strain can be divided into two effects: the curvature term, named κ_{curv}, and the strain term a as shown by the following expression [10]:

$$\kappa = \kappa_{curv} + a \tag{2.2}$$

where $\kappa_{curv} = -s_d(1/\mathscr{R}_1 + 1/\mathscr{R}_2)$ and $a = (\delta_{ij} - n_i n_j)\partial_{x_j} u_i$, s_d is the displacement speed, $\mathscr{R}_1$ and $\mathscr{R}_2$ are the flame surface radii of curvature, δ_{ij} is the Kronecker delta, n_i and n_j are components of the unit vector normal to the flame surface, u the flow velocity and x the coordinates.
These definitions will be our starting point. They can be considered in any configuration as long as there is a flame or even without flame, for a quenched diffusion flame for example.

[2] For example in a constant density theory [87], and in a stagnation point flame configuration, χ is expressed directly as a function of strain: $\chi = a\ f(z)$

However, despite the elegance of the upper definition, its evaluation during simulation can be tricky. Theoreticians indicate that Eq. (2.2) is valid only for infinitely thin flames. In the real world, flames have a thickness and using equation (2.2) requires another piece of information: "where" it is measured. This can only be done by having velocity profiles through the front, something which needs either a experiment or a simulation. This explains why models for a using only boundary values have been developed during the past decades. For example, for a stagnation diffusion point, flame created by two impinging jets separated by a distance H, a is often evaluated as $(|v_{fuel}| + |v_{oxy}|)/H$ where v_{fuel} and v_{oxy} are the two jet velocities. This is only a "model" and not a true evaluation of Eq. (2.2). Many of these models are linked to a specific flame configuration. In this thesis, we focus on non-premixed flames. Thus, models shown below are specific to these flames and more specifically to laminar stagnation point flames schematized in figure 2.2.

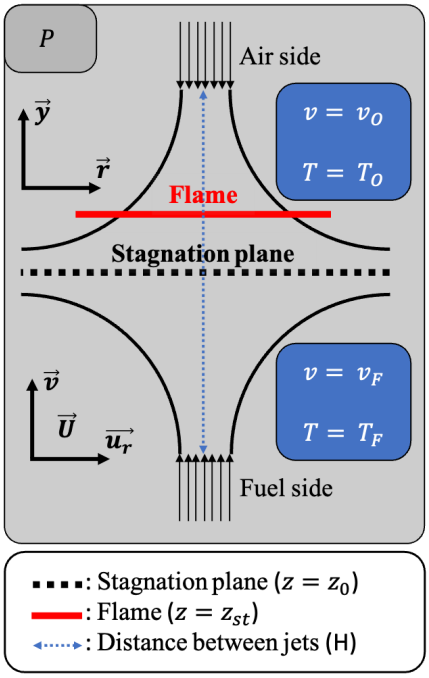


Figure 2.2: Steady, axisymmetric, strained diffusion flame (CANTERA)

The literature proposes multiple definitions for the strain of a planar diffusion flame where $\kappa_{curv} = 0$ and, it seems unclear which one is the best as no consensus has been clearly established. The following definitions present the most common ones found in the literature or implemented in codes such as CANTERA. They are all based on the profile of the velocity v normal to the flame and are summarized in Table 2.1 (along the y axis represented in fig. 2.2) [14, 42, 76, 87]:

Table 2.1: Models and definitions of strain from the literature.

Name	Reference	Location	Type	Definition
$\mathscr{A}_1$	[87]	Global	Model	$\frac{\lvert v_{fuel}\rvert + \lvert v_{oxy}\rvert}{H}$ (2.3)
$\mathscr{A}_2$	[14, 76]	Global	Model	$\frac{2v_{oxy}}{H}\left(1 + \frac{v_{fuel}\sqrt{\rho_{fuel}}}{v_{oxy}\sqrt{\rho_{oxy}}}\right)$ (2.4)
$\mathscr{A}_3$	[42]	Local	Definition	$\max\left[\left\lvert\frac{\partial v}{\partial y}\right\rvert\right]$ (2.5)
$\mathscr{A}_4$	[42]	Local	Definition	$\left\lvert\frac{\partial v}{\partial y}\right\rvert\Big\rvert_{z=z_{st}}$ (2.6)

where v refers to the velocity along the y axis (y is the flame normal direction), z the mixture fraction, H the distance between the jets and ρ the density. All of them try to obtain the exact definition of stretch (2.1). These models will be compared to true definitions in the next section. It is relevant to note that the two first are extracted from the literature and the two last ones are implemented, by default, in CANTERA.

2.1.2 True definition of strain in an axisymmetric counterflow configuration

First Eq. (2.1) is developed to obtain exact expressions of stretch which are easier to use. Note that theoreticians point out that stretch is properly defined for infinitely thin flames. In DNS and LES, flames are not thin and we need to choose where Eq. (2.1) will be evaluated. For diffusion flames, this will be the stoichiometric surface where $z = z_{st}$ (CANTERA has the same thought with Eq. 2.6).
In an axisymmetric counterflow configuration (such as in CANTERA) the flame surface can be seen as a disk that will grow with time under the effect of strain. Figure 2.3 below shows the evolution of the flame surface due to strain close to the flame axis ($r \simeq 0$):

We can have a first estimation of the temporal variation of the flame area:

$$d_t A = \frac{A\left[r(t+\Delta t)\right] - A\left[r(t)\right]}{t+\Delta t - t} = \frac{\pi\left(r+\Delta r\right)^2 - \pi r^2}{\Delta t} \approx 2\pi r \frac{\Delta r}{\Delta t} \tag{2.7}$$

Since diffusion flames have no intrinsic speed and are propagated only by the flow:

$$u_t = \frac{\Delta r}{\Delta t}$$

Noting that u_r/r is equal to $\partial_r u_r$ for small r values, this expansion gives a first true expression for the strain a in an axisymmetric stagnation point flame:

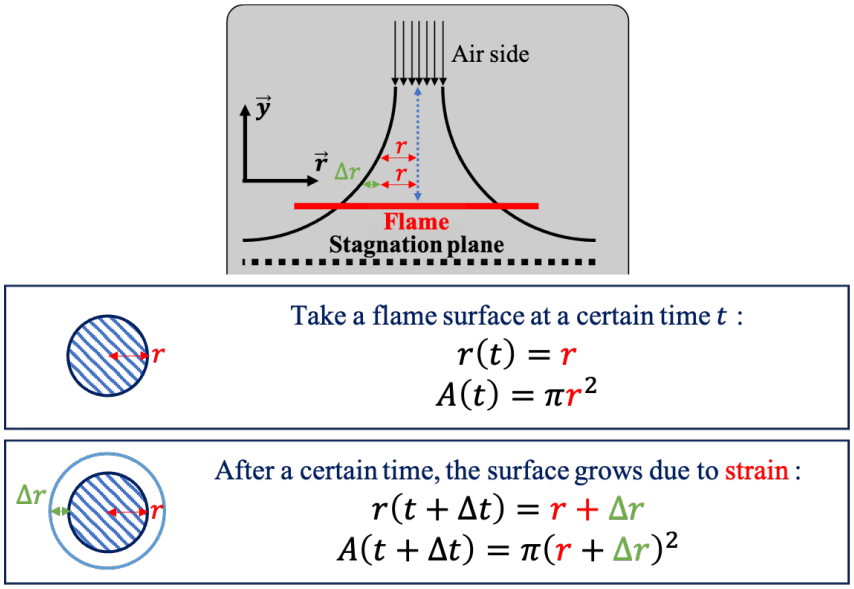


Figure 2.3: Flame strain in axisymmetric flows.

$$a_1 = 2\partial_r(u_r)$$

It is not the only true definition that we can establish. Using mass conservation, strain can also be expressed as a function of v, the velocity normal to the flame:

$$\frac{1}{r}\partial_r(\rho r u_r) + \partial_y(\rho v) = 0 \tag{2.8}$$

so that,

$$\partial_y(\rho v) = -\frac{1}{r}\left[r u_r \partial_r(\rho) + \rho r \partial_r(u_r) + \rho u_r\right] = -\rho\left[\partial_r(u_r) + \frac{u_r}{r}\right] = -2\rho\partial_r(u_r)$$

because $\partial_r(\rho) = 0$ in a planar stagnation point flame. This leads to a second true definition called a_2:

$$a_2 = -\frac{\partial_y(\rho v)}{\rho}$$

Note that the same derivation can be done for 2D symmetric flames (by replacing $(\vec{r}, \vec{y})$ by $(\vec{x}, \vec{y})$) and $\vec{U} = (u_r, v)$ by $\vec{U} = (u_x, v)$ in Fig. 2.2) which leads also to two definitions.

Table 2.2: Local definitions of strain based on Eq. (2.1).

System	Axisymmetric (u_r, v)	2D symmetric (u_x, v)
Solver	CANTERA	AVBP
a_1	$2\partial_r(u_r)$ (2.9)	$\partial_x(u_x)$ (2.10)
a_2	$-\frac{\partial_y(\rho v)}{\rho}$ (2.11)	$-\frac{\partial_y(\rho v)}{\rho}$ (2.12)

Table 2.2 summarizes strain definitions for axisymmetric and 2D flames. When strain is expressed as a function of the normal velocity v to the flame the same expression is obtained for all flames. These definitions can be expressed everywhere in the simulation. However, a non-premixed flame at a stagnation plane is defined by a single value of strain. If we assume that density is constant, strain can be assumed constant along the flame surface [87]. In fact, under the constant density assumption, the velocity field is well-known, for example $u_x = ax$ and $v = -ay$ in a 2D symmetric flame. This is compatible with equations (2.10) and (2.12) at any point in the flame only if density is constant. In a flow where density changes, this consideration is no longer valid so that the place where $\partial_y \rho v$ is measured controls the strain value which is obtained. In this context, we must take our estimation where the flame burns because the gradients of v or of ρv change everywhere in the flow. Here, strain is evaluated at stoichiometry $z = z_{st}$ (Tab. 2.3).

Table 2.3: Final definitions of strain based on Eq. (2.1) evaluated at stoichiometry.

System	Axisymmetric	2D symmetric
Solver	CANTERA	AVBP
a_2	$-\frac{\partial_y(\rho v)\|_{z=z_{st}}}{\rho_{st}}$ (2.13)	$-\frac{\partial_y(\rho v)\|_{z=z_{st}}}{\rho_{st}}$ (2.14)

2.2 Comparison of models for strain rate in stagnation point laminar diffusion flames

2.2.1 Comparison between exact definitions

We now have models and definitions for strain (Eqs. (2.3) to (2.4): $\mathscr{A}_1$ and $\mathscr{A}_2$) and exact definitions (a_1 and a_2 in Tab. 2.3 and $\mathscr{A}_3$ and $\mathscr{A}_4$ in equations (2.5) and (2.6)). In this section we focus on definitions under two prisms:

1. Make sure that our exact definitions a_1 and a_2 are strictly equivalent under the same system for CANTERA solutions.

2. Determine the accuracy of the CANTERA definitions $\mathscr{A}_3$ and $\mathscr{A}_4$ (Eqs. (2.5) & (2.6)).

The models $\mathscr{A}_1$ and $\mathscr{A}_2$ will be treated in section 2.2.2 and not here. The expression of $\mathscr{A}_3$ and $\mathscr{A}_4$ used in CANTERA requires a full solution including the velocity profile v. They are not 'models' like $\mathscr{A}_1$ and $\mathscr{A}_2$ which depend only on boundary conditions.
To do so, we simulate several axisymmetric counterflow steady, strained diffusion flames with CANTERA over a wide range of strain, from very low to extinction. The theory used for CANTERA is described in Kee *et al.* [52]. Figure 2.4 displays inlets parameters required to perform a CANTERA simulation.

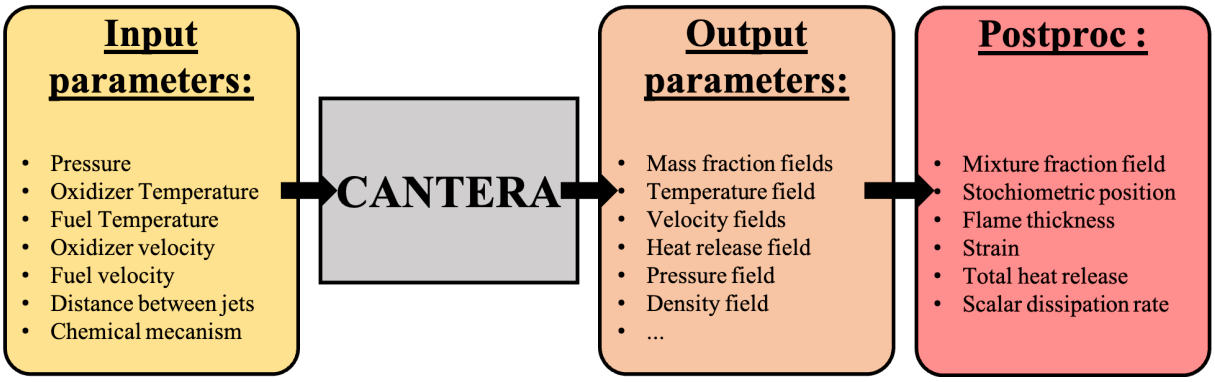


Figure 2.4: Scheme of inputs, outputs and postproc on CANTERA

Figures 2.5 to 2.8 show examples of axisymmetric strained CH_4-Air and H_2-Air at a stagnation point flame. Table 2.4 displays the parameters to obtain these flames. For each figure, key variables are plotted as a function of the normalized position such as normal velocity to the front v, product of the latter with the density ρ (which is relevant for strain determination) and temperature, mixture fraction z, mass fraction of the fuel Y_{fuel} and mass fraction of the oxidizer Y_{O_2}. Note that even if flames are similar small differences have a strong impact on strain estimation especially because the position of the stoichiometric front is critical to evaluate velocity gradients.

Table 2.4: Parameters for the test cases for CANTERA

Case	$P\ [bar]$	$T_{fuel}\ [K]$	$T_{oxy}\ [K]$	$v_{fuel}\ [m.s^{-1}]$	$v_{oxy}\ [m.s^{-1}]$	$H\ [mm]$
CH_4	1	300	300	0.5	0.5	80
H_2	1	300	300	1	5	5

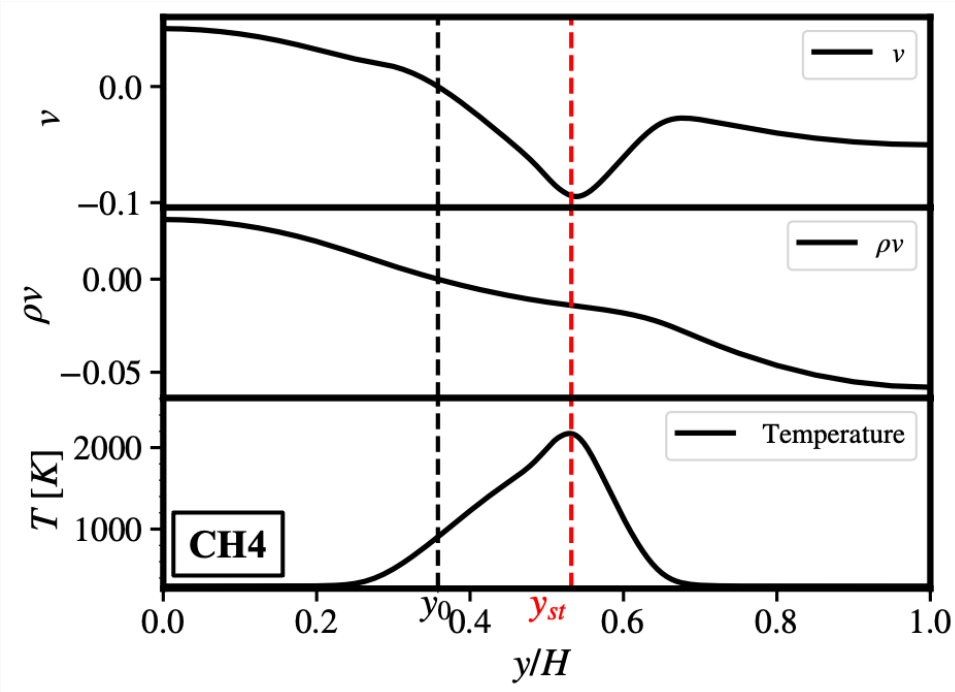


Figure 2.5: Normal velocity v, mass flux ρv and temperature T as function of the position of a CH_4-Air axisymmetric strained at a stagnation point flame using CANTERA

Figures 2.5 and 2.7 provide profiles of v and ρv for strain evaluations. Figures 2.9 and 2.10 display how v and ρv profile are used for strain evaluations based on equations from tables 2.1 and 2.3. First, ρv profiles are monotonic for both fuels so that definitions a_1 and a_2 make sense everywhere. However, v profiles are not, so that $\mathscr{A}_3$ and $\mathscr{A}_4$ do not make sense: without the absolute value in equations (2.5) and (2.6), $\mathscr{A}_3$ & $\mathscr{A}_4$ would indicate negative strain values, ie. "compressed flames", for H_2 (Fig. 2.10). We have continued the analysis for $\mathscr{A}_3$ & $\mathscr{A}_4$ but these definitions should not be used. $\mathscr{A}_1$ (the mean velocity gradient) in Eq. (2.3) gives more reasonable values. $\mathscr{A}_2$ can not be plotted in figures 2.9 and 2.10 because it uses densities in air and fuel streams (Tab. 2.1).
After looking at one flame, we now study the effects of strain and compare quantitatively a_1, a_2,$\mathscr{A}_1$, $\mathscr{A}_2$, $\mathscr{A}_3$ and $\mathscr{A}_4$.

When aiming to increase strain, there are two approaches. On one hand it is possible to change the distance H between the two inlets but a good control of strain can be difficult and performing high strain simulations can be arduous if not impossible if one relies solely on that. On the other hand, a variation of inlet velocities allows to vary the strain value and is a more docile parameter to work on. Moreover we will extend the study over a wide range of pressures and inlets temperatures summarized in table 2.5.

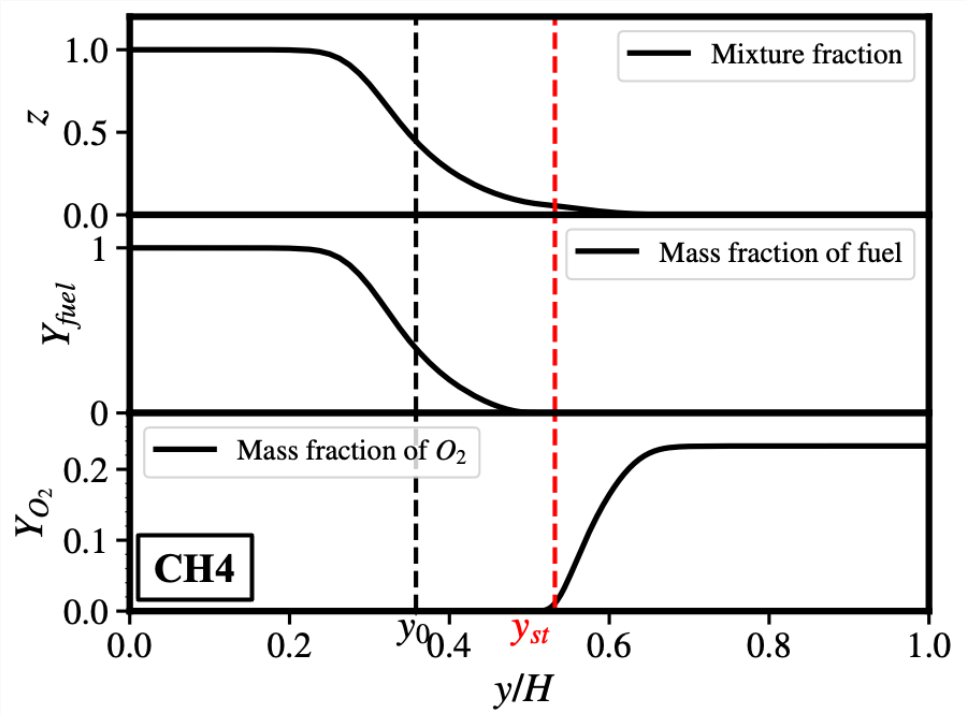


Figure 2.6: Mixture fraction z, H_2 mass fraction and O_2 mass fraction as function of the position of a CH_4-Air axisymmetric strained at a stagnation point flame using CANTERA

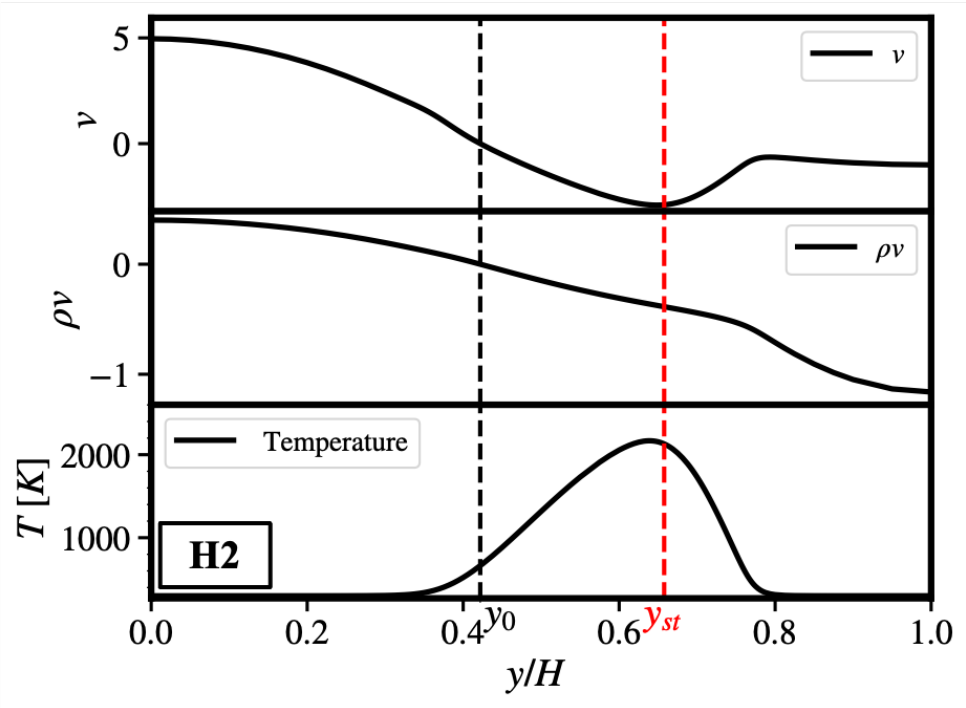


Figure 2.7: Normal velocity v, mass flux ρv and temperature T as function of the position of a H_2-Air axisymmetric strained at a stagnation point flame using CANTERA

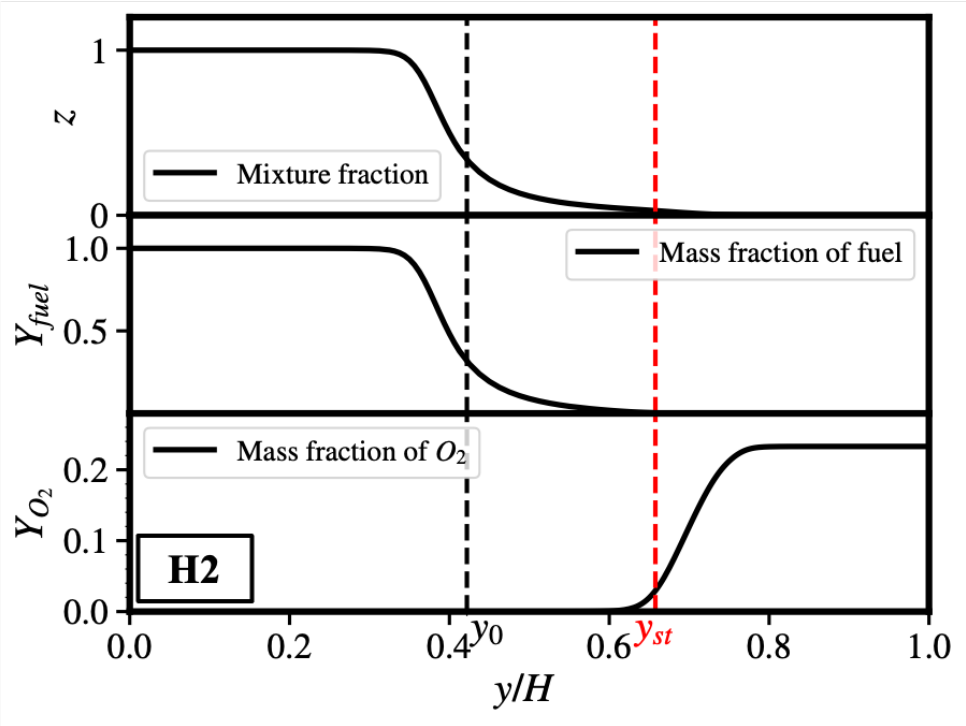


Figure 2.8: Mixture fraction z, H_2 mass fraction and O_2 mass fraction as function of the position of a H_2-Air axisymmetric strained at a stagnation point flame using CANTERA

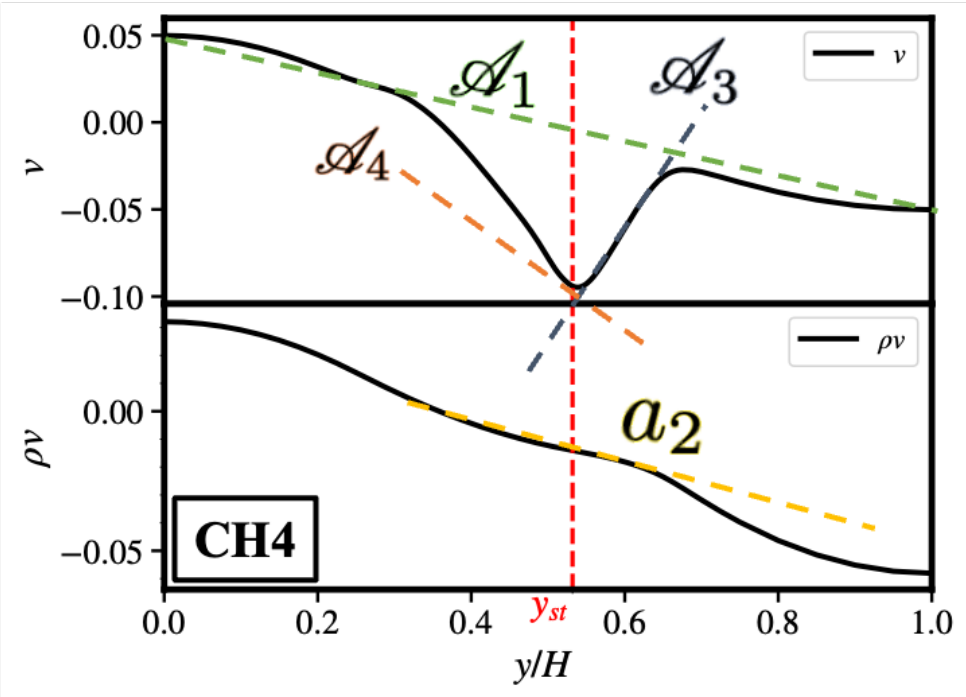


Figure 2.9: Graphical interpretations of strain estimations based on tables 2.1 and 2.3 by plotting v and ρv as function of the position (solid black curves) and stoichiometric position (red dashed line) of a CH_4-Air axisymmetric strained at a stagnation point flame using CANTERA

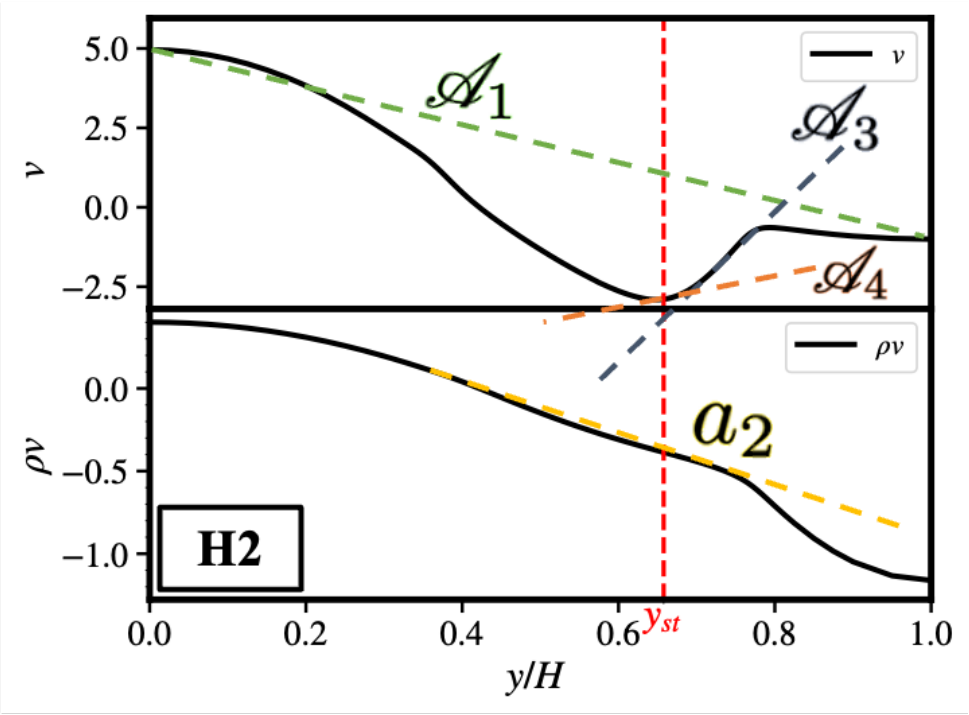


Figure 2.10: Graphical interpretations of strain estimations based on tables 2.1 and 2.3 by plotting v and ρv as function of the position (solid black curves) and stoichiometric position (red dashed line) of a H_2-Air axisymmetric strained at a stagnation point flame using CANTERA

In addition, to be sure that our observations are not specific to H_2 we will do the same study with CH_4 since the goal here is to define strain in a general definition (CANTERA setup is described in Fig. 2.2).

Table 2.5: Test cases to test strain models for stagnation point diffusion flames

Test cases	P $[bar]$	T_f $[K]$	T_o $[K]$	Chemical Mechanism	Transport Model
Ref	1	300	300	San Diego mechanism for H_2 [100] or GRI-30 for CH_4 [44]	Complex [6, 30, 110] [48, 70]
B_{cant}	10	300	300		
C_{cant}	20	300	300		
D_{cant}	1	500	500		
E_{cant}	1	700	500		
F_{cant}	1	300	500		
G_{cant}	1	300	700		

Once those parameters have been defined we start from a low strain value and raise it following the work of T. Fiala *et al.* [32] in order to reach extinction. Figures 2.11 and 2.12 display for each test case, the evaluation of a_2 as a function of a_1 (respectively refer to Eq. (2.11) & (2.9)). This type of figure will be recurrent in this chapter: first the type

of fuel is indicated in the top left (methane for fig. 2.11 & hydrogen for fig. 2.12). Second, each row corresponds to a variation from the reference case (Tab. 2.5) described by the title on the left and they share the same abscissa range. For example, the first row shows a variation of pressure from 1 to 20 bar (from left to right respectively case ref, B_{cant} and C_{cant} of Tab. 2.5). Last, the legend, at the bottom, describes the curve that is present in all graphics. In the rest of this chapter we will add new curves, on each graphic, to focus on key questions linked to the notion of strain so that each graphic will always present the same curves.
We may now analyze figures 2.11 (CH_4) and 2.12 (H_2). Independently of the configuration (type of fuel, pressure or inlets temperature) these figures illustrate the perfect equality between a_1 and a_2 (by plotting $y = x$), which was expected since the definition of a_2 is based on the mass balance equation. Thus, in the future when we will report strain values, it will be evaluated with a_2 in Tab. 2.3 (which is the same for axisymmetric and two dimensional flames).

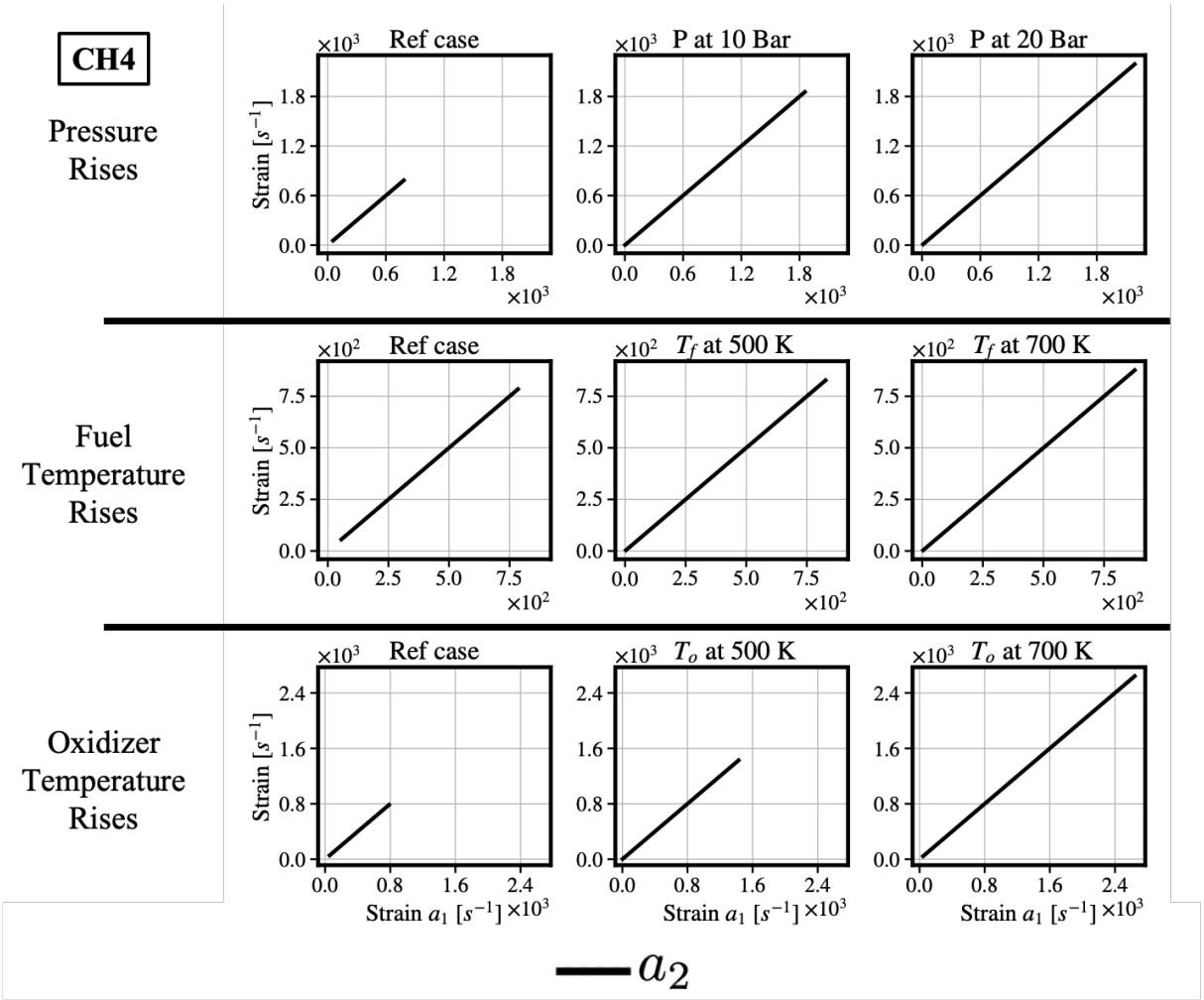


Figure 2.11: Verification of equality between definitions a_1 and a_2 (Eqs (2.9) and (2.11)) from low strain to extinction over a wide range of pressure, fuel and oxidizer temperature (Tab. 2.5) for CH_4-Air counterflow diffusion flames using CANTERA.

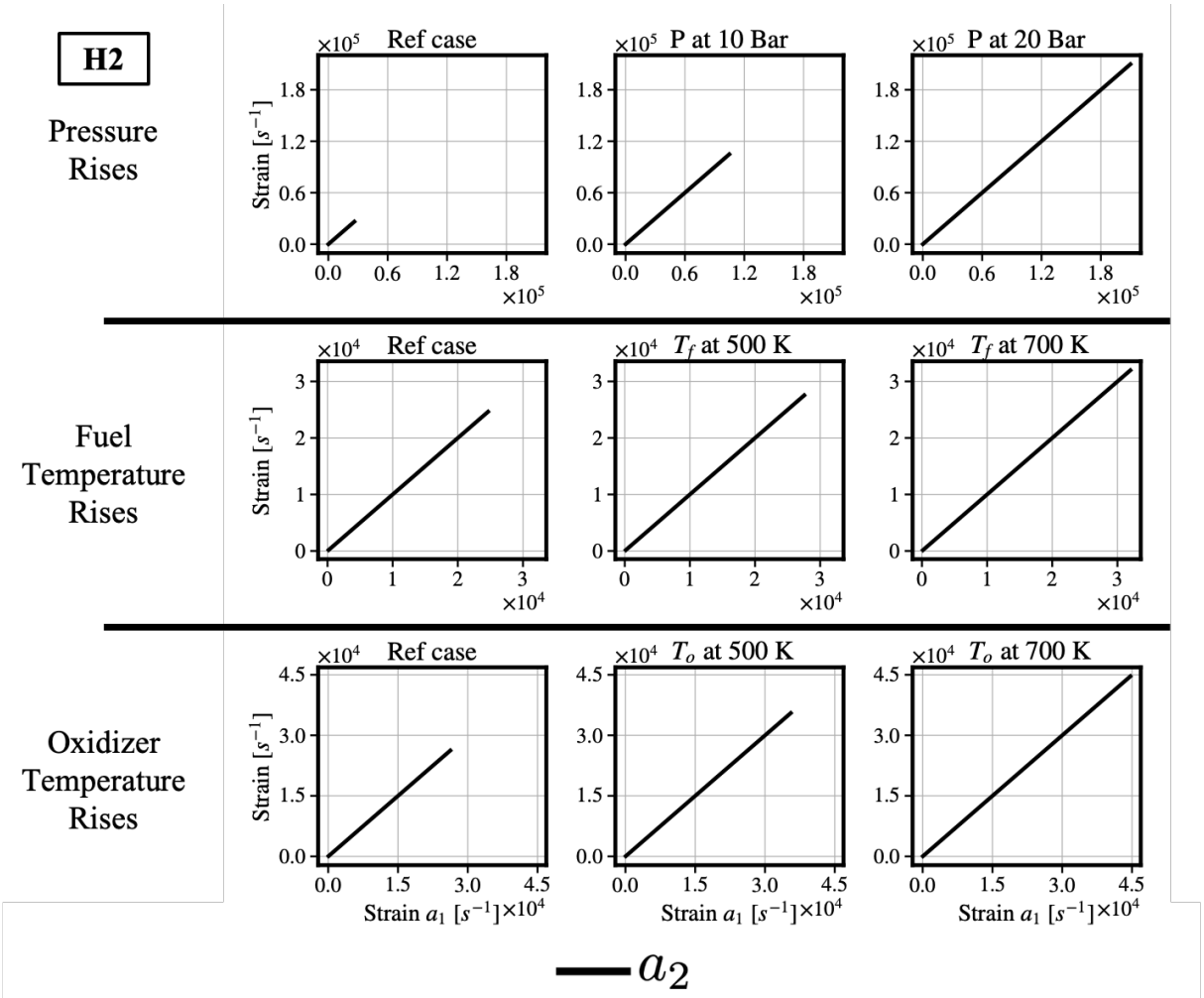


Figure 2.12: Verification of equality between definitions a_1 and a_2 (Eqs. (2.9) and (2.11)) from low strain to extinction over a wide range of pressure, fuel and oxidizer temperature (Tab. 2.5) for H_2-Air counterflow diffusion flames using CANTERA.

Starting from there, we consider now $\mathscr{A}_3$ and $\mathscr{A}_4$ given by CANTERA (Eqs. (2.5) & (2.6)). The goal is to evaluate the pertinence of what is often considered against our true definition especially because this is the CANTERA standard expressions. Keep in mind that, the main difference between them is the consideration of the density for one of them a_1 (and a_2) and not for $\mathscr{A}_4$: $a_1 = a_2 = 1/\rho_{st}\ \partial_y(\rho v)\,|_{z=z_{st}}$ while $\mathscr{A}_4 = \left|\frac{\partial v}{\partial y}\right|\Big|_{z=z_{st}}$. Of course, $\mathscr{A}_4$ degenerates naturally to a_1 (or a_2) when density changes are small. This is often the case for CH_4 (or most hydrocarbon fuels) near the stoichiometry line where temperature peaks, density is minimum and has a zero derivative. First, we shall focus on methane: figure 2.13 shows the same result as figure 2.11 except that we add definitions of CANTERA $\mathscr{A}_3$ (green circles and Eq. (2.5)) and $\mathscr{A}_4$ (purple squares and Eq. (2.6)). It is clear that the proximity between a_2 and $\mathscr{A}_4$ strikes and this for all cases. On the opposite, $\mathscr{A}_3$ depicts an unsuitable definition. Therefore, in a context where the constant density hypothesis is acceptable, the definition of CANTERA $\mathscr{A}_4$ is valid.

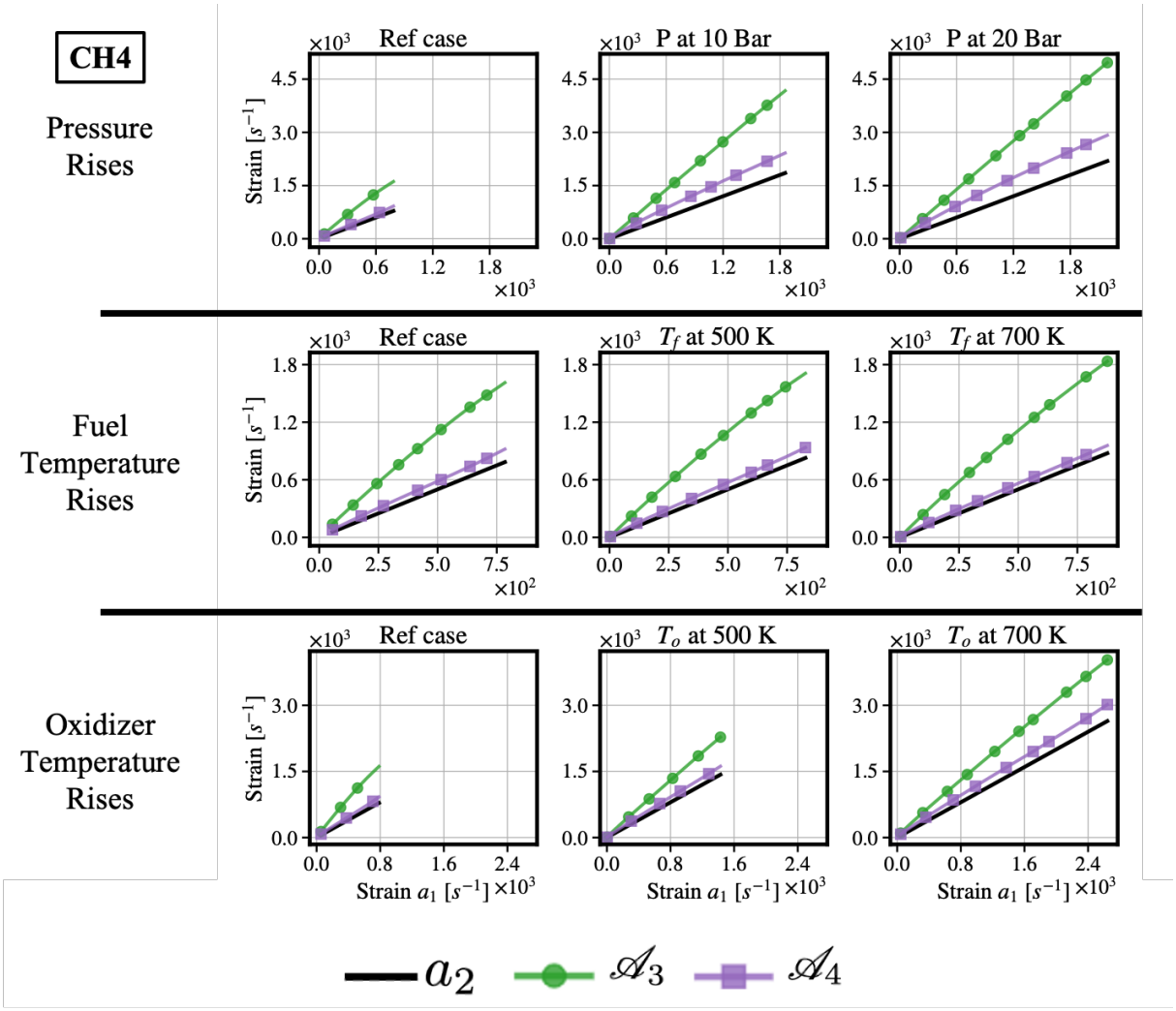


Figure 2.13: Verification of Cantera strain models $\mathscr{A}_3$ and $\mathscr{A}_4$ (Eqs. (2.5) & (2.6)) from low strain to extinction over a wide range of pressure, fuel and oxidizer temperature (Tab. 2.5) for CH_4-Air counterflow diffusion flames using CANTERA

If we focus now on hydrogen (Fig. 2.14) the conclusion differs: the definition of CANTERA $\mathscr{A}_4$ strongly underestimates the value of the strain given by a_1 and a_2. Thus, when density has a non-negligible variation, its role can no longer be ignored. Furthermore, the same conclusion on $\mathscr{A}_3$ is observed, thus it can be seen as a fully unsatisfactory candidate (remember that we already observed that $\mathscr{A}_3$ and $\mathscr{A}_4$ did not even have the correct sign for H_2-Air flames).

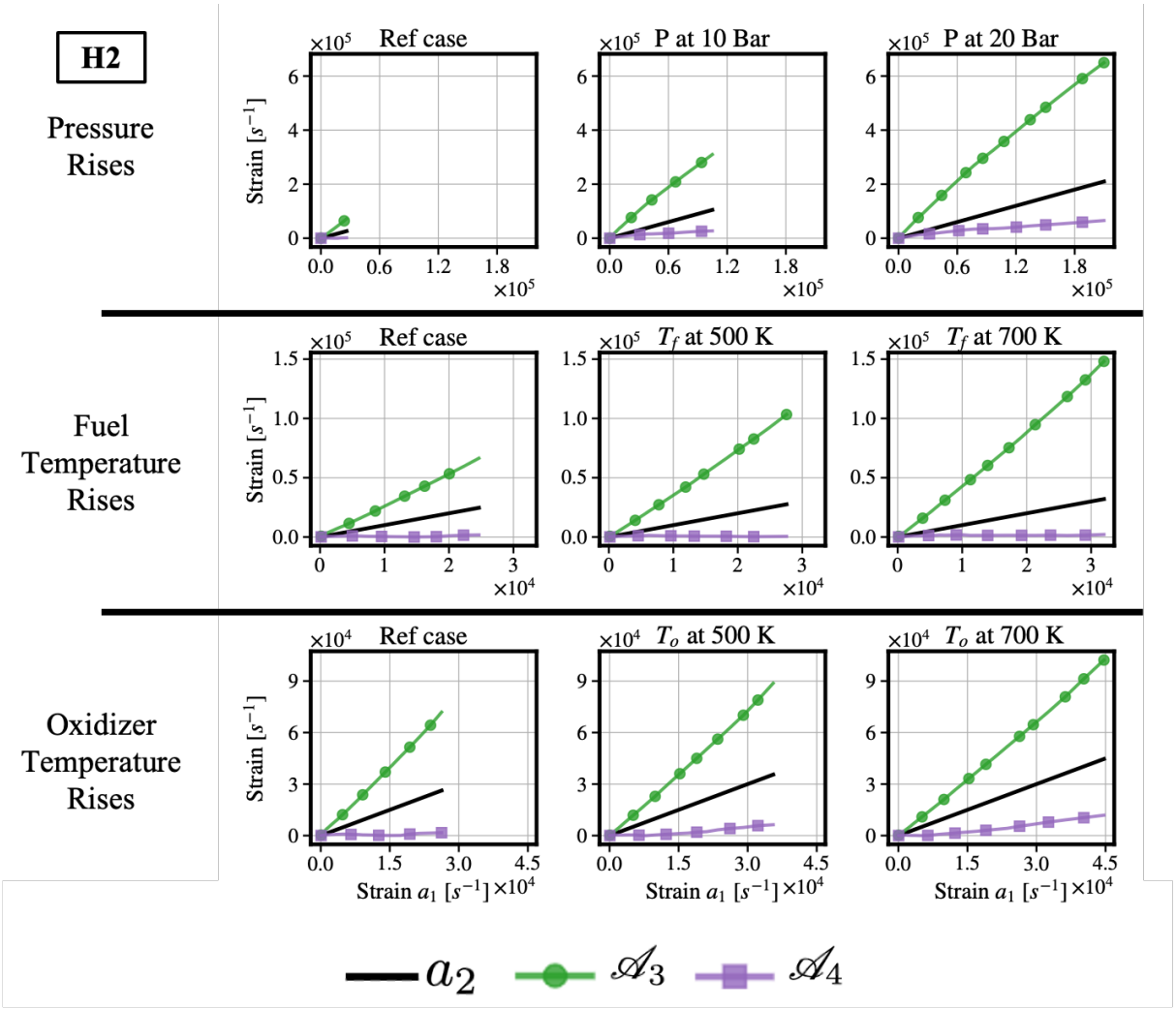


Figure 2.14: Verification of Cantera strain models $\mathscr{A}_3$ and $\mathscr{A}_4$ (Eqs. (2.5) & (2.6)) from low strain to extinction over a wide range of pressure, fuel and oxidizer temperature (Tab. 2.5) for H_2-Air counterflow diffusion flames using CANTERA

This section confirms that multiple definitions of strain are used but are not necessarily right for H_2-air flames: for all flames, a_1 and a_2 are equal. $\mathscr{A}_3$ and $\mathscr{A}_4$ are approximate for methane and wrong for hydrogen. Since the heart of this manuscript is hydrogen we will rely on the definition a_2 to define strain. This closes the discussion on true definition of strain where density and velocity profiles are available. However, in contrast to true definitions, the scientific community often uses models (such as $\mathscr{A}_1$ and $\mathscr{A}_2$) in order to obtain rapidly and easily an estimation of strain. The next section is dedicated to measure their validity.

2.2.2 Comparison between real definitions and models

Two additional models $\mathscr{A}_1$ and $\mathscr{A}_2$ are presented in this chapter (Eqs. (2.3) & (2.4)). They are trivial to determine in all simulations and experiments since they are only based on boundary conditions and do not need density or velocity profiles along the

flame front. Figure 2.15 compares evaluations of strain based on models ($\mathscr{A}_1$ and $\mathscr{A}_2$) for methane against a_1. In a context of pressure or fuel temperature variation the model $\mathscr{A}_2 = \frac{2v_{oxy}}{H}\left(1 + \frac{v_{fuel}\sqrt{\rho_{fuel}}}{v_{oxy}\sqrt{\rho_{oxy}}}\right)$ (red curve) presents good agreement with a_2 and can be considered as reference (with an error of 10% to 20%). For the case of oxidizer temperature variation it is even better. On the opposite, $\mathscr{A}_1 = \frac{|v_{fuel}|+|v_{oxy}|}{H}$ (based on the mean velocity gradient) is unsuitable. Nevertheless, both models always underestimate strain.

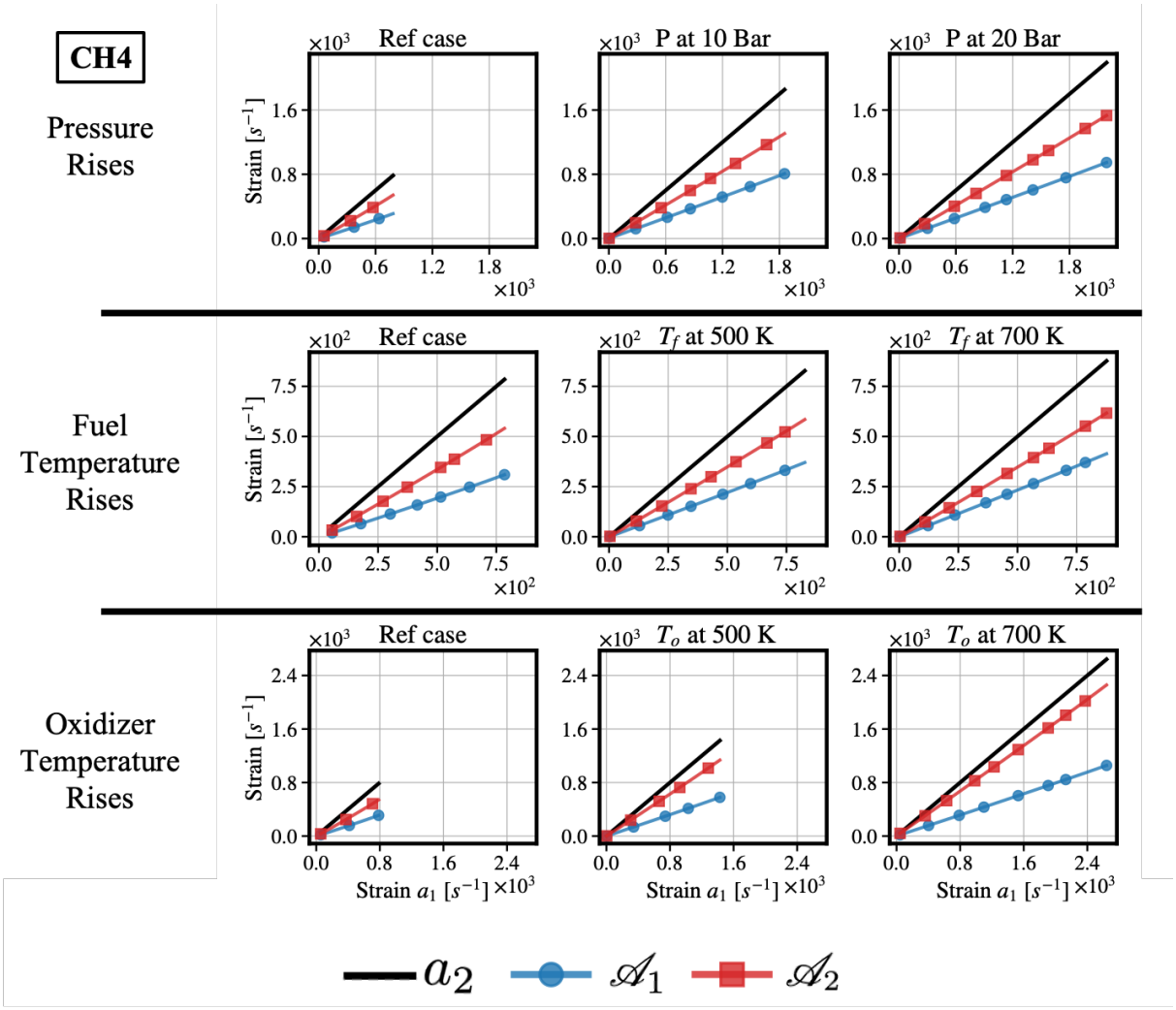


Figure 2.15: Test of strain models $\mathscr{A}_1$ and $\mathscr{A}_2$ (Eqs. from (2.3) and (2.4)) from low strain to extinction over a wide range of pressure, fuel and oxidizer temperature (Tab. 2.5) for CH_4-Air counterflow diffusion flames using CANTERA

Again, the conclusion is not the same with hydrogen: figure 2.16 shows that no model stands out directly. On one hand, both evaluations, $\mathscr{A}_1$ and $\mathscr{A}_2$, can be extremely close. This observation is illustrated by ref case, or when the oxidizer temperature varies. On the other hand, for other cases, $\mathscr{A}_1$ presents the best result with an error between 10% to 20%. Additionally, as for methane, both models always underestimate strain values. In conclusion, $\mathscr{A}_1$ is acceptable for H_2-Air flames for a quick estimation of strain with the reminder that $\mathscr{A}_1$ always underestimates by 10% to 20%.

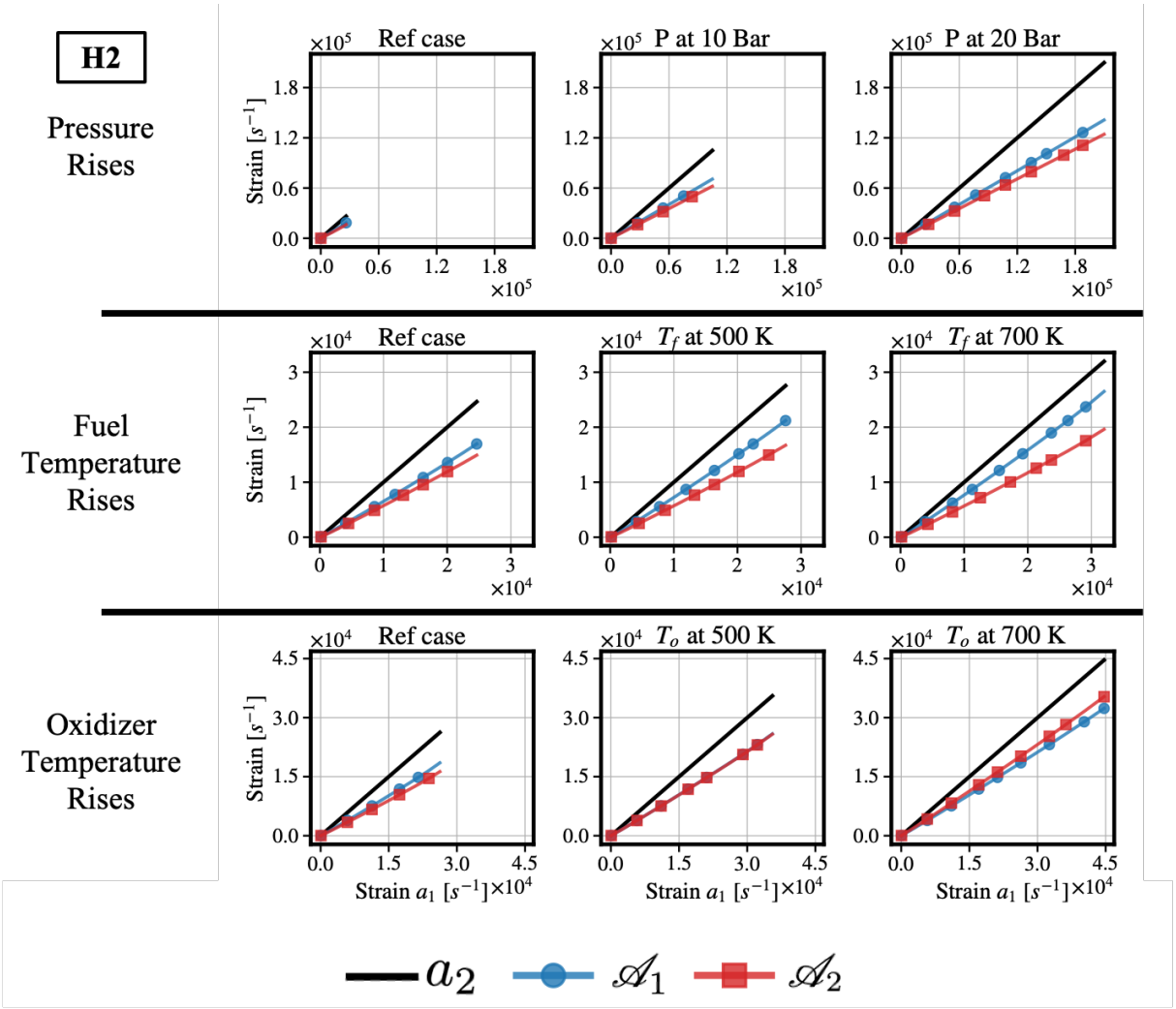


Figure 2.16: Test of strain models $\mathscr{A}_1$ and $\mathscr{A}_2$ (Eqs. from (2.3) and (2.4)) from low strain to extinction over a wide range of pressure, fuel and oxidizer temperature (Tab. 2.5) for H_2-Air counterflow diffusion flames using CANTERA.

2.3 Conclusions

This chapter has presented exact expressions of strain for diffusion flames using normal velocity profiles a_1, a_2, $\mathscr{A}_3$ and $\mathscr{A}_4$ and compared them to models $\mathscr{A}_1$ and $\mathscr{A}_2$ using only boundary values (jet velocities, distance between jets and densities of inlet streams). Our definitions have been tested on several configurations of pressure and temperature from low strain to extinction for both CH_4-Air and H_2-Air flames. We have seen a perfect match between the exact definition a_1 (2.9) and a_2 (2.11) in axisymmetric cases simulated by CANTERA. Additionally, CANTERA can directly give an evaluation if the constant density assumption is respected, for CH_4 for example, close to the stoichiometric line.
Next to true definitions, two models have been tested against the exact expression a_1 and a_2 in order to understand which one presents the best fit since models propose a significantly simpler approach to estimate the strain. In both cases, the best model,

$\mathscr{A}_2$ for CH_4 and $\mathscr{A}_1$ for H_2, always presents an underestimation between 10% to 20%. Thus, there is no perfect model that fits correctly all cases. The two models are the standard ones for strain estimation in the community: $\mathscr{A}_2 = \frac{2v_{oxy}}{H}\left(1 + \frac{v_{fuel}\sqrt{\rho_{fuel}}}{v_{oxy}\sqrt{\rho_{oxy}}}\right)$ and $\mathscr{A}_1 = \frac{|v_{fuel}|+|v_{oxy}|}{H}$. It is important to note that they are approximate and that definitions a_1 and a_2 should be preferred whenever possible.
In the end, this exercise, depending on the type of fuel studied (or the context or pressure and temperature variation at the boundaries), highlights two observations. First, strain is a laborious key parameter to study. Based on its definition expressed in this chapter, it can not be perfectly gauged in advance. Choosing H, v_f and v_o does not say a priori what a will be. Only a definition based on gradients of velocity and density profiles such as a_1 or a_2 can provide a reliable value for strain. Second, if we still want to capture the strain beforehand we recommend to perform a CANTERA study in a pre-processus part to test all models. In the rest of this manuscript we will always use a_2 to evaluate strain.

$\mathscr{A}_2 = \frac{2v_{oxy}}{H}\left(1 + \frac{v_{fuel}\sqrt{\rho_{fuel}}}{v_{oxy}\sqrt{\rho_{oxy}}}\right)$ $\mathscr{A}_1 = \frac{|v_{fuel}|+|v_{oxy}|}{H}$

Chapter 3

Variable density extension of theory for diffusion flames

This chapter recalls the classical theory presented in textbooks to evaluate the mixture fraction z and other key variables in stagnation point flames and presents an extension of the theory introducing variable density. Our first objective is to verify the validity of the historical $1D$ model of non-premixed stagnation point flames under the hypothesis of constant density ('CD' model) when hydrogen is the fuel [87]. Because of the mismatch of density between air and H_2 (at $P = 1\ bar,\ T = 300K$, $\rho_{Air}/\rho_{H_2} \approx 15$), the reliability of this model is not clear so that the logical next question is: could we build a model that accounts for variable density ('VD' model) ? This will be the second objective of this chapter. The purpose of the model will still be the estimation of the mixture fraction z and the evaluation of the integral of fuel consumption $\dot{\Omega}_F$, the flame thickness δ_f, the maximum of temperature T^{max} and the scalar dissipation rate χ.

3.1 Introduction

The goal of this chapter is to present an analytical approach to evaluate scalar dissipation rate χ, total fuel consumption rate $\dot{\Omega}_F$ and flame thickness δ_f in H_2 - air diffusion flames, including variable density. The model will be called VD for the rest of the chapter. This new approach will be compared to historical constant density theory [87] (named CD) and to one-dimensional fully resolved simulations using CANTERA [42]. After detailing the context, the numerical setup and the principles of the CD model, we will derive variable density equations for diffusion flames with infinitely fast chemistry. This part is built on the derivation of the mixture fraction balance equation. Once we succeed in evaluate it we will also directly determine physical quantities of interest.
To summarize, we want our model to :

- evaluate the local H_2 consumption rate as function of flame strain evaluated using the a_2 definition (Eq. (2.14)).
- evaluate local scalar dissipation on the stoichiometric plane as function of flame strain and vice versa [76].

- evaluate local flame thickness (based on the gradients of the mixture fraction z) to determine whether the spatial resolution is sufficient.
- be valid at any strain, even close to quenching, any pressure and operating conditions.

Even though we will relax the constant density assumption, we will still assume infinitely fast chemistry to allow a priori analytical resolution and use San Diego [100] flames in CANTERA for validations.

3.1.1 Target H_2-Air flames

This chapter will focus on an axisymmetric laminar strained non-premixed flame at a stagnation point on a counterflow system described in Fig. 3.1. We base our simulation on the H2 sub-mechanism from San Diego adapted for H2/air combustion [100].

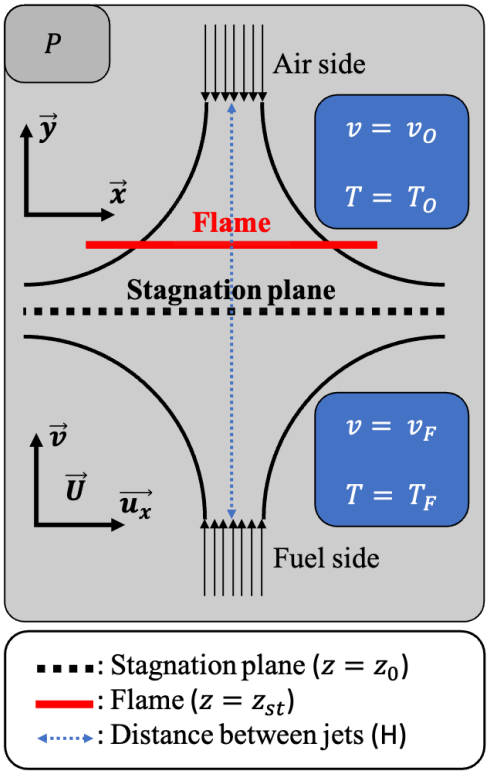


Figure 3.1: Steady, axisymmetric, strained diffusion flame in a 2D symmetric system

In order to examine local and global quantities two groups of study are defined. The first, called group A, focuses on a simulation of a single flame where all characteristics are summarized in table 3.1. Second, in a group called B, several flames are considered for various ratios between inlet velocity, defined by $r_v = v_{oxy}/v_{fuel}$, so that the flame is close to the center to prevent any potential influence of the inlet condition. Once a flame is computed for a given strain, simulations are repeated over a range of conditioned scalar information extracted from each flame. Their evolution versus strain is studied. The simulation starts at a very low strain and continues to evolve until extinction (or at least really high strain in high pressure cases).

Table 3.1: Characteristics of group A flames

CASES	T_f $[K]$	T_o $[K]$	P $[bar]$	v_f $[m.s^{-1}]$	v_o $[m.s^{-1}]$	H $[mm]$
CASES A.1	300	300	1	5	1	5
CASES A.2	300	300	10	5	1	5
CASES A.3	300	700	1	5	2.5	5
CASES A.4	300	700	10	5	2.5	2

Table 3.2: Characteristics of group B flames

CASES	T_f $[K]$	T_o $[K]$	P $[bar]$	r_v	H $[mm]$
CASES B.1	300	300	1	0.2	5
CASES B.2	300	300	10	0.2	5
CASES B.3	300	700	1	0.5	5
CASES B.4	300	700	10	0.5	2

where H is the burner separation. Table 3.3 presents the list of variables of interest and their dependence on location.

Table 3.3: Table of flames variables, their definitions and dependence on location

Variables	Definitions	Location
A	Global strain	Global
a	Local strain	Local
χ	Scalar Dissipation rate	Local
ρ	Density	Local
D_{H_2}	Diffusivity of H_2	Local
D_{th}	Thermal diffusivity	Local
δ	Flame thickness	Global
$\dot{\Omega}_F$	Total fuel consumption	Global
$\dot{\omega}_F$	Local fuel consumption	Local
$\dot{\Omega}_T$	Total heat released	Global
$\dot{\omega}_T$	Local heat released	Local
Y_k	Mass fraction of the species k	Local
z	Mixture fraction	Local

Table 3.4 summarizes variables of interest. The gray rows highlight the set of flame variables used to compare CD and VD models with fully resolved simulations.

Table 3.4: Flame variables and definitions

Variables	Definitions
A	Global strain
a_{st}	Local strain at stoichiometry
χ_{st}	Scalar dissipation rate at stoichiometry
χ_{max}	Maximum value of the scalar dissipation rate
$\dot{\Omega}_F$	Total fuel consumption
δ_f	Flame thickness
$\dot{\omega}_{F,max}$	Maximum value of the local fuel consumption
$\dot{\Omega}_T$	Total heat released
$\dot{\omega}_{T,max}$	Maximum value of local heat released
T_{max}	Maximum value of temperature
z_0	Mixture fraction at the stagnation plan

3.1.2 State of the art: Constant density model (CD)

The analysis of diffusion strained flames is often performed using an important simplification where the density ρ is constant. Starting from the z balance equation and the mass conservation equation [87] :

$$\begin{cases} \partial_t[\rho(z)z] + \partial_{x_i}[\rho u_i z] - \partial_{x_i}[\rho D \partial_{x_i}(z)] &= 0 \\ \partial_t \rho + \nabla_i(\rho u_i) &= 0 \end{cases} \tag{3.1}$$

Assuming that the diffusion flame is one-dimensional, steady and strained that density ρ is constant as well as the diffusivity D, system (3.1) can be written as (following the notation of Fig. 3.1) :

$$\begin{cases} \rho v \partial_y z = & \rho D \partial_{yy} z \\ \rho \partial_y v = & -\rho \partial_x u_x = -\rho a_1 \end{cases} \tag{3.2}$$

Since, by definition in a constant density strained flow, $a_1 = \partial_x u_x$ (Eq. 2.10). For the rest of the section we will use a_2 since it is equal to a_1 and it is evaluated along the flame. The mass equation gives the velocity field as follows:

$$\begin{cases} u_x &= a_2 x \\ v &= -a_2 y \end{cases} \tag{3.3}$$

Note that a_2 here is constant over the whole field, a property which is not true for variable density flames. The z balance equation (3.2) can be solved using the change of variable $\zeta = y\sqrt{\dfrac{a_2}{2D}}$:

$$\partial_{\zeta\zeta}(z) + 2\zeta \partial_\zeta(z) = 0 \tag{3.4}$$

leading, with $z(-\infty) = 1$ and $z(\infty) = 0$, to the following z expression solution:

$$z = \frac{1}{2}[1 - erf(\zeta)] \tag{3.5}$$

where

$$erf(\zeta) = \frac{2}{\pi}\int_0^{\zeta} \exp(-\psi^2)d\psi$$

The estimation of z given by Eq. (3.5) allows to obtain key variables such as $\dot{\Omega}_F$, the total fuel consumption rate (for infinitely fast chemistry), χ the scalar dissipation rate and δ_{flame} the flame thickness.

Total fuel consumption rate $\dot{\Omega}_F$

Assuming that the diffusion flame is one-dimensional, steady, strained and with a constant density, the conservation equation for the fuel is the following :

$$\partial_y(\rho v Y_F) = \partial_y(\rho D \partial_y Y_F) + \dot{\omega}_F \tag{3.6}$$

Defining $\dot{\Omega}_F$ as fuel reaction rate per unit area. We can integrate equation (3.6) between y_{st}^- and y_{st}^+ under the assumption of infinitely fast chemistry where y_{st}^- and y_{st}^+ are the closest point to the flame front (at stoichiometry):

$$\dot{\Omega}_F = \int_{y_{st}^-}^{y_{st}^+} \dot{\omega}_F dy = -\left[\rho D \partial_y(Y_F)\right]_{y_{st}^-}^{y_{st}^+} = \rho D \partial_y(Y_F)|_{y=y_{st}^-} = \rho D\ \partial_z(Y_F)|_{z=z_{st}^-}\ \partial_y(z)|_{y=y_{st}^-}$$

or, based on equation (3.13), $\partial_z(Y_F)|_{z=z_{st}^-} = Y_{fuel}^0/(1 - z_{st})$. Moreover, based on the definition of ζ and equation (3.5) :

$$\partial_y(z) = -\sqrt{\frac{a_2}{2D}}\frac{1}{\sqrt{\pi}}\exp[-\zeta^2] \tag{3.7}$$

so that a fully explicit expression for the reaction rate is:

$$\dot{\Omega}_F = -\rho\frac{Y_{fuel}^0}{1 - z_{st}}\sqrt{\frac{a_2 D}{2\pi}}\exp[-\zeta_{st}^2] \tag{3.8}$$

Scalar dissipation rate χ

By definition, the scalar dissipation rate χ of one-dimensional flames is characterized as follows:

$$\chi = 2D(\partial_y z)^2 \tag{3.9}$$

Thus, based on equation (3.7) :

$$\chi = \frac{a_2}{\pi}\exp[-\zeta^2] \tag{3.10}$$

or as a function of the mixture fraction z (based on equation (3.5)) :

$$\chi = \frac{a_2}{\pi} \exp[-(erf^{-1}(1-2z))^2] \tag{3.11}$$

Flame thickness δ_{flame}

A simple method to measure δ_L in a diffusion flame is $\delta_L = 1/\max(\nabla z)$ in flames where the mixture fraction z varies from 0 in the oxidizer stream to 1 in the fuel stream. Figure 3.2 displays $g(z) = \exp[-(erf^{-1}(1-2z))^2]$ as a function of z, where $z \in [0,1]$. Based on the curve of figure 3.2, it appears that the maximum of g is always obtained at $z = 0.5$ and its equal to one. Thus :

$$\delta_f = \frac{1}{\max(\nabla z)} = \frac{1}{\max\left(\frac{\chi}{2D}\right)^{1/2}} = \frac{1}{\max\left(\frac{a_2}{2D\pi} g(z)\right)^{1/2}} = \sqrt{\frac{2\pi D}{a_2}} \frac{1}{\max(g(z))}$$

and then:

$$\delta_f = \sqrt{\frac{2\pi D}{a_2}} \tag{3.12}$$

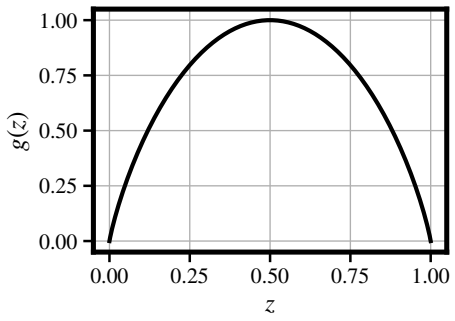


Figure 3.2: g as a function of z

3.2 Stagnation point laminar diffusion flames with variable density: VD model

This section describes a new formulation for infinitely fast chemistry diffusion flames where density as well as molecular diffusion changes are accounted for. It requires a preliminary analysis to specify changes in molecular weight (which are large in H_2 - air diffusion flames), temperature profile and molecular diffusion coefficient.

3.2.1 Preliminary analysis

Building a model for variable density requires multiple steps described here to express molecular weights W, temperature profile T and molecular diffusivity D. This is done assuming infinitely fast chemistry and the validity of the z-diagram given in figure 3.3.

Modeling of the local molecular weight W

The local molecular weight is given by $\frac{1}{W} = \sum_k \frac{Y_k}{W_k}$. In an H_2-Air diffusion flame, W changes from 2 $g.mol^{-1}$ to $30g.mol^{-1}$, something which should be accounted for and is not in the CD model. The VD model uses an evaluation of W as a function of z. To obtain W we assume infinitely fast chemistry so that only four species (H_2, O_2, H_2O and N_2) are sufficient to describe the composition [87] :

$$Y_{H_2} = \begin{cases} Y^0_{fuel} \ \dfrac{z - z_{st}}{1 - z_{st}} & \text{if } z > z_{st} \\ 0 & \text{if } z \leqslant z_{st} \end{cases} \tag{3.13}$$

$$Y_{H_2O} = \begin{cases} Y^{max}_{H_2O} \ \dfrac{1 - z}{1 - z_{st}} & \text{if } z > z_{st} \\ Y^{max}_{H_2O} \ \dfrac{z}{z_{st}} & \text{if } z \leqslant z_{st} \end{cases} \tag{3.14}$$

$$Y_{O_2} = \begin{cases} 0 & \text{if } z > z_{st} \\ Y^0_{oxy} \ \left(1 - \dfrac{z}{z_{st}}\right) & \text{if } z \leqslant z_{st} \end{cases} \tag{3.15}$$

$$Y_{N_2} = Y^0_{N_2} \ (1 - z) \tag{3.16}$$

where $Y_{H_2O,max} = W_{H_2O}/(W_{H_2O} + \nu'_{N_2} W_{N_2}) \approx 0.25$, $Y^{\text{oxy side}}_{N_2} = 1 - Y^{\text{oxy side}}_{air} \approx 0.77$ and $z_{st} = 0.028$. Figure 3.3 displays the evaluation of mass fraction following equations from (3.13) to (3.16) for respectively, Y_{H_2}, Y_{H_2O}, Y_{O_2} and Y_{N_2}.

Using this simplification W is given by :

$$\frac{1}{W} = \begin{cases} \dfrac{\dfrac{z - z_{st}}{1 - z_{st}}}{W_{H_2}} + \dfrac{0}{W_{O_2}} + \dfrac{Y^{max}_{H_2O} \dfrac{1 - z}{1 - z_{st}}}{W_{H_2O}} + \dfrac{Y^0_{N_2}(1 - z)}{W_{N_2}} & \text{if } z > z_{st} \\ \dfrac{0}{W_{H_2}} + \dfrac{Y^0_{O_2}(1 - \dfrac{z}{z_{st}})}{W_{O_2}} + \dfrac{Y^{max}_{H_2O} \dfrac{z}{z_{st}}}{W_{H_2O}} + \dfrac{Y^0_{N_2}(1 - z)}{W_{N_2}} & \text{if } z \leqslant z_{st} \end{cases}$$

which is expressed in a more compact form as:

$$\boxed{\frac{1}{W} = \begin{cases} z \left[\alpha^{sup}\right] + \left[\beta^{sup}\right] & \text{if } z > z_{st} \\ z \left[\alpha^{inf}\right] + \left[\beta^{inf}\right] & \text{if } z \leqslant z_{st} \end{cases}} \tag{3.17}$$

where :

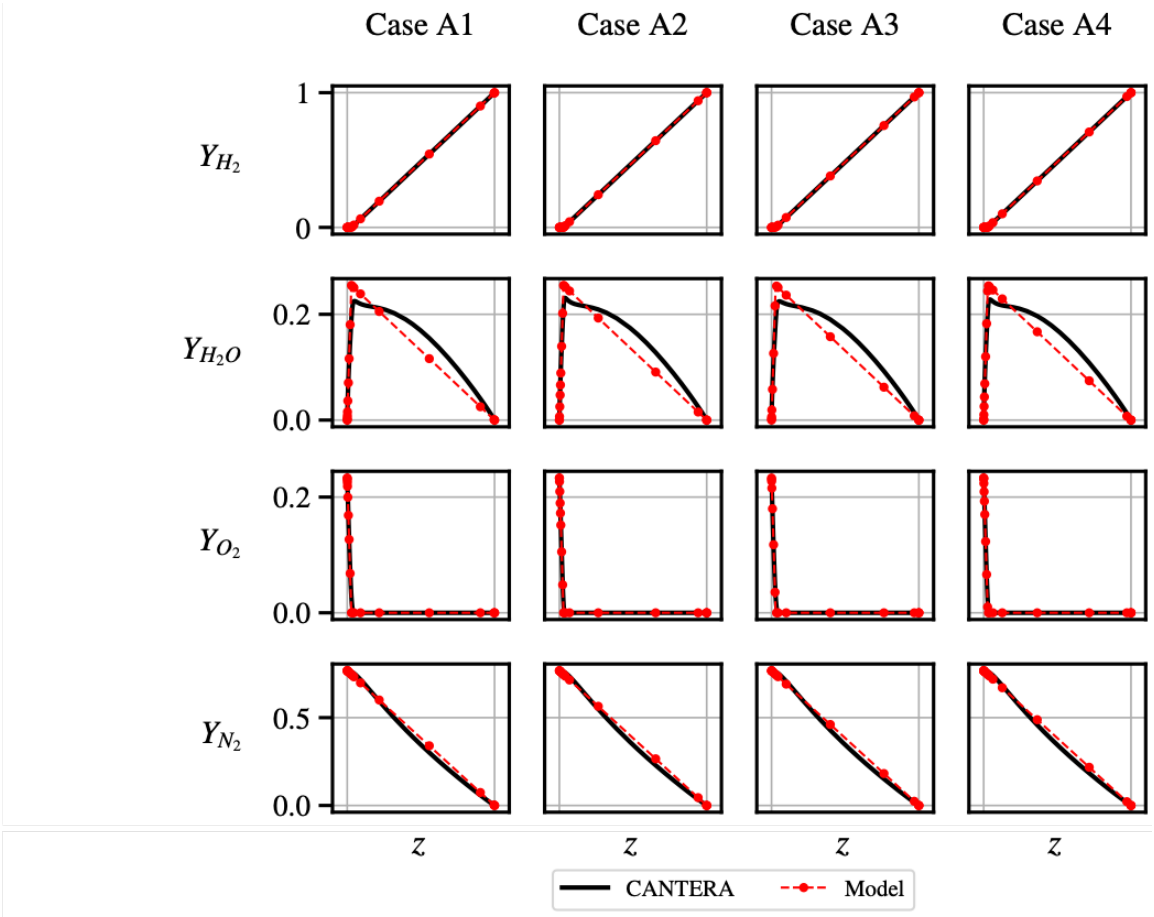


Figure 3.3: Comparison of models, from top to bottom : Y_{H_2} (Eq. (3.13)), Y_{H_2O} (Eq. (3.14)), Y_{O_2} (Eq. (3.15)) and Y_{N_2} (Eq. (3.16)). From left to right, each flame of the group A (Tab. 3.1) is plotted.

$$\begin{cases} \alpha^{sup} = \left[\dfrac{1}{W_{H_2}(1-z_{st})} - \dfrac{Y^{max}_{H_2O}}{W_{H_2O}(1-z_{st})} - \dfrac{Y^0_{N_2}}{W_{N_2}}\right] & \text{if } z > z_{st} \\ \beta^{sup} = \left[-\dfrac{z_{st}}{W_{H_2}(1-z_{st})} + \dfrac{Y^{max}_{H_2O}}{W_{H_2O}(1-z_{st})} + \dfrac{Y^0_{N_2}}{W_{N_2}}\right] & \text{if } z > z_{st} \\ \alpha^{inf} = \left[\dfrac{Y^{max}_{H_2O}}{W_{H_2O}z_{st}} - \dfrac{Y^0_{O_2}}{W_{O_2}z_{st}} - \dfrac{Y^0_{N_2}}{W_{N_2}}\right] & \text{if } z \leqslant z_{st} \\ \beta^{inf} = \left[\dfrac{Y^0_{O_2}}{W_{O_2}} + \dfrac{Y^0_{N_2}}{W_{N_2}}\right] & \text{if } z \leqslant z_{st} \end{cases}$$

The mixture molecular weight W (normalized by its maximum) as a function of z is plotted in figure 3.4 for each case of the group A flames and matches CANTERA results very well.

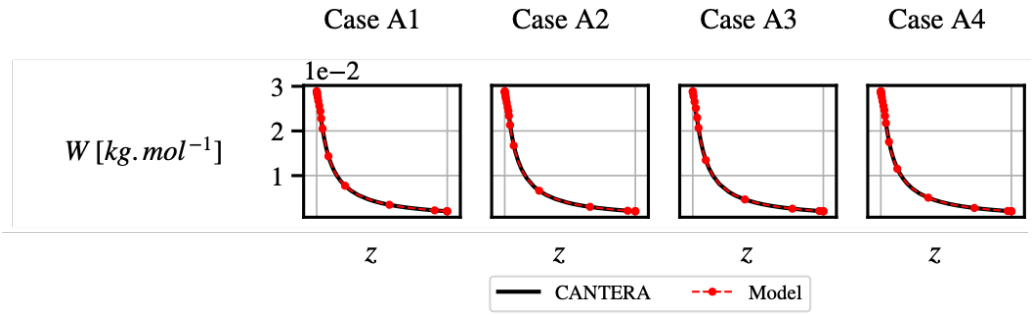


Figure 3.4: Comparison of model for W (Eq. (3.17)). From left to right, each flame of the group A (Tab. 3.1) is plotted.

Modeling of the heat capacity C_p

By definition, $C_p = \sum_k Y_k C_{p,k}$. Using the model (3.17), and under the assumption that coefficients $C_{p,k}$ are constant, C_p is defined by :

$$\boxed{C_p = \begin{cases} z\eta^{sup} + \lambda^{sup} & \text{if } z > z_{st} \\ z\eta^{inf} + \lambda^{inf} & \text{if } z < z_{st} \end{cases}} \tag{3.18}$$

With

$$\begin{cases} \eta^{sup} = \dfrac{C_{p,H_2}}{1-z_{st}} - \dfrac{C_{p,H_2O}Y^{max}_{H_2O}}{1-z_{st}} - C_{p,N_2}Y^0_{N_2} \\ \eta^{inf} = \dfrac{-Y^0_{O_2}C_{p,O_2}}{z_{st}} + \dfrac{Y^{max}_{H_2O}C_{p,H_2O}}{z_{st}} - C_{p,N_2}Y^0_{N_2} \\ \lambda^{sup} = \dfrac{-z_{st}C_{p,H_2}}{1-z_{st}} + \dfrac{C_{p,H_2O}Y^{max}_{H_2O}}{1-z_{st}} + C_{p,N_2}Y^0_{N_2} \\ \lambda^{inf} = Y^0_{O_2}C_{p,O_2} + C_{p,N_2}Y^0_{N_2} \end{cases}$$

The heat capacity C_p (normalized by its maximum) as a function of z is plotted in figure 3.5 for each flame of group A (Tab. 3.1).

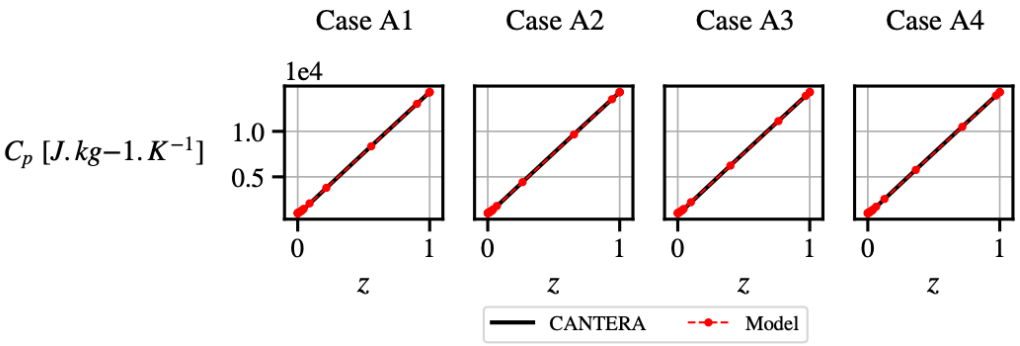


Figure 3.5: Comparison of model for C_p (Eq. (3.18)). From left to right, each flame of the group A (Tab. 3.1) is plotted.

Modeling of the temperature profile T

For infinitely fast chemistry, the temperature T can be obtained from z using two different expressions in rich ($z > z_{st}$) and lean ($z < z_{st}$) zones:

$$\boxed{T = \begin{cases} T^{sup} = z\Delta T^0 + T_O^0 + \dfrac{1-z}{C_p}\gamma^{sup} & \text{if } z > z_{st} \\ T^{inf} = z\Delta T^0 + T_O^0 + \dfrac{z}{C_p}\gamma^{inf} & \text{if } z < z_{st} \end{cases}} \tag{3.19}$$

where:

$$\begin{cases} \Delta T^0 = & T_F^0 - T_O^0 \\ \gamma^{sup} = & \dfrac{Qz_{st}}{1-z_{st}} \\ \gamma^{inf} = & Q \end{cases}$$

Note that the model for W does not take flame strain into account since chemistry is supposed to be fast. This is acceptable for W but leads to discrepancies for the maximum temperature: $T^{max,CANTERA}$ decreases as a function of a_2 [82, 98, 108] showing that the infinitely fast chemistry assumption can not be used here, we want to capture the temperature correctly. This $T^{max,CANTERA}$ variation is due to complex chemistry mechanisms as strain increases. It can be mimicked by introducing a correction factor p, estimated by the following equation :

$$p = \begin{cases} \dfrac{T^{max,CANTERA}(a_2) - z\Delta T^0 - T_O^0}{\dfrac{1-z}{C_p}\gamma^{sup}} & \text{if } z_{max} > z_{st} \\ \dfrac{T^{max,CANTERA}(a_2) - z\Delta T^0 - T_O^0}{\dfrac{z}{C_p}\gamma^{inf}} & \text{if } z_{max} < z_{st} \end{cases} \tag{3.20}$$

where z_{max} denotes the z values position at maximum temperature. The model (Eq. (3.20)) expresses the temperature (Eq. (3.19)) with an additional correction factor p on the third term of equation 3.19. Fig. 3.6 displays $T(x)$ from CANTERA and Eq. (3.19) without p correction. Even if the modeled temperature matches the behavior of

CANTERA there is a mismatch between CANTERA and modeled temperature for each flame of the group A (Tab. 3.1). This issue is resolved under the addition of the p factor based on equation (3.20). Figure 3.7 displays the same couple of curves for each case A (Tab. 3.1) but with the addition of p correction. With it, the maximum of temperature is recovered. Thus the combination of equations (3.19) and (3.20) matches CANTERA results efficiently. In the rest of the paper, the correction factor p is included in γ (sup or inf from equation 3.19).

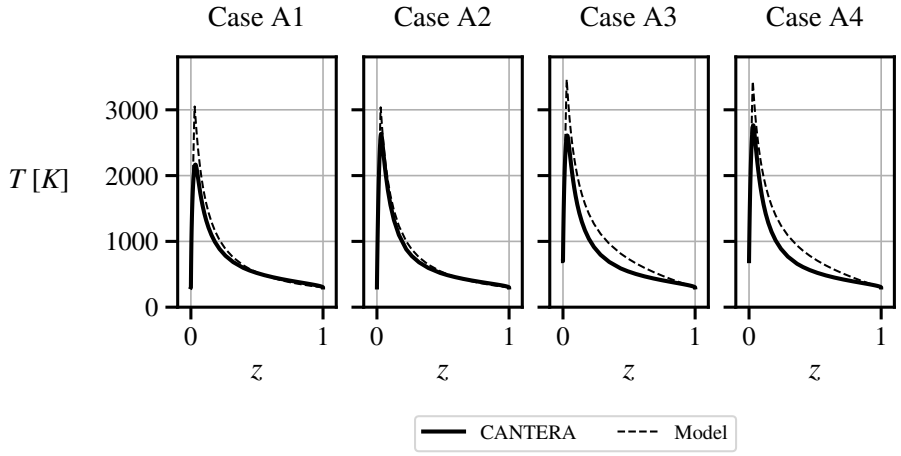


Figure 3.6: Comparison of temperature profiles as function of mixture fraction z for each flame of the group A (Tab. 3.1). Here, the modeled temperature is based on equation (3.19) without p correction.

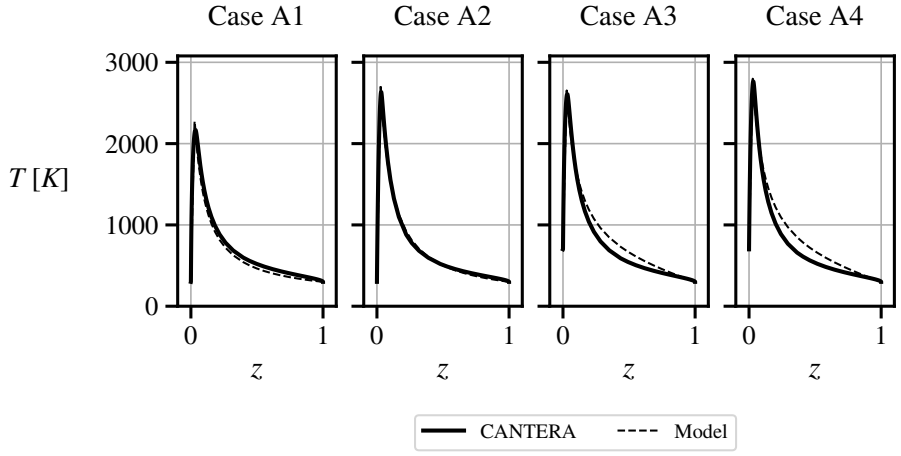


Figure 3.7: Comparison of temperature profiles as function of mixture fraction z for each flame of the group A (Tab. 3.1). Here, the modeled temperature takes into account p correction based on equation (3.20).

In order to verify the convergence between the model and CANTERA figure 3.8 displays the evaluation of T_{max} as function of strain for each B case (Tab. 3.2) with or

without p correction. It is clear that the addition of the p factor allows to capture perfectly the maximum compared to CANTERA. Figure 3.9 shows the evaluation of p for each B case. Unfortunately, p depends on the pressure/temperature configuration (which is expected since it follows the behavior of T_{max} as function of the strain). A temperature model more directly strain dependent could be an interesting improvement. Still, our model presents satisfying results and will be henceforth considered as the reference.

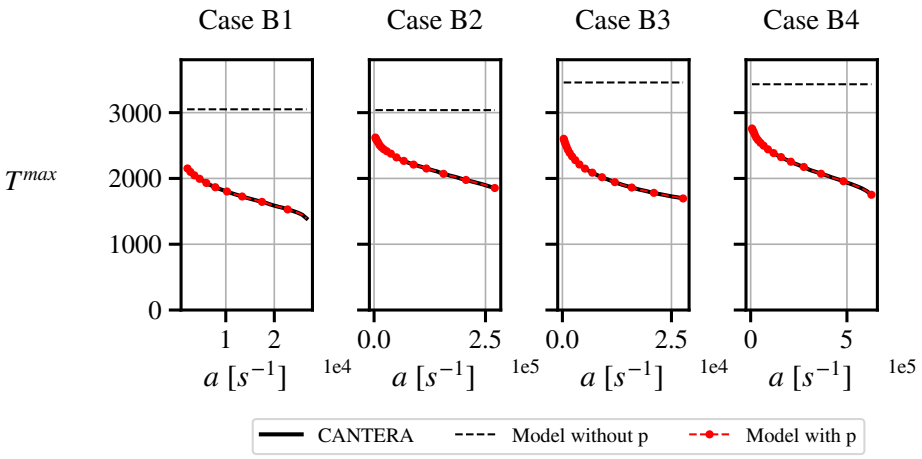


Figure 3.8: Maximum temperature as function of strain a for each case of the group B (Tab. 3.2. CANTERA result is displayed (solid black curve) against the modeled temperature with (dashed red dotted curve) or without (dashed black curve) p correction.

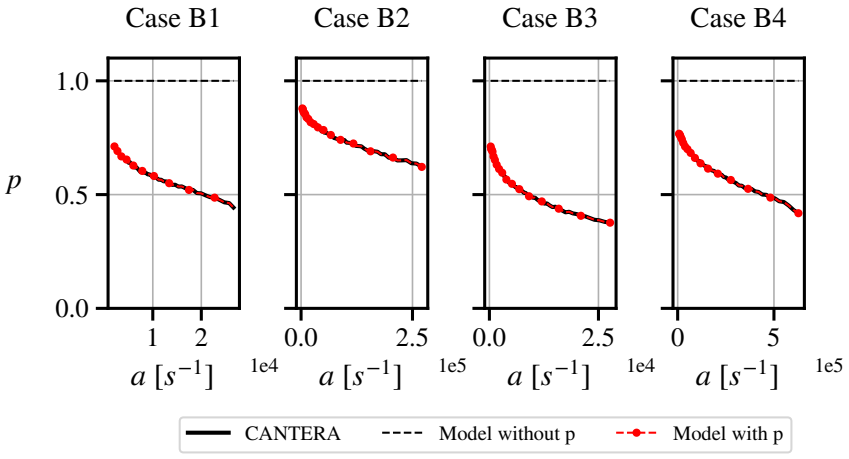


Figure 3.9: Factor p as function of strain a for each case of the group B (Tab. 3.2. CANTERA result is displayed (solid black curve) against the modeled temperature with (dashed red dotted curve) or without (dashed black curve) p correction.

Modeling of the molecular diffusivity of species D

A last quantity required for the VD model is the diffusivity of species D (supposed to be equal to the heat diffusivity D_{th}). Kinetic theory indicates that the diffusivity D is proportional to temperature to the power b at constant pressure [5]:

$$D \propto T^b \tag{3.21}$$

The state of the art testifies that $b = 1.5$ if we consider a perfect gas and $b \in [1.5; 2]$ otherwise (if we consider the interaction of particles) [67]. To find the best value of b in our context, several test were performed following cases B (Tab. 3.2). For each flame we estimate the ratio r_{hyp}, in order to check under which b value the relation (3.21) is correct, defined as follows:

$$r_{hyp} = \frac{D_{st}}{T_{st}^b} \frac{T_0^b}{D_0} \tag{3.22}$$

where subscripts st and 0 correspond respectively to stoichiometry and stagnation position. Note that, if Eq. (3.21) is correct for a given b value, then $r_{hyp} = 1$. Figure 3.10 displays r_{hyp} as function of strain for each B case. It is clear that in each case the value $b = 1.75$ strikes as the reference value which guarantees that $r_{hyp} = 1$.

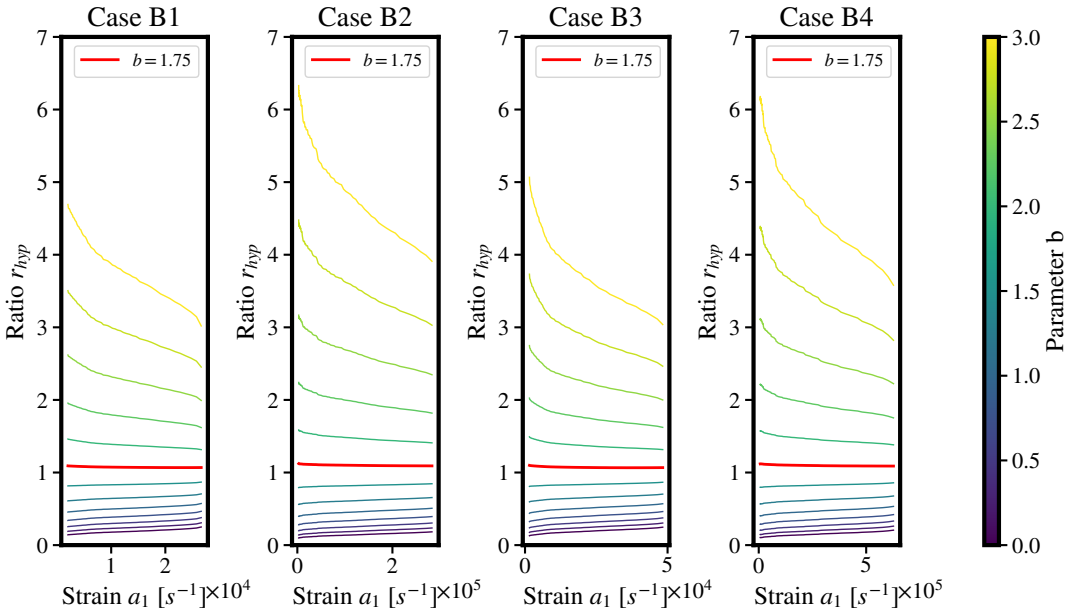


Figure 3.10: Evaluation of ratio r_{hyp} as function of strain a_1 from case B1 (on the left) to case B4 (on the right) (Tab 3.2). Thin solid lines represent the ratio r_{hyp} for a range of b from 0 to 3. Thick red line highlights the case where $b = 1.75$, the closest to one

Since pressure is constant, the perfect gas law $\rho = \frac{PW}{RT}$ shows that :

$$\frac{\rho}{\rho_0} = \frac{WT_0}{W_0T} \tag{3.23}$$

so that the $D\rho^2$ quantity, which we will be used for future notice, because it appears naturally in the z equation, can be written:

$$\frac{D\rho^2}{\rho_0^2} = D_0 \left(\frac{W}{W_0}\right)^2 \left(\frac{T_0}{T}\right)^{(2-b)} \tag{3.24}$$

This expression for $D\rho^2$ will be used in the next section because it appears in the z equation

3.2.2 Mixture fraction balance equation with variable density

For a diffusion, one-dimensional strained and steady flame under the system coordinate used in figure 3.1, the z-balance equation is defined :

$$\rho v \partial_y z = \partial_y(\rho D \partial_y z) \tag{3.25}$$

Solving this equation requires a change of spatial variable and a few manipulations presented now.

3.2.2.1 Change of coordinate Y to account for density changes

Defining a "mass-weighted coordinate" Y :

$$Y = \frac{1}{\rho_0} \int_{y_0}^{y} \rho(\widetilde{y}, t) d\widetilde{y} \tag{3.26}$$

and knowing that $\partial_y Y = \dfrac{\rho}{\rho_0}$ and that mass conservation is written [87]:

$$\partial_y = \frac{\rho}{\rho_0}\partial_Y \implies \partial_y(\rho v) = \frac{\rho}{\rho_0}\partial_Y(\rho v)$$

thus, continuity becomes:

$$\frac{\rho}{\rho_0}\partial_Y(\rho v) = -\partial_x(\rho u_x) = -\rho\partial_x(u_x) - u_x\partial_x(\rho) = -\rho a_1$$

then:

$$\rho v = -\rho_0 a_2 Y \tag{3.27}$$

The strain a_2 was expressed in the previous chapter and we will consider the definition of the equation (2.11). Thus, the z-balance equation (3.25) becomes :

$$\begin{aligned} & -\rho_0 a_2 Y \partial_y z = \partial_y[\rho D \partial_y(z)] \\ \Leftrightarrow \quad & -\rho_0 a_2 Y \frac{\rho}{\rho_0}\partial_Y z = \frac{\rho}{\rho_0}\partial_Y \left[\rho D \frac{\rho}{\rho_0}\partial_Y(z)\right] \\ \Leftrightarrow \quad & -a_2 Y \partial_Y z = \partial_Y \left[\frac{\rho^2 D}{\rho_0^2}\partial_Y(z)\right] \end{aligned}$$

The $\rho^2 D$ term can be expressed using Eq. (3.24) to give:

$$-a_2 Y \partial_Y z = \partial_Y \left[D_0 \left(\frac{W}{W_0} \right)^2 \left(\frac{T_0}{T} \right)^{(2-b)} \partial_Y(z) \right]$$

Finally, the z equation with the variable density can be written:

$$\boxed{\frac{a_2}{D_0} Y \partial_Y z + \partial_Y \left[\left(\frac{W}{W_0} \right)^2 \left(\frac{T_0}{T} \right)^{(2-b)} \partial_Y(z) \right] = 0} \tag{3.28}$$

3.2.2.2 Change of coordinate ζ to account for strain

Furthermore, it is useful to add one last change of variable with $\zeta = Y\sqrt{a_2/2D_0}$ (so that $\partial_Y = \sqrt{a_2/2D_0}\ \partial_\zeta$). With this addition:

$$\frac{a_2}{D_0}\sqrt{\frac{2D_0}{a_2}}\zeta\sqrt{\frac{a_2}{2D_0}}\partial_\zeta(z) + \sqrt{\frac{a_2}{2D_0}}\partial_\zeta \left[\left(\frac{W}{W_0} \right)^2 \left(\frac{T_0}{T} \right)^{(2-b)} \sqrt{\frac{a_2}{2D_0}}\partial_\zeta(z) \right] = 0$$

leading to the following expression:

$$\boxed{2\zeta\partial_\zeta(z) + \partial_\zeta \left[K_0 W^2 T^{b-2} \partial_\zeta(z) \right] = 0} \tag{3.29}$$

with $K_0 = W_0^{-2} T_0^{2-b}$.

Eq. (3.29) is a generic equation controlling any diffusion flame. It gives an estimation of the mixture fraction field in ζ-space which encompasses the strain ($\zeta = Y\sqrt{a_2/2D_0}$) that represents, at given operating point of pressure and temperature, any flame from low strain to extinction. However the solution of this equation is not trivial and can not be solved analytically unlike what was done at constant density in section 3.1.2. Here we use a numerical method to solve Eq. (3.29) and provide the solution for the z field. This is done using the library "*scipy*" and the function "*solvebvp*" on PYTHON. Note that equation (3.29) is independent of strain a_2 which appears only in the reduced variable $\zeta = Y\sqrt{a_2/2D_0}$. To do so some additional work is mandatory.

3.2.2.3 Evaluation of $\partial_\zeta W$

Based on Eq. (3.17), we can write $W = (\alpha z + \beta)^{-1}$. Then we can evaluate $\partial_\zeta W$ as:

$$\partial_\zeta W = \partial_\zeta \left[(\alpha z + \beta)^{-1} \right] = -(\alpha z + \beta)^{-2} \partial_\zeta (\alpha z + \beta) = -(\alpha z + \beta)^{-2} \alpha \partial_\zeta z$$

so that, using the definition of W (3.17) :

$$\boxed{\partial_\zeta W = -W^2 \alpha \partial_\zeta z} \tag{3.30}$$

3.2.2.4 Evaluation of $\partial_\zeta T$

Based on equations (3.18) and (3.19), we can write :

$$T = \begin{cases} z\Delta T^0 + T_O^0 + \dfrac{1-z}{\eta^{sup} z + \lambda^{sup}}\gamma^{sup} & \text{if } z > z_{st} \\ z\Delta T^0 + T_O^0 + \dfrac{z}{\eta^{inf} z + \lambda^{inf}}\gamma^{inf} & \text{if } z < z_{st} \end{cases}$$

On one hand, we can evaluate the 'sup' derivative of T as follows :

$$\begin{aligned}
\partial_\zeta T^{sup} &= \partial_\zeta \left(z\Delta T^0\right) + \partial_\zeta \left(\frac{1-z}{\eta^{sup} z + \lambda^{sup}}\gamma^{sup}\right) \\
&= \Delta T^0 \partial_\zeta z + \gamma^{sup}\left[\partial_\zeta (1-z)\right](\eta^{sup} z + \lambda^{sup})^{-1} + \gamma^{sup}\left[\partial_\zeta \left[(\eta^{sup} z + \lambda^{sup})^{-1}\right]\right](1-z) \\
&= \Delta T^0 \partial_\zeta z + \gamma^{sup}\left[-\partial_\zeta(z)(\eta^{sup} z + \lambda^{sup})^{-1} - (1-z)\,\eta^{sup}(\eta^{sup} z + \lambda^{sup})^{-2}\partial_\zeta(z)\right] \\
&= \partial_\zeta z \left[\Delta T^0 + \gamma^{sup}\left[-(\eta^{sup} z + \lambda^{sup})^{-1} - \eta^{sup}\frac{(1-z)}{(\eta^{sup} z + \lambda^{sup})^2}\right]\right] \\
&= \partial_\zeta z\ \widetilde{T}^{sup}
\end{aligned}$$

$$\text{with } \widetilde{T}^{sup} = \left[\Delta T^0 + \gamma^{sup}\left[-(\eta^{sup} z + \lambda^{sup})^{-1} - \eta^{sup}\frac{(1-z)}{(\eta^{sup} z + \lambda^{sup})^2}\right]\right]$$

On the other hand, we can evaluate the 'inf' derivative of T as follows :

$$\begin{aligned}
\partial_\zeta T^{inf} &= \partial_\zeta \left(z\Delta T^0\right) + \partial_\zeta \left(\frac{z}{\eta^{inf} z + \lambda^{inf}}\gamma^{inf}\right) \\
&= \Delta T^0 \partial_\zeta z + \gamma^{inf}\left[\partial_\zeta (1-z)\right]\left(\eta^{inf} z + \lambda^{inf}\right)^{-1} + \gamma^{inf}\left[\partial_\zeta \left[(\eta^{sup} z + \lambda^{sup})^{-1}\right]\right](z) \\
&= \Delta T^0 \partial_\zeta z + \gamma^{inf}\left[\partial_\zeta(z)\left(\eta^{inf} z + \lambda^{inf}\right)^{-1} - \eta^{inf} z\left(\eta^{inf} z + \lambda^{inf}\right)^{-2}\partial_\zeta(z)\right] \\
&= \partial_\zeta z \left[\Delta T^0 + \gamma^{inf}\left[\left(\eta^{inf} z + \lambda^{inf}\right)^{-1} - \eta^{inf}\frac{z}{(\eta^{inf} z + \lambda^{inf})^2}\right]\right] \\
&= \partial_\zeta z\ \widetilde{T}^{inf}
\end{aligned}$$

$$\text{with } \widetilde{T}^{inf} = \left[\Delta T^0 + \gamma^{inf}\left[\left(\eta^{inf} z + \lambda^{inf}\right)^{-1} - \eta^{inf}\frac{z}{(\eta^{inf} z + \lambda^{inf})^2}\right]\right]$$

so that :

$$\boxed{\partial_\zeta T = \begin{cases} \partial_\zeta z\ \widetilde{T}^{sup} & \text{if } z > z_{st} \\ \partial_\zeta z\ \widetilde{T}^{inf} & \text{if } z < z_{st} \end{cases}} \tag{3.31}$$

3.2.2.5 Evaluation of $\partial_\zeta[W^2T^{b-2}\partial_\zeta z]$

Since $(f\ g\ h)' = f'\ gh + g'\ fh + h'\ fg$ then :

$$\begin{aligned}\partial_\zeta[W^2T^{b-2}\partial_\zeta z] &= \partial_\zeta\left(W^2\right)T^{b-2}\partial_\zeta z\ +\ \partial_\zeta\left(T^{b-2}\right)W^2\partial_\zeta z\ +\ \partial_\zeta\left(\partial_\zeta z\right)W^2T^{b-2}\\ &= 2W\partial_\zeta\left(W\right)T^{b-2}\partial_\zeta z\ +\ (b-2)T^{b-3}\partial_\zeta\left(T\right)W^2\partial_\zeta z\ +\ \partial_{\zeta\zeta}(z)W^2T^{b-2}\\ &= -2\alpha WW^2\partial_\zeta\left(z\right)T^{b-2}\partial_\zeta z\ +\ (b-2)T^{b-3}\widetilde{T}\partial_\zeta\left(z\right)W^2\partial_\zeta z\ +\ \partial_{\zeta\zeta}(z)W^2T^{b-2}\end{aligned}$$

so that :

$$\boxed{\partial_\zeta[W^2T^{b-2}\partial_\zeta z] = \partial_{\zeta\zeta}(z)W^2T^{b-2} + (\partial_\zeta z)^2\left[-2\alpha W^3T^{b-2} + (b-2)T^{b-3}\widetilde{T}W^2\right]} \tag{3.32}$$

3.2.2.6 Final system of equation for z

Using equations (3.30), (3.31), (3.32) and setting $h = \partial_\zeta z$, we can develop a system of first order ODE's (Ordinary Differential Equation) starting from Eq. (3.29) :

$$\begin{aligned}2\frac{\zeta}{K_0}h + \partial_\zeta\left[W^2T^{b-2}h\right] &= 0\\ 2\frac{\zeta}{K_0}h + \partial_\zeta(h)W^2T^{b-2} + h^2\left[-2\alpha W^3T^{b-2} + (b-2)\widetilde{T}W^2\right] &= 0\\ 2\frac{\zeta}{K_0}\frac{T^{2-b}}{W^2}h + h^2\left[-2\alpha\frac{W^3T^{b-2}T^{2-b}}{W^2} + (b-2)\widetilde{T}\frac{W^2T^{b-3}T^{2-b}}{W^2}\right] &= -\partial_\zeta(h)\\ 2\frac{\zeta}{K_0}\frac{T^{2-b}}{W^2}h + h^2\left[-2\alpha W + (b-2)\frac{\widetilde{T}}{T}\right] &= -\partial_\zeta(h)\end{aligned}$$

so that:

$$\boxed{\begin{cases}\partial_\zeta h = & -h\left[2\dfrac{\zeta}{K_0}\dfrac{T^{2-b}}{W^2} + h\left[-2\alpha W + (b-2)\dfrac{\widetilde{T}}{T}\right]\right]\\ \partial_\zeta z = & h\end{cases}} \tag{3.33}$$

where $b = 1.75$ and $K_0 = W_0^{-2}T_0^{2-b}$. This set of equations is solved PYTHON and provides the solution $z(\zeta)$ which can then be used to obtain all fields of interest.

3.2.3 Evaluation of key physical quantities

3.2.3.1 Evaluation of diffusivity

Given the invalidity of Lewis's unitary hypothesis, we must choose the appropriate diffusivity at this point (thermal or species). The present theory assumes unity lewis numbers but still requires to choose a (single) value for diffusivities. This unique diffusivity value can be the thermal diffusivity D_{th} but also one of the species diffusivity D_k (k subscript for species) or a combination of them. Since CANTERA resolves Eq. (3.25), we can actually determine for which value of D between the thermal and the species (with H_2) has

the best fit by evaluate $\rho u \partial_y z$ against $\partial_y[\rho D \partial_y z]$ as function of position. An additional curve is present with perfect diffusivity, named D_{fit}, with the following evaluation :

$$D_{fit} = \frac{\int_{y_0}^{y} (\rho u \partial_y z)\, dy}{\rho \partial_y z} \tag{3.34}$$

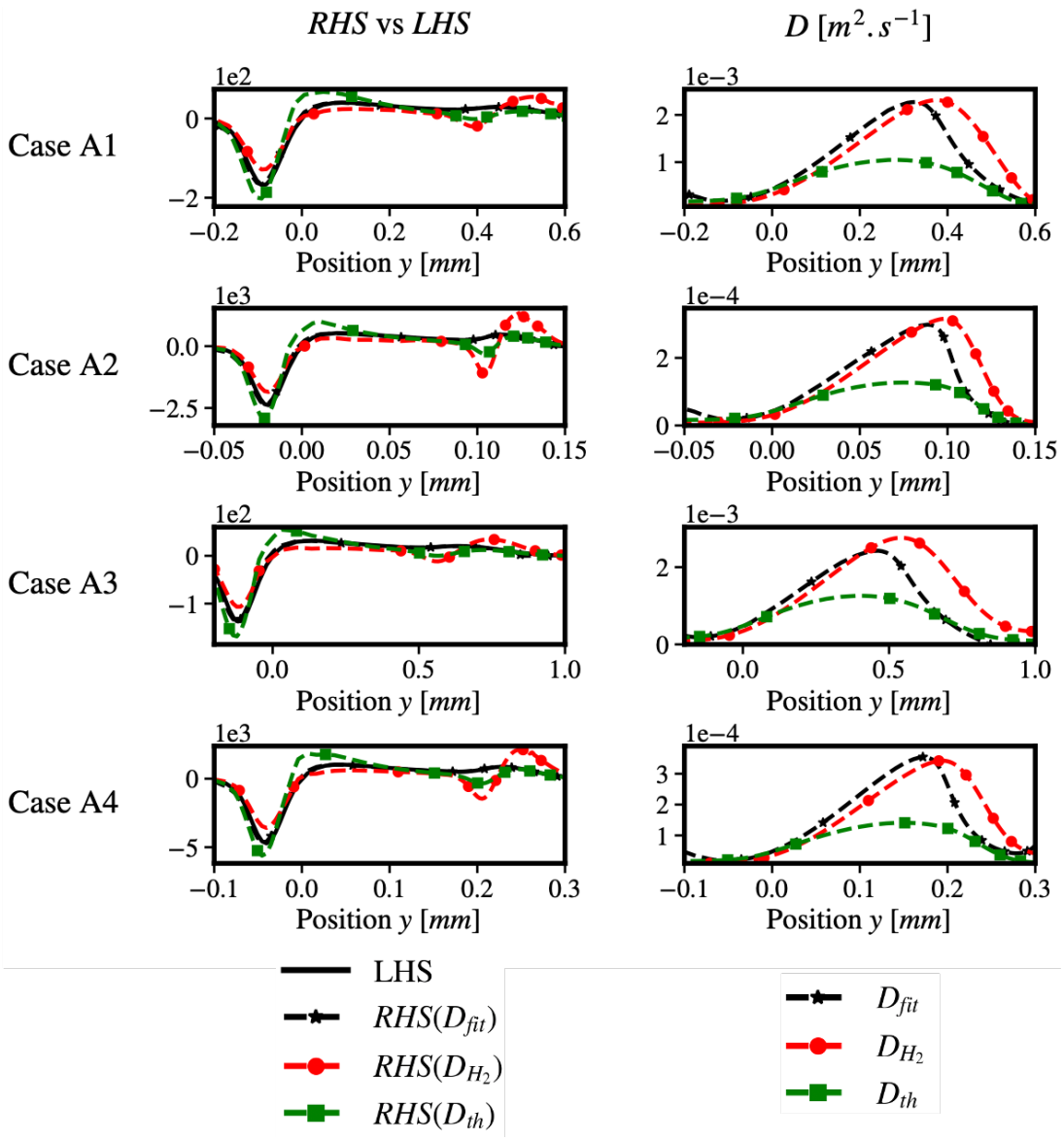


Figure 3.11: Determination of the best diffusivity to use in the VD z equation by evaluating Eq. (3.25) with D_{H_2}, D_{th} and D_{fit} as function of position on left. On right, diffusivities as function of position are displayed for each flame of group A (Tab. 3.1).

Figure 3.11 left displays the comparison between the RHS of equation (3.25) and the LHS depending on the diffusivity chosen and on the right diffusivities as function of the position zoomed on the reactive part. The black dashed curve on left with star symbol represents the best fit from the black solid curve, which is the goal. There is a remarkable agreement between diffusivities profiles of H_2 (red curve with circle marker) and the

numerical but fitted one (black dashed curve with star symbol): this shows that D_{H_2} is the best choice and will be the diffusivity value ued in the rest of the chapter under the variable D. In these H_2-air flames, the diffusivity of H_2 appears to be the controlling one.

3.2.3.2 Evaluation of the scalar dissipation rate χ

Once the z equation of Eq. (3.33) is solved using the diffusivity of H_2, we can evaluate all quantities of interest. We begin by expressing the scalar dissipation rate χ. By definition:

$$\chi = 2D(\partial_y z)^2 \tag{3.35}$$

Furthermore:

$$\partial_y z = \partial_y Y \ \partial_Y \zeta \ \partial_\zeta z$$

Using:

$$\begin{cases} \partial_y Y & = \dfrac{\rho}{\rho_0} \\ \partial_Y \zeta & = \sqrt{\dfrac{a}{2D_0}} \\ \partial_\zeta z & = \text{solution of (3.29)} \end{cases} \tag{3.36}$$

Then:

$$\partial_y z = \frac{\rho}{\rho_0}\sqrt{\frac{a}{2D_0}}\partial_\zeta z \tag{3.37}$$

So that χ is obtained by :

$$\chi = 2D\left(\frac{\rho}{\rho_0}\sqrt{\frac{a}{2D_0}}\partial_\zeta z\right)^2 = \frac{2a}{2}\frac{D\rho^2}{D_0\rho_0^2}(\partial_\zeta z)^2 = a\left(\frac{W}{W_0}\right)^2\left(\frac{T_0}{T}\right)^{(2-b)}(\partial_\zeta z)^2$$

Then:

$$\boxed{\chi = \frac{a}{K_0}W^2T^{(b-2)}(\partial_\zeta(z))^2} \tag{3.38}$$

3.2.3.3 Evaluation of the total fuel consumption $\dot{\Omega}_F$

By definition the integrated fuel consumption rate per square meter of flame is [87]:

$$\dot{\Omega}_F = \rho_{st}D_{st}\partial_y Y_F\Big|_{y=y_{st}^-} \tag{3.39}$$

Furthermore:

$$\partial_y Y_F\Big|_{y=y_{st}^-} = \partial_y Y \ \partial_Y \zeta \ \partial_\zeta z \ \partial_z Y_F\Big|_{z=z_{st}^-}$$

Using:

$$\begin{cases} \partial_y Y\Big|_{z=z_{st}^-} & = \dfrac{\rho_{st}}{\rho_0} \\ \partial_Y \zeta\Big|_{z=z_{st}^-} & = \sqrt{\dfrac{a}{2D_0}} \\ \partial_z Y_F\Big|_{z=z_{st}^-} & = \dfrac{Y_F^{\text{fuel side}}}{1-z_{st}} \\ \partial_\zeta z\Big|_{z=z_{st}^-} & = \text{solution of (3.29) evaluated at "minus" stoichiometry} \end{cases} \tag{3.40}$$

So that:

$$\partial_y Y_F\Big|_{y=y_{st}^-} = \frac{\rho_{st}}{\rho_0}\sqrt{\frac{a}{2D_0}}\frac{1}{\alpha^{inf}}\frac{Y_F^{\text{fuel side}}}{1-z_{st}}\partial_\zeta y\Big|_{z=z_{st}^-} \tag{3.41}$$

And:

$$\dot{\Omega}_F = \rho_{st} D_{st}\frac{\rho_{st}}{\rho_0}\sqrt{\frac{a}{2D_0}}\frac{1}{\alpha^{inf}}\frac{Y_F^{\text{fuel side}}}{1-z_{st}}\partial_\zeta y\Big|_{z=z_{st}^-}$$

So that:

$$\boxed{\dot{\Omega}_F = \frac{\rho_{st}^2 D_{st}\sqrt{a}}{\rho_0(1-z_{st})\sqrt{2D_0}}\ \partial_\zeta z\Big|_{z=z_{st}^-}} \tag{3.42}$$

where $\partial_\zeta z$ can be obtained from the field of $z(\zeta)$ (result of Eq. (3.33)).

3.2.3.4 Evaluation of the thickness flame δ_{flame}

By definition :

$$\delta_{flame} = \frac{1}{\max(\nabla z)} \tag{3.43}$$

Eq. (3.37) gives $\nabla(z)$:

$$\boxed{\delta_{flame} = \max\left(\frac{\rho}{\rho_0}\sqrt{\frac{a}{2D_0}}\partial_\zeta z\right)^{-1}} \tag{3.44}$$

With the last equation and the use of the model VD, it is possible to determine all the key variables for characterizing the flames in table 3.4 based on the results of the EDO equation (3.33). Table 3.5 summarizes major differences between constant density model (literature), variable density model (our model) and CANTERA (fully resolved simulation). All relevant variables are expressed directly and we can now compare the VD and CD models on reference cases extract from tables 3.1 and 3.2.

Table 3.5: Summary of the differences between CD, VD models and CANTERA

	Constant Density (CD)	Variable density (VD)	CANTERA
Density ρ	$\rho = cst$	$\rho = \frac{PW}{RT}$	$\rho = \frac{PW}{RT}$
Molecular Weight W	$W = cst$	$\frac{1}{W} = \sum_k \frac{Y_k}{W_k}$ (4 species)	$\frac{1}{W} = \sum_k \frac{Y_k}{W_k}$ (all species)
Diffusivity D	$D_k = D_{th} = cst$	$D_k = D_{th} = D_{H_2} = D^0_{H_2} \left(\frac{T}{T_0}\right)^b$	Mixture average [6, 30, 48, 70, 110]
Lewis Le	$Le = 1$	$Le = 1$	$Le_k = \frac{D_{th}}{D_k}$
Temperature T	Independent of strain	Dependent of strain	Transported
Chemistry	Infinitely fast	Infinitely fast	Detail chemistry : San Diego [100]

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3.3 Comparison of constant density and variable density flame structures with numerical results

We now have three methods to compute steady strained diffusion H_2-air flames:

- the classical CD (constant density) analytical solution described in section 3.1.2;
- the new VD (variable density) approach described in section 3.2;
- CANTERA with full chemistry (documentation Sec. 1.4).

We compare these three methods above.
For the VD model, we will also explore various values for b which controls the change of D versus T: $D = D_0(T/T_0)^b$.

3.3.1 Comparison on one flame at a given strain

The group A of table 3.1 defines different types of flames in order to see if the model VD can be used for steady, strained, one-dimensional diffusion flames. Figure 3.12 displays evaluations of key variables (the mixture fraction z, its derivative with respect to ζ and the scalar dissipation rate χ, from top to bottom) as function of ζ for case A1. First, several thin solid curves are plotted following the VD model and are based on the value of the temperature exponent of diffusivities b. Once again, the value $b = 1.75$ represents the best candidate in the VD model: the evaluation of z is close to CANTERA and the shape of χ follows CANTERA (in term of position and height). Figures 3.13, 3.14 and 3.15 present the same conclusion for, respectively, cases A2, A3 and A4 from table 3.1.
Figure 3.16 summarizes all results of A cases with the addition of temperature profile. The VD model with $b = 1.75$ (blue dashed curve with stars) presents better agreement with CANTERA (solid black line) against CD model (red dashed curve with squares). This observation is validated for all pressures and temperatures. Notice that the slight discrepancy on z evaluation between CANTERA and VD model tends to increase on χ evaluation (which makes sense since χ depends on $(\nabla z)^2$), thus we must be more stringent with z than for other variables.
Moreover, we still need to challenge our model over a wide range of strain to ensure its validity.

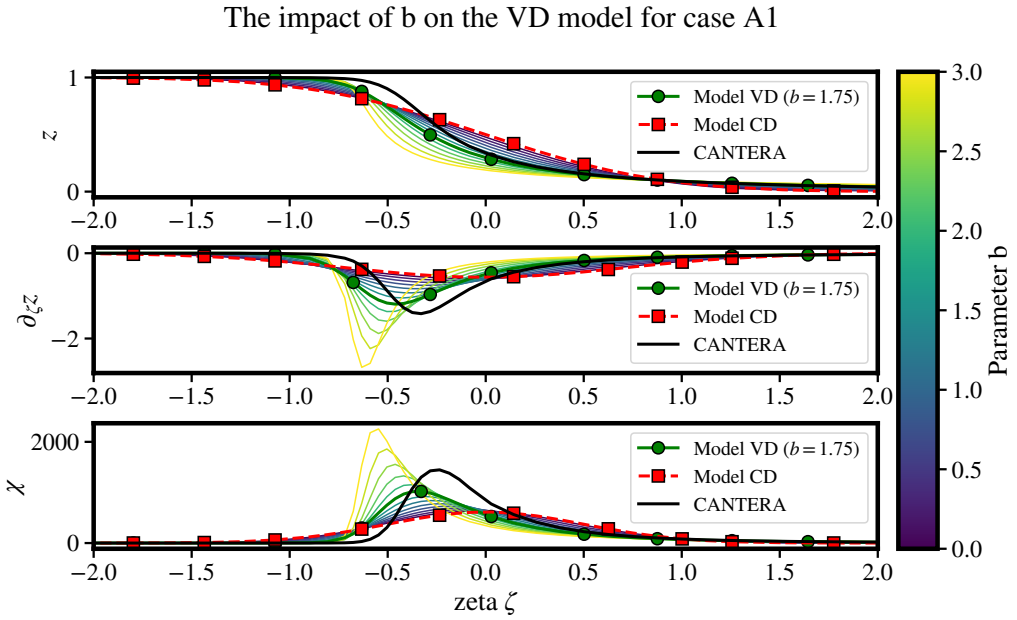


Figure 3.12: From top to bottom, mixture fraction z, its derivative with respect to ζ and scalar dissipation rate χ versus ζ. Thin solid lines represent the model VD for a range of b from 0 to 3. Thick lines represent the main curves which are the result of CANTERA (solid black), model CD(dashed red with squares) and model VD for $b = 1.75$ (dashed green curve with circles) for the case A1 (cf Tab. 3.1)

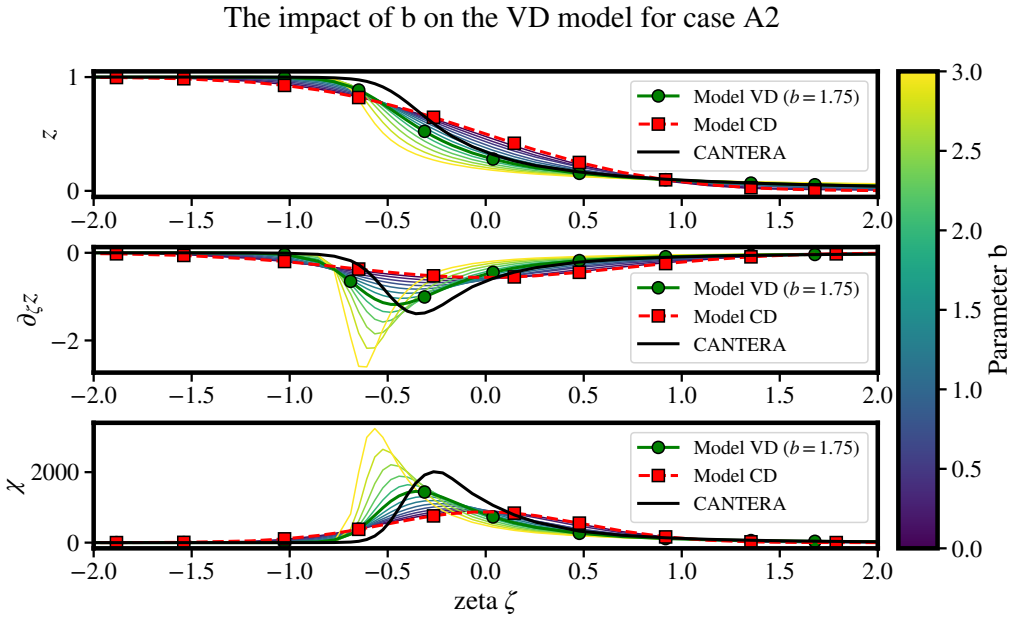


Figure 3.13: From top to bottom, mixture fraction z, its derivative with respect to ζ and scalar dissipation rate χ versus ζ. Thin solid lines represent the model VD for a range of b from 0 to 3. Thick lines represent the main curves which are the result of CANTERA (solid black), model CD (dashed red with squares) and model VD for $b = 1.75$ (dashed green curve with circles) for the case A2 (cf Tab. 3.1)

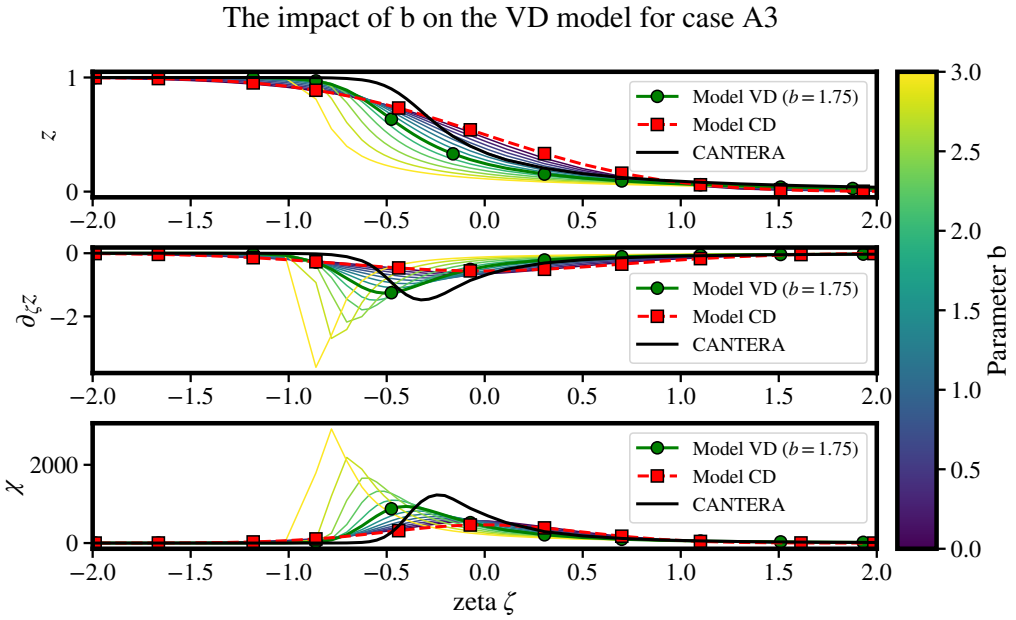


Figure 3.14: From top to bottom, mixture fraction z, its derivative with respect to ζ and scalar dissipation rate χ versus ζ. Thin solid lines represent the model VD for a range of b from 0 to 3. Thick lines represent the main curves which are the result of CANTERA (solid black), model CD (dashed red with squares) and model VD for $b = 1.75$ (dashed green curve with circles) for the case A3 (cf Tab. 3.1)

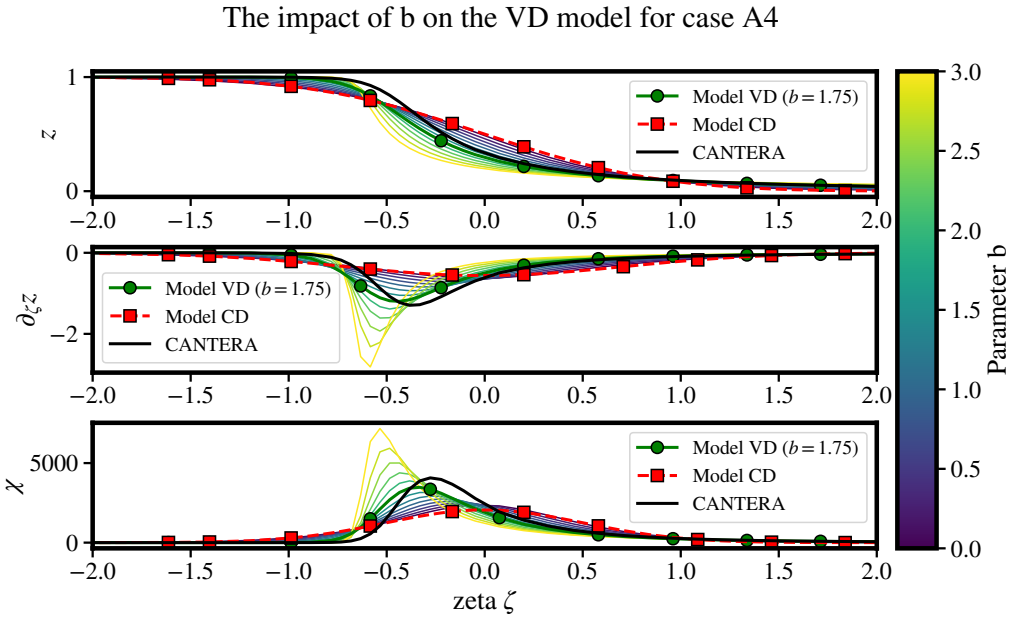


Figure 3.15: From top to bottom, mixture fraction z, its derivative with respect to ζ and scalar dissipation rate χ versus ζ. Thin solid lines represent the model VD for a range of b from 0 to 3. Thick lines represent the main curves which are the result of CANTERA (solid black), model CD (dashed red with squares) and model VD for $b = 1.75$ (dashed green curve with circles) for the case A4 (cf Tab. 3.1)

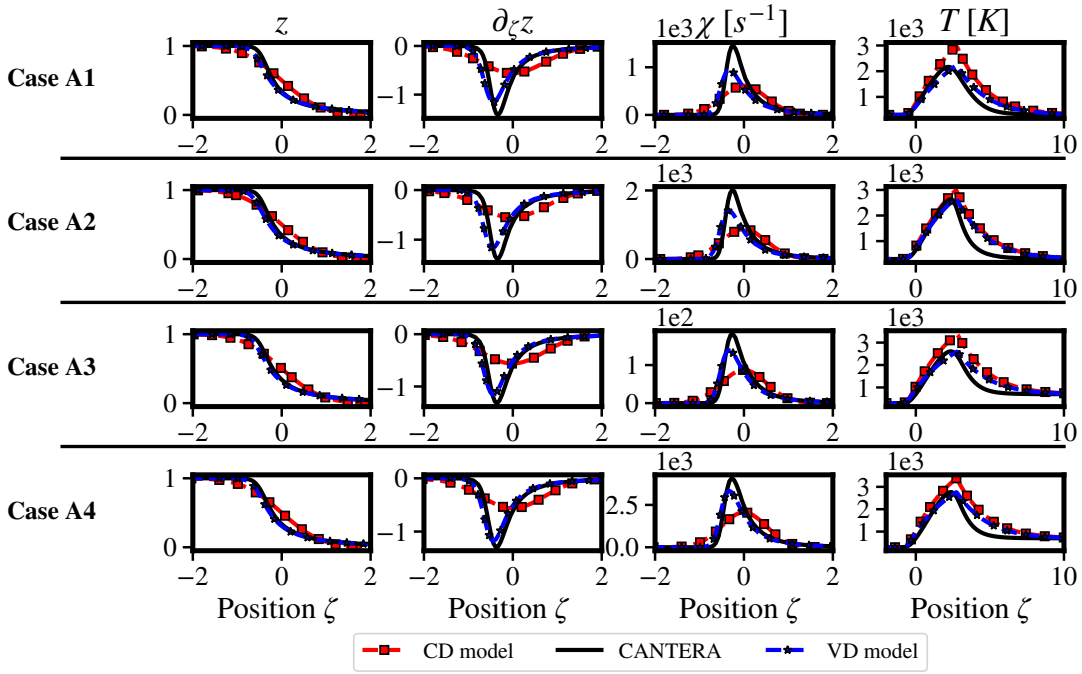


Figure 3.16: From left to right, key variables: mixture fraction z, its derivative, scalar dissipation rate χ and temperature profile T as function of position ζ for VD model (blue dashed line with stars), CD model (red dashed line with squares) and CANTERA (solid black line). from top to bottom, all A cases are plotted (Tab. 3.1)

3.3.2 Effect of strain a

The group B of the table 3.2 defines different type of flame in order to see if the model VD can be use for all steady, strained, one dimensional diffusion flame over a wide range of strain value. Figure 3.17 displays evaluations of key variables for the VD model: from left to right, total fuel consumption $\dot{\Omega}_F$, thickness of the flame δ_f, maximum of scalar dissipation rate χ_{max} and maximum of temperature as function of normalized strain a_1/a_{max} for each B cases. The VD model still presents a real improvement against the classical CD model.

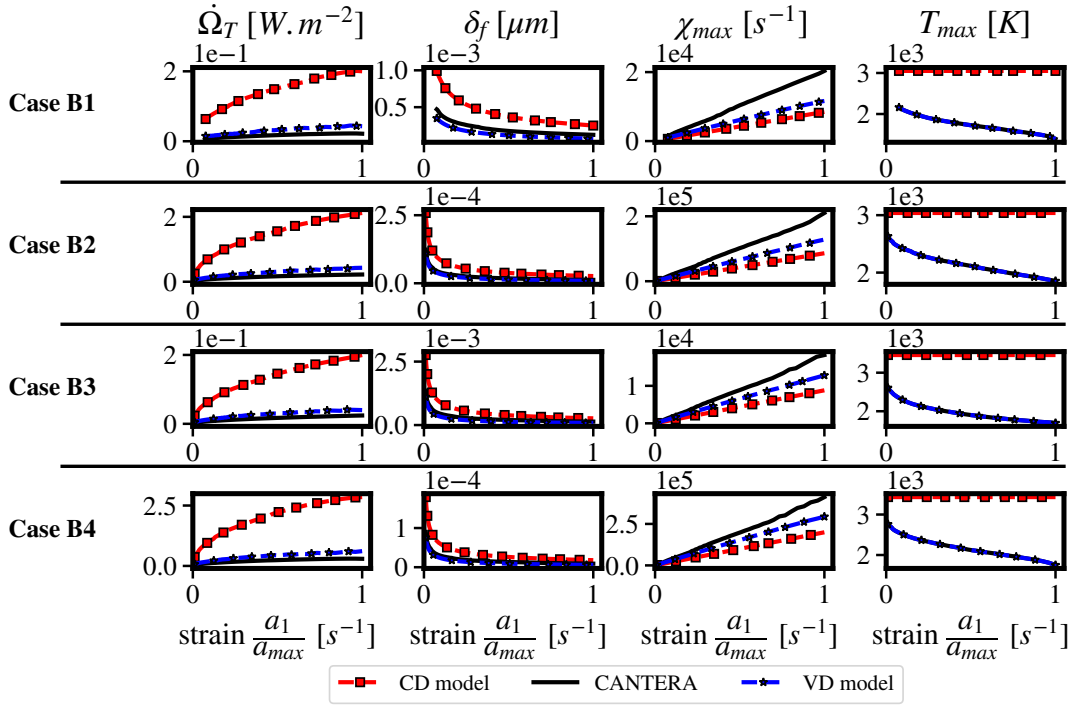


Figure 3.17: From left to right, key variables: total fuel consumption $\dot{\Omega}_F$, thickness of flame δ_f, maximum of scalar dissipation rate χ_{max} and maximum of temperature T_{max} as function of normalized strain a_1/a_{max} for VD model (blue dashed line with stars), CD model (red dashed line with squares) and CANTERA (solid black line). from top to bottom, all B cases are plotted (Tab. 3.2)

3.4 Conclusion and perspective

In conclusion, it is possible to build a theoretical model accounting for variable density for hydrogen-air diffusion flames which can be solved almost fully analytically and matches exact results (obtained with CANTERA) better than the usual constant density (CD) approach described in textbooks. One reason for this is that this VD method, like CD, assumes infinitely fast chemistry, something which is true for most of hydrogen flames. The variable density model has been tested over a wide range of strain, pressure and temperature values and still captures clearly CANTERA results in all cases. The VD model can be a major improvement in H_2 LES (or any other simulation where density exhibits huge variations) especially under the assumption of flamelet where a 1D model is a must [29, 91].
However, the CD model exhibits one important quality. It has direct analytical formula and relies solely on one parameter: the strain (even if it is not straight forward to capture in realistic 3D simulations). The VD model requires either to use Eq. (3.33) directly or to tabulate z over operating points determined by the ambient temperature, pressure and strain. These differences lead to further difficulties for a practical development of a full LES model for diffusion flame. Thus, the constant density formula will be used whenever it is needed in the next chapter and the VD model application will be a future improvement for the model presented in PART III.

Finally this is the first generation of the VD model and several enhancements can be proposed. First, since only the H_2 species diffusivity is considered, one could consider to incorporate the mixture diffusivity instead [48]. Second, concerning the temperature profile, our first trivial approach to capture the strain dependency (using the 'p' factor of Eq. 3.20) can be upgraded following Liñán theory [63] where temperature can be modeled directly as function of strain and tends to a better agreement with T_{max}. Third, an investigation to estimate key variables on the stagnation plane (such as T, ρ or D) can be a great path to follow. With these improvements it will be sure that the VD model can be a better model than CD for H_2 diffusion flames LES and with the last point it will have no longer any additional application issue.

Part III

Construction of an LES model for turbulent diffusion flames

The previous part allowed us to understand the behavior of H_2-air steady strained diffusion flames, the role of key parameters and evaluate them in a one dimensional context. In addition, consistent definitions of strain that can be used for all different flames were presented, that give more precise estimation of it in an LES simulation (under the flamelet assumption). This knowledge will be our foundation to move on to the next couple of objectives. This part is split into two chapters :

- Chapters 4 determines how many mesh points are needed inside a diffusion flames to ensure a proper spatial resolution. In fact, resolution issues of the resolved scales must be solved before any model can be proposed for unresolved scales. Using CANTERA and AVBP, chapter 4 focuses on resolution issues and establishes a simple criterion. Based on the thickness of the diffusion front compared to mesh size [17], this criterion will be tested over a wide range of configurations. To assure the veracity of our criterion, several test cases are presented from one-dimensional laminar steady strained diffusion H_2-air flames to two dimensional flame/turbulence interaction by way of two dimensional flame/vortex interaction. Moreover, to guarantee the transition from CANTERA to AVBP an additional study has been performed to clarify the consistency in the results across the two solvers.

- Chapter 4 provides a spatial resolution criterion and chapter 5 shows how thickening can be used in a model called DFTC to satisfy this criterion. First, a clear list of objectives that our model should respect will be presented. Starting from the resolution aspect, the model must conserve key parameters (such as quenching at the right strain or provides the correct maximum of temperature) and then try to recover global quantities (such as the total heat release $\dot{\Omega}_F$) that are strongly affected by poorly resolved simulations. The so-called Diffusion Flame Thickening Control (DFTC) model relies on a dynamic adaptation of the classical TFLES (Thickened flame LES) for premixed flames [17] linked to a temporal monitoring of the flame resolution during the simulation. If the strain of the diffusion flame is high and the resolution is not sufficient, the model is turned on by thickening the diffusion front. As for TFLES, a 'mapping' operation can be done on diffusion and/or reaction terms in the transport equations in order to recover the objectives of the DFTC model. Each parameter of the 'mapping' will be investigated and a discussion about whether they may or may not show improvement against the list of model requirements will be risen. To do so, the impact of 'mapping' on global and local behavior is examined. Here, the ambition of the model is to be self-sufficient under a dynamic adaptation following local conditions without any additional help such as tabulated variables or other transported equations.

Chapter 4

Resolution of diffusion flame fronts

This chapter presents a study of spatial resolution in Large eddy simulation of diffusion H_2-air flames, based on the following points:

- Observe and analyze the impact of the mesh size on key variables.
- Determine, over a wide range of mesh sizes, the frontier between resolved front flame and unresolved one.
- Extend these conclusions over several configurations from steady to turbulent.

Once these issues are solved, this knowledge will be the foundation to build a new model dedicated to diffusion flames in chapter 5. On top of that, this chapter will demonstrate that resolution is a key parameter, that any simulation needs to look into and that a full comprehension of it unlocks the possibility to develop a model based on it. This approach is also orthogonal to the methods which rely on artificial viscosity in many codes in order to solve the under resolution issue. We will actually show that the role of that artificial viscosity and why it should be avoided.
A naive view of resolution would be to say that if the front flame is not resolved enough, additional mesh points should solve the issue. This approach may work at low pressure and low Reynolds numbers. However it fails for many real flames: in engine-relevant configurations, the combination of elevated pressures (ranging from several tens of bar in automotive and aero-engines to several hundreds of bar in rocket propulsion systems) and intense strain rates leads to extremely thin instantaneous flame thicknesses (Cf. evolution of flame thickness as function of strain and pressure in appendix A.1). Although the exact flame thickness depends on the local thermochemical state and transport properties, its proper numerical resolution would typically require grid spacings of the order of a few micrometers.
Thus, while many studies focus on flame-turbulence interaction (FTI) models, an issue which is often ignored is the spatial resolution of the flame. A simple metric for spatial resolution is a quantity called FRI (Flame Resolution Index), which measures the number of points used to resolve the laminar flame thickness δ_L:

$$FRI = \delta_L/\Delta \tag{4.1}$$

where Δ is the local grid size. This definition applies to both premixed and diffusion flames, using an appropriate definition of flame thickness δ_L in each case. FRI indicates

how many mesh points are used to solve the conservation equations describing chemical reactions and diffusion processes within the flame. Most DNS and one-dimensional flame solvers like CANTERA [41] use high FRI values (typically 20 for DNS and more than 100 using mesh adaptation in 1D solvers for laminar flames).

For premixed configurations, flamelet-based approaches such as TFLES [17, 57, 89] and F-TACLES [28, 34] explicitly address the need to control the FRI of premixed fronts, either through artificial thickening or by filtering the flame structure. In this framework, the underlying mechanisms are well established. Within TFLES, a laminar premixed flame can be thickened by enhancing molecular diffusivities, while theoretical arguments provide a straightforward scaling to preserve the laminar flame speed, which is a key intrinsic property [87]. Moreover, a correct turbulent flame speed can, at least in principle, be recovered by introducing an efficiency function that compensates for the reduced wrinkling of the artificially thickened flame [13, 17].

For diffusion flames, the situation is fundamentally different. Chapter 4 and chapter 5 address respectively the evaluation of FRI and the development of flamelet-based models that ensure adequate resolution within LES frameworks applied to diffusion flames. This issue is generic and does not depend on the specific fuel considered. Hydrogen flames are adopted here as a representative test case, as they allow to avoid two-phase flow effects and rely on comparatively simple chemical kinetics compared to hydrocarbons. In addition, hydrogen–air combustion naturally promotes diffusion-type flame structures and provides a particularly stringent framework for model development and assessment.

The question of flame resolution for diffusion flames is often ignored as LES users assume that the numerical method somehow will adjust if the spatial resolution is insufficient, i.e. if FRI is small. When FRI becomes close to 3, the flame is resolved on two or three mesh points: it is difficult to imagine that the numerical method can handle such steep fronts accurately. The usual solution is to add numerical viscosity [18] or to use upwind schemes for spatial differentiation, in order to avoid code crashes, hoping that this numerical adjustment does not interfere with the solution.

The fact that most studies linked to resolution focus on premixed flames and ignore diffusion fronts is due to a crucial difference between premixed and diffusion fronts when it comes to evaluate $FRI = \delta_L/\Delta$. For premixed flames, the thickness δ_L is easy to estimate using the usual definition $\delta_L = (T_b - T_u)/\max(\nabla T)$ [87]. Moreover, δ_L is only weakly sensitive to strain so that this estimate can be used before the simulation to determine resolution needs. For non-premixed flames, however, the situation is more complicated. A simple method to measure δ_L in a diffusion flame is $\delta_L = 1/\max(\nabla z)$ in flames where the mixture fraction z varies from 0 in the oxidizer stream to 1 in the fuel stream. In the flamelet regime, the thickness of the diffusion flames varies like $1/\sqrt{a}$ where a is the local strain [77]. Chapter 2 presents strain definitions that show directly that a can change over a wide range of values in a turbulent diffusion flame, and that it is unknown a priori. So, estimating resolution requirements for a diffusion flame is difficult and depends strongly on the flow. However, simple estimates show that the thickness of diffusion fronts can become very small at high strains and high pressures (Fig. 4.1 and appendix A.1) so that highly strained diffusion fronts are the zones controlling resolution in most high pressure simulations. Note that non-reacting fronts (such as quenched or non ignited flames) raise the same problem.

An additional difficulty for diffusion flames is that two values of FRI must be distin-

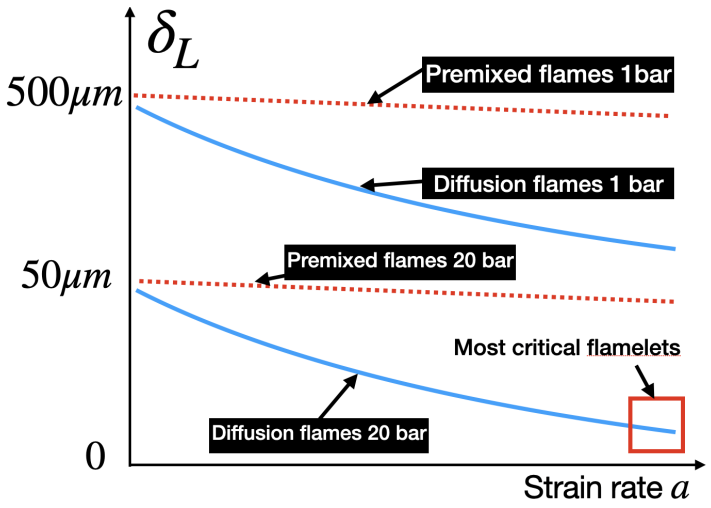


Figure 4.1: Typical H_2-air flame thicknesses as a function of local strain rate a and pressure for premixed and diffusion flamelets.

guished:

1. the true value of FRI called FRI_t is:

$$FRI_t = \delta_L^t/\Delta \tag{4.2}$$

   where δ_L^t is the exact flame thickness;

2. the resolved value of FRI, called FRI_{res} is:

$$FRI_{res} = \delta_L^{res}/\Delta \tag{4.3}$$

   where δ_L^{res} is the flame thickness measured during the LES, as resolved on its mesh.

Obviously, even if δ_L^t is very small, δ_L^{res} will never be less that approximately two times the mesh size Δ when the flame is resolved on two mesh points so that FRI_{res} is always larger than two. When the flame is well resolved, FRI_{res} and FRI_t are large and equal. When the flame becomes under resolved, FRI_{res} never goes below two while FRI_t can become arbitrarily small on coarse meshes for highly strained flames.

In this context, the present chapter focuses on two issues. First, the resolution needs in terms of FRI are discussed using laminar stagnation point flames (Section 4.2) and flame / vortex interaction (Section 4.3) studies. In the flamelet regimes limit, these laminar flames are proper configurations to investigate the effects of resolution (measured by FRI) on the solution. However, in order to initiate each simulation, CANTERA profile is used. For the sake of consistency, a check of consistency between CANTERA and AVBP is presented before looking into qualitative observations (Section 4.1) and proves that CANTERA and AVBP provide similar results. Figure 4.2 schematically represents the temperature, species and heat release profiles typically found in a canonical diffusion flame.

The heat release (like the mass source terms for most species) follows a bell curve with three inflection points. A simple piecewise linear representation that preserves the extrema of the curve and inflection points requires at least 5 points. Then, it is clear

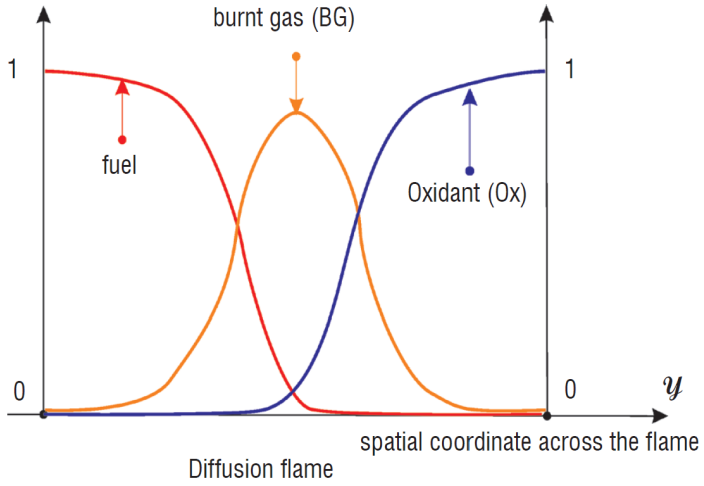


Figure 4.2: Schematic representation of profiles (fuel mass fractions in red, oxidizer mass fractions in blue, heat release in orange) in a diffusion flame.

that it would therefore be difficult to use fewer than 5 points to discretize this curve ($FRI_t = 5$). However, the primary question that naturally arises concerns the minimum number of points required to preserve the essential features of the flame like quenching strain rates a_Q or maximum temperature T_{max}. A second - and more fundamental - question is to what extent the minimum resolution criterion established for the physical flame, formulated in terms of FRI_t, can be transposed to FRI_{res}, based on the resolved flame thickness.

This chapter shows that, for hydrogen-air flames solved using the San Diego chemical scheme [100], $FRI_{res} \approx FRI_t = 5$ is indeed the threshold under which the computation becomes wrong and that it is not necessary to go beyond. More generally, a direct examination of many recent LES papers at atmospheric pressure, including during the recent HYLON campaign of the TNF workshop (tnfworkshop.org/hylon2/), shows that this condition is usually reached in most codes: these flames have thicknesses of the order of 500 microns and many solvers today can have meshes of the order of 100 microns so that FRI_t is of the order of 5. The situation changes drastically at high pressures where flame thicknesses decrease to a few microns (appendix A.1) leading to a systemic under resolution and a significant risk for the community: using codes correctly validated at 1 bar, at high pressures where they become systematically under resolved. A proposition to address this under-resolution issue is presented in the next chapter 5.

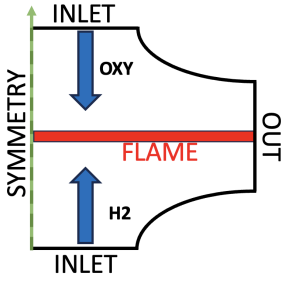
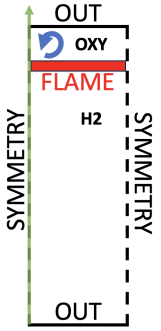
The resolution issue interferes with Flame Turbulence Interaction (FTI) models. The approach followed here is to consider that no proper FTI model can be built on an under resolved solution and that the resolution issue must be solved before introducing any FTI model. For example, many FTI models for diffusion flames are based on scalar dissipation χ i.e. on ∇z [80, 84, 87]. If the z front is resolved on only three points separated by a distance equal to the mesh size Δ, ∇z will be of the order of $(1-0)/\Delta$: it will be imposed by the mesh size Δ and have no link with the real gradients of mixture fraction so that the FTI model may work but will lead to wrong subgrid effects and non-physical responses.

Two test cases are then used to examine the FRI criterion in table 4.1:

- Strained stagnation Point Laminar Flame (SPLF) (Section 4.2).

- Flame/Vortex Interaction (FVI) (Section 4.3).

Table 4.1: Schemes of test cases

Case Acronym	Scheme
SPLF	
FVI	

4.1 Migration from CANTERA to AVBP

In the previous part, several flames were studied with CANTERA. In order to move on multi-dimensional simulation AVBP is used. Five hydrogen-air flames have been selected from low to high strain at atmospheric pressure in order to perform a parametric study. The name and key parameters of these flame are summarize in table 4.2.

To initiate the flame in all configurations described in table 4.1 an AVBP tool is used called *CAN2AV*. This tool extracts, from a 1D CANTERA solution, mass fraction Y_k profiles for each species k, temperature profile and density profile then convert it into AVBP formalism. Unlike premixied flames, where CANTERA and AVBP are both operated in one dimension, strained diffusion falmes are calculated in one dimension flames are

Table 4.2: Strained stagnation point laminar flames used to determine minimum FRI_t values for stable and consistent solutions.

Case Name	Strain $[s^{-1}]$	Thickness $[\mu m]$	Maximum Temperature $[K]$
A	398	1045	2410
B	1603	526	2201
C	5279	289	1962
D	10165	205	1802
E	23868	133	1503

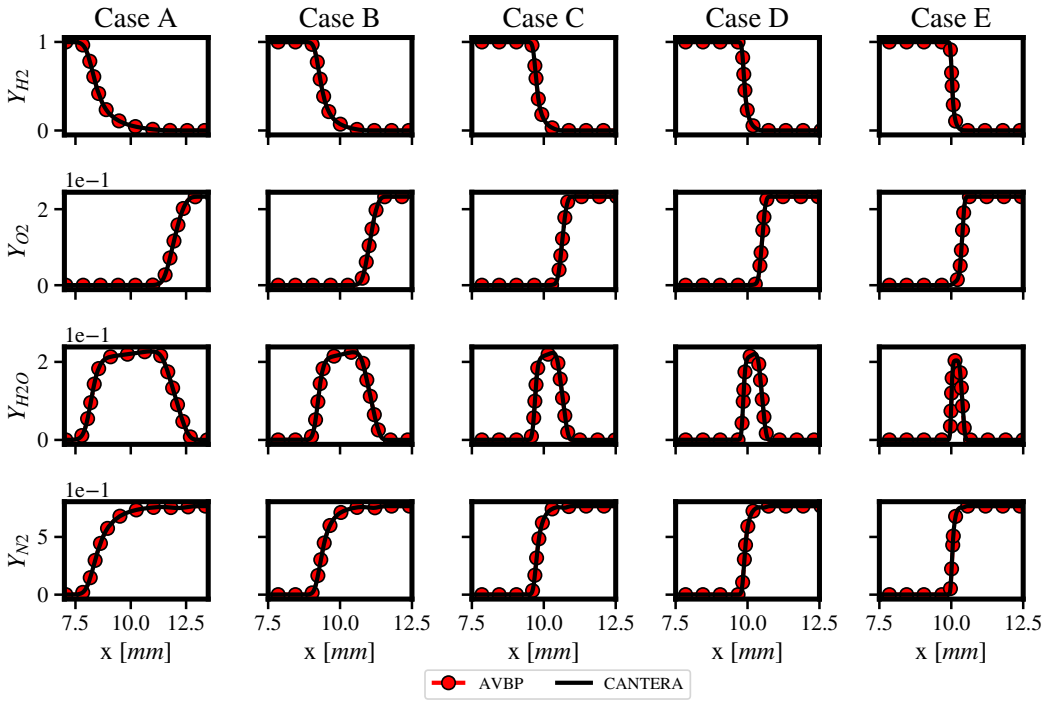


Figure 4.3: Comparison of AVBP initial solution (dashed red line with circle) against CANTERA (solid black line) of mass fraction Y_k for major species (from top to bottom H_2, O_2, H_2O and N_2) for each case from table 4.2

calculated in one dimension in CANTERA (using an axisymmetric approximation which is valid on the axis of symmetry) but must be ported to AVBP two dimensional strained configuration of opposed streams. In order to migrate from one dimensional CANTERA solution to two dimensional AVBP solution (on an AVBP mesh which is different from the CANTERA one) an interpolation along a first axis is performed and all values are set constant along a second axis parallel to the flame. The velocity field was not transferred from CANTERA to AVBP but these velocities are imposed from the stagnation point configuration and adjust very fast. Yet, before running simulations in AVBP following the table 4.1, the continuity between CANTERA and the latter must be verified. Since these two solvers are not based on the same governing equations we must check that the initialization is satisfactory. For all cases, the transport model is the same in AVBP and CANTERA: complex [6, 30, 48, 70, 110]. AVBP is first initialized with the CANTERA solution and then run until convergence. First, figures 4.3 and 4.4 display the CANTERA steady solution and the initial solution introduces in AVBP by *CAN2AV*. These figures prov that AVBP is correctly initialized. Figure 4.3 displays spatial evaluations of AVBP profiles of H_2, O_2, H_2O and N_2 at t=0 (dashed red line with circle) against CANTERA (solid black line). For each case of the table 4.2. AVBP and CANTERA exhibit perfect match which assures the validity of *CAN2AV*. Figure 4.4 displays the same conclusion but for local species consumption rates $\dot{\omega}_k$.

Note that AVBP is initialized with CANTERA profiles, it is run until convergence. There are major differences between CANTERA and AVBP. On one hand, CANTERA

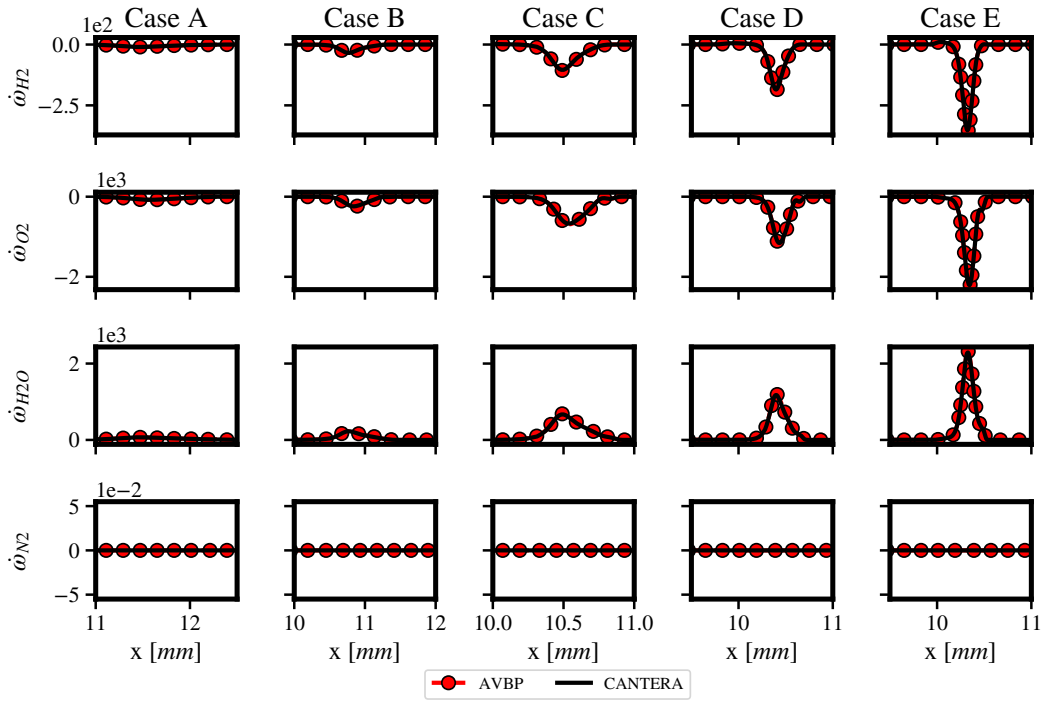


Figure 4.4: Comparison of AVBP initial solution (dashed red line with circle) against CANTERA (solid black line) of species consumption rates $\dot{\omega}_k$ for major species (from top to bottom H_2, O_2, H_2O and N_2) for each case from table 4.2

is used to simulate resolved 0D or 1D configurations such as SPLF (Tab. 4.1) and is based on simplified governing equations dedicated to a context (axysimetric for example with SPLF) without any additional filters. On the other hand, AVBP is an LES solver dedicated to two or three dimensional systems. It is built to simulate realistic and highly complex configurations, be it combustion or aerodynamic phenomena. Thus, these two solvers do not solve exactly the same equations. To ensure the migration between CANTERA and AVBP, one last test has been performed. The goal on this last part is to compare CANTERA and AVBP on the same configuration : SPLF (with the sole exception that CANTERA is axisymmetric and AVBP 2D symmetric). To do so several simulations have been performed on AVBP. Each run was initialized with the tool *CAN2AV* and was kept running until the flame stabilizes. To use *CAN2AV* the system has the same length as CANTERA solution ($L_x = L_y = 20mm$). Table 4.3 summarizes boundary conditions used for simulations also with the study axis (green arrow) where the data will be compared to CANTERA. Five flames are simulated (characteristics detailed in table 4.2). Concerning the mesh, for each case a constant size of cell have been set in order to conserve constant resolution all over the domain. The size has been chosen following the strain of the flame in order to obtain a sufficient resolution, for case A: $\Delta_x = 80\mu m$, for case B: $\Delta_x = 40\mu m$ and $\Delta_x = 20\mu m$ for the rest of the cases.

Table 4.3: Boundary conditions of SPLF (Strained stagnation Point Laminar Flame)

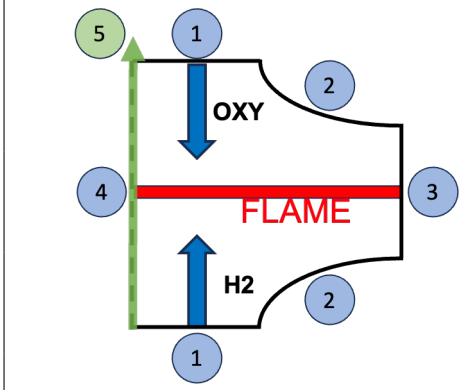
Scheme	Reference	Boundary Conditions
	1	Inlet
	2	Wall
	3	Outlet
	4	Symmetry
	5	Study Axis

Figure 4.5: Comparison of steady solution AVBP (dashed red line with circles) against CANTERA (solid black line) of mixture fraction z, velocity v and temperature T for each case from table 4.2

Figure 4.5 displays spatial local evaluations along the study axis for the converged solution of AVBP against CANTERA of key variables such as mixture fraction z, velocity field v and temperature profile T for each case. Results show satisfactory agreement between CANTERA and AVBP. We can mention minor differences in the velocity field for the most strained case E. Another key point is interesting to look at, with *CAN2AV*,

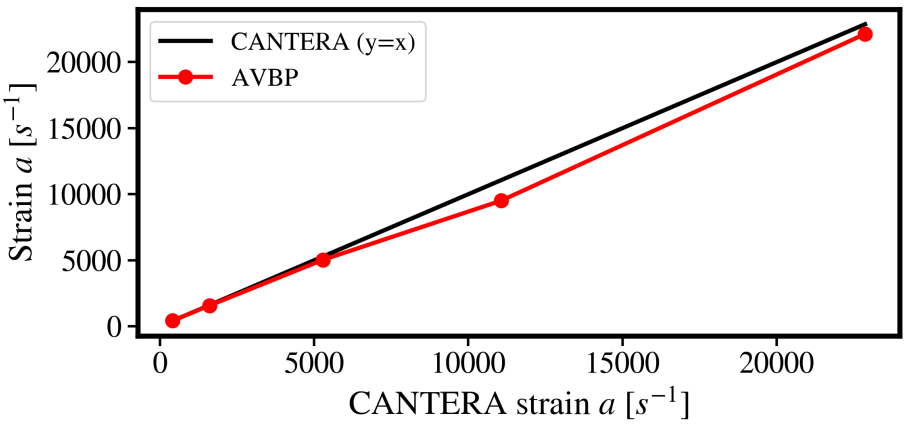


Figure 4.6: On the left axis - Comparison of steady solution AVBP (solid red line with circles) against CANTERA (solid black line) of strain evaluation as function of the strain of CANTERA for each case from table 4.2. On the right axis, the relative differences between CANTERA and AVBP (dashed black line)

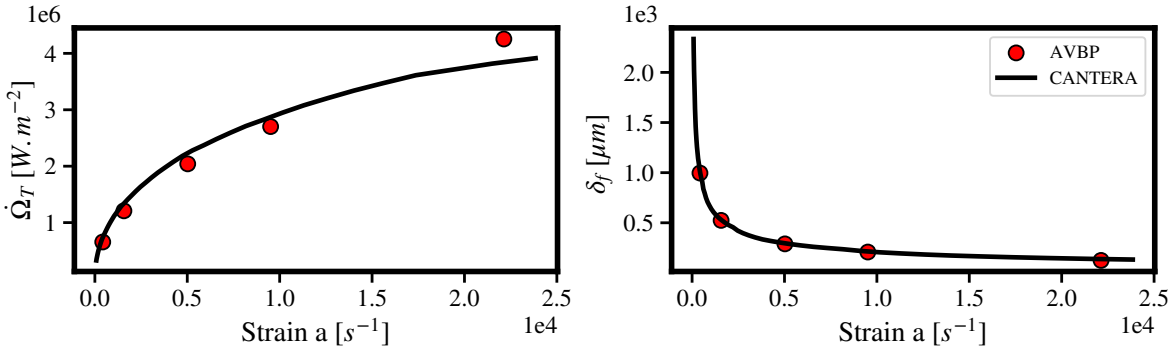


Figure 4.7: Comparison of steady solution AVBP (dashed red line with circles) against CANTERA (solid black line) of the integral of heat release $\dot{\Omega}_T$ (on the left) and thickness flame δ_f (on the right) for each case from table 4.2

a CANTERA flame is transferred into AVBP. However it is not exactly the same SPLF configuration, one is 3D axisymmetric (CANTERA) and the other one is 2D symetric (AVBP). Even if strain definitions, in both configuration, are similar (Eqs. (2.13) & (2.14)) it is not orrect to assume that the initial strain from CANTERA will be the same as the final strain in AVBP. Figure 4.6 displays the evaluation of the AVBP final strain (red solid) next to CANTERA strain represented by $y = x$ function. Remarkably, both of them presents a good agreement. Therefore, it is reasonable to consider CANTERA strain as reference in SPLF cases, even in AVBP SPLF cases. Figure 4.7 displays global evaluations of key parameters such as integrated heat release $\dot{\Omega}_T$ over the study axis for AVBP (and over the whole domain for CANTERA) and flame thickness δ_f in micrometers for each case for AVBP against estimation across the entire range of strain for CANTERA. Most results present excellent agreement especially concerning flame thickness estimations. However, the most strained case shows a small deviation from CANTERA.

This section confirms that CANTERA can be used as an initial solution for AVBP in two dimensional configurations, that *CAN2AV* can be trusted and that AVBP and CANTERA steady solution are the same. Therefore, the following section moves on the main problem of this chapter: resolution.

4.2 Minimum resolution in a strained stagnation point laminar flame (SPLF)

The goal of this section is to determine the frontier between resolved and unresolved simulations of SPLF. To do so, for hydrogen-air flames, a parametric study was first performed for H_2-air, atmospheric stagnation point flames where both streams are injected at 300 K (Table 4.2). The paramount issue of resolution appears at high pressure (Fig. 4.1). In fact, recent comparison exercises for example on the HYLON burner of TNF workshop with 20 CFD codes worldwide show that nowadays we can afford, in realistic 3D LES, cell sizes of the order of $100\mu m$ (tnfworkshop.org/hylon2/). In this situation, key variables, such as heat release, are contained within a cell ($FRI_t < 1$) for high pressure diffusion flames (Fig. 4.8)

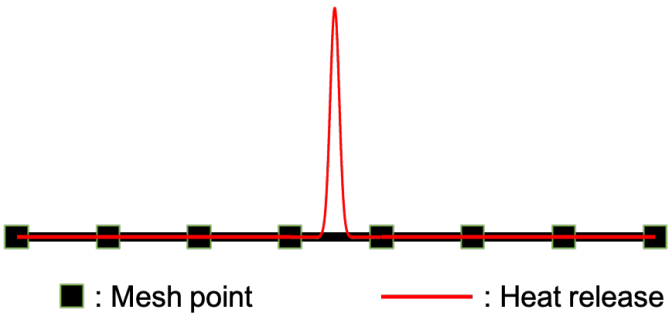


Figure 4.8: Characteristic ratio between mesh size (Black rectangles) and reactive length scale of highly-strained diffusion flame at high pressure (red solid curve).

To counter this, two options are available: thickening (for example TFLES [17]) or artificial viscosity (for example [58]). Each of them will smooth the gradients and this will result in their numerical diffusion on the grid. Figure 4.9 describes this situation.

Due to calculation limitations we will perform our study at atmospheric pressure because it is easier to compute an under resolved atmospheric flame with only a few thousand points than under resolved high pressure flame with millions of points. Ultimately only the fact that FRI_t is small matters and it can be studied at one bar. Therefore, to mimic the situation described by figure 4.8, coarse meshes have been used in order to obtain relevant FRI_t. The parametric study uses different grids (Table 4.4) ensuring values of FRI_t from 50 (DNS-like resolution) to 0.1 (coarse grid). They are computed using CANTERA [41] in a one-dimensional formulation to initialize the flame in AVBP [94] in a two-dimensional stagnation point flow (Scheme in Tab. 4.3).

For each grid/case combination, once the flame is initialized, simulations keep running until the flame stabilizes. Figure 4.10 describes the three possible behaviors depending on the simulation resolution:

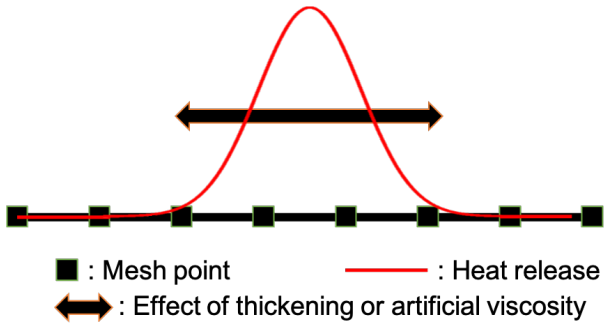


Figure 4.9: Characteristic ratio between mesh size (Black rectangles) and reactive length scale of highly-strained diffusion flame at high pressure (red solid curve) under the effect of thickening or artificial viscosity

Table 4.4: FRI_t values reached in the meshes used for stagnation point flames. These flames A to E are those of Table 4.2 with increasing strain from 398 to 23868 s^{-1}

Case Name	Mesh 2 ($20\mu m$)	Mesh 3 ($40\mu m$)	Mesh 4 ($80\mu m$)	Mesh 5 ($160\mu m$)	Mesh 6 ($320\mu m$)	Mesh 7 ($640\mu m$)	Mesh 8 ($1280\mu m$)
A	52.25	26.13	13.06	6.53	3.27	1.63	0.82
B	26.30	13.15	6.58	3.29	1.64	0.82	0.41
C	14.45	7.23	3.61	1.81	0.90	0.45	0.26
D	10.25	5.13	2.56	1.28	0.64	0.32	0.16
E	6.95	3.48	1.74	0.87	0.43	0.22	0.11

- stable: fully resolved simulation.
- unstable: key variables such as heat release $\dot{\omega}_T$ presents either global instabilities (on the left in unstable cases) or noncontinuous variation along the flame front (on the right in unstable cases). In either case, the flame does not succeed in stabilizing (regardless of the transition period at the start).
- crash: the simulation can result in non-physical responses (on the left in crash cases), the flame extinguishes at the wrong strain or the simulation crashes directly by presenting non-physical responses locally in the flame front (ex: at one point $T > 5000K$).

The goal is to determine is to determine the barrier between unresolved and resolved simulations. To do so a first group of simulations is performed following each mesh/case combination of table 4.4 without any additional model (no thickening or artificial viscosity) that can help the simulation to run smoothly. Figure 4.11 displays the sorting of each result in the three categories described in Fig. 4.10 as function of FRI_t. This study seems to indicate that the barrier is located at $FRI_t = 5$. Moreover it is unclear to determine when the simulation switches between unstable and crash situations. However, when it starts to be close to $FRI_t = 1$ then the simulation will systematically crash. Thus, a partial conclusion is to consider that under $FRI_t = 5$ a simulation should either be not trusted or even crash due to local under-resolution.

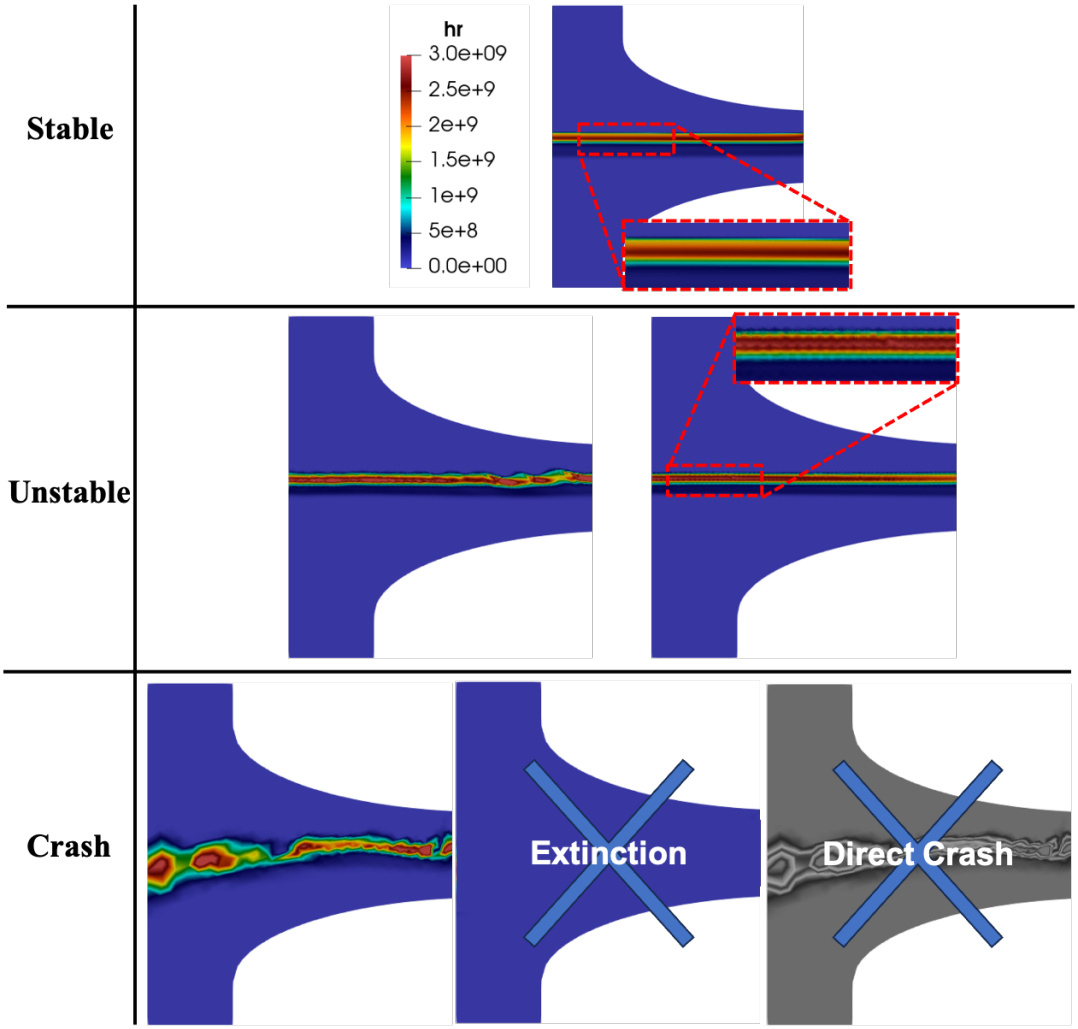


Figure 4.10: Typical behavior of stable (top), unstable (middle) or crash (bottom) simulation of SPLF (Strained stagnation point laminar flames) in AVBP

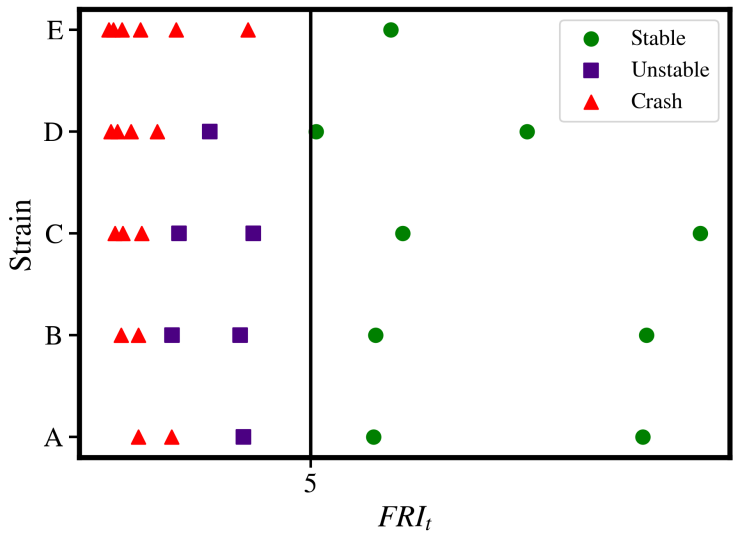


Figure 4.11: Stabilization mapping of SPLF cases (Tab. 4.4) as function of FRI_t where stable cases are separated from the rest when $FRI_t > 5$ (solid black line)

We can also study how key variables are affected by an under resolved context. Figure 4.12 displays evolution of the integral of heat release $\dot{\Omega}_T$ $[W.m^{-2}]$ over the study axis (Tab. 4.3) as function of FRI_t. First, above $FRI_t = 5$, the estimation of $\dot{\Omega}_T$ is independent of the mesh, so that the conclusion of the barrier of $FRI_t = 5$ to separate resolved and under resolved simulation is still valid. Second, when the barrier $FRI_t = 5$ is reached we can see an over-estimation of the integral. This observation can be explained by looking into the variation of the thickness of flame with resolution (ie the characteristic cell size of the mesh since its supposed to be the same flame). When the mesh becomes coarser, the flame hardly stabilizes and tries to "survive" on at least 3 cells. In the situation when $FRI_t < 5$, the flame naturally grow over the mesh and turn on a physical object directly linked to the mesh (which should not be the case). Then, a 'larger' flame (due to the poor mesh quality) shows a 'larger' distribution of heat release which explains the overestimation of the integral of it. Figure 4.13 displays evolution of the thickness of the flame δ_f $[\mu m]$ over the study axis (Tab. 4.3) as function of FRI_t.

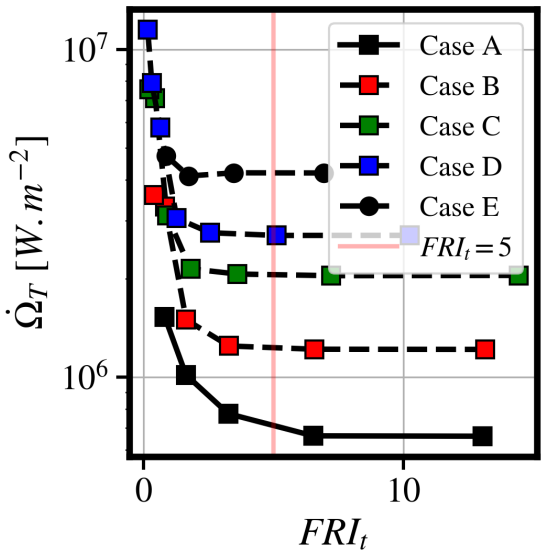


Figure 4.12: Integral of heat release $\dot{\Omega}_T$ (Tab. 4.3) as function of the mesh quality FRI_t for each case of table 4.4. Vertical red solid line: $FRI_t = 5$, the barrier between resolved and under resolved simulations

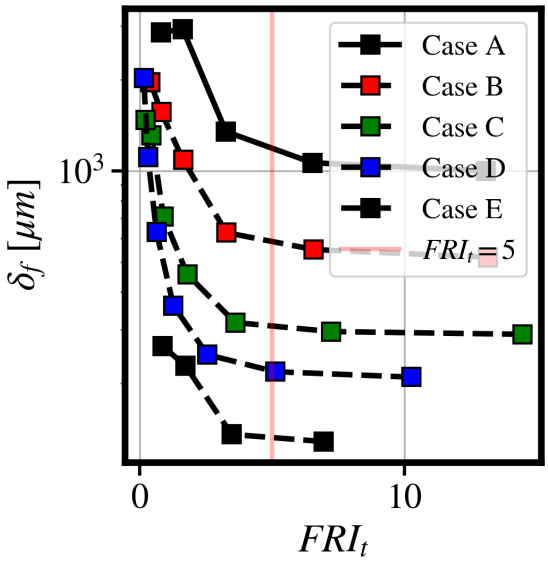


Figure 4.13: Thickness of diffusion flame δ_f as function of the mesh quality FRI_t for each case of table 4.4. Vertical red solid line: $FRI_t = 5$, the barrier between resolved and under resolved simulations

A key issue in the control of resolution using FRI is that its resolved value, FRI_{res} can be measured during an LES but its true value FRI_t cannot because it would require to know the true flame thickness δ_L^t. Fortunately, Fig. 4.14 shows that both definitions provide the same value when the resolution is sufficient: $FRI_t = FRI_{res}$ as soon as $FRI_t > 5$. Therefore in the model which will be presented in chapter 5, the resolution criterion which will be used to determine whether a simulation is well resolved or not will be $FRI_{res} > 5$. Note that when FRI_{res} is less than 5, it is not possible to evaluate the exact value of FRI_t; however, in this situation, it is possible to state that the simulation is under resolved and that $FRI_t < 5$. Figure 4.14 displays estimations of FRI_t and FRI_{res} for each case of Table 4.2 as function of mesh quality. As expected, a correlation can be done between FRI_t and FRI_{res} when the criterion is above five. When FRI is lower than five, then the correlation is no longer respected due to:

- the theoretical barrier of FRI_{res} around 3 or 2.
- The under resolved context that imposes an overestimation of the thickness of the flame δ_f (Fig. 4.13).

Under this observation, if a model is created following a simple, yet critical, goal of resolution the criteria $FRI_{res} > 5$ is critical. Due to uncertainties present within the definition FRI_{res} (linked to uncertainties of δ_f under resolved) a safety should be included (for example take $FRI_{res} = 6$ instead of $FRI_{res} = 5$). Besides, the FRI criterion and the estimation of the barrier value between resolved and under resolved has been tested only on laminar and stable cases. The next section exposes a similar study but on unsteady cases by adding a couple of vortices that will interact with the flame.

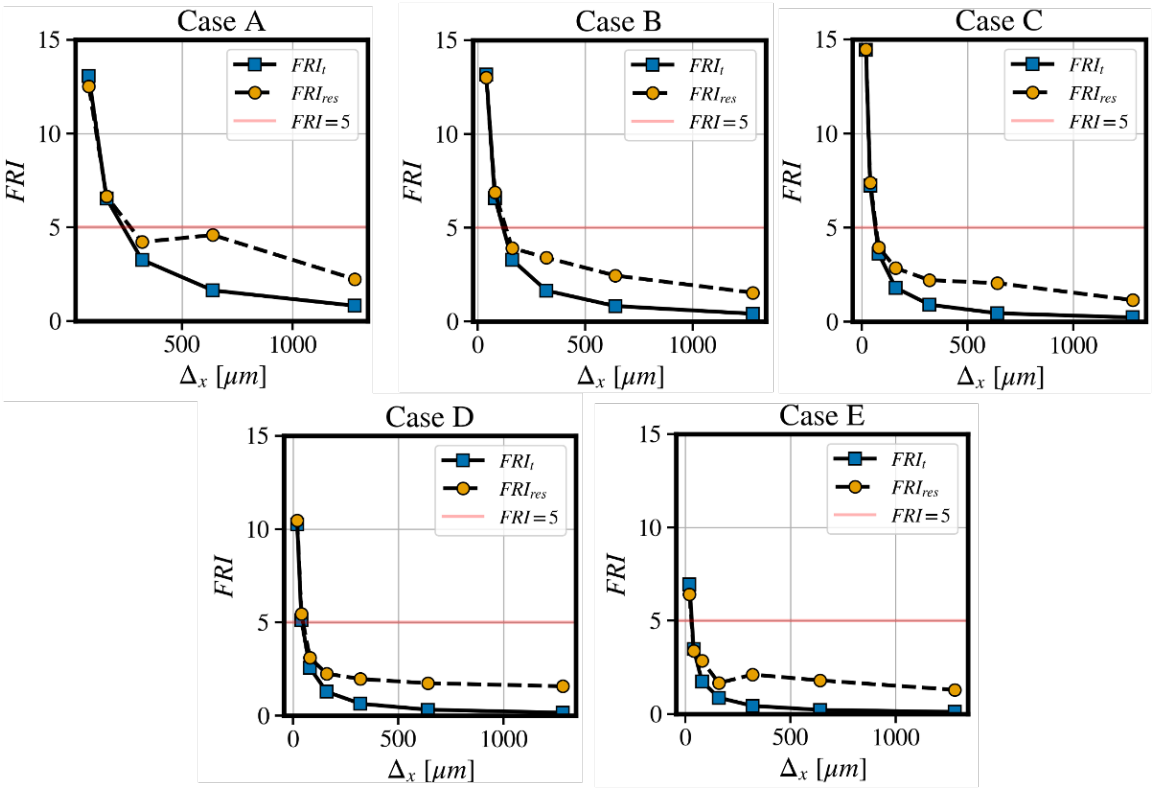


Figure 4.14: Comparison between true FRI (FRI_t) and resolved FRI (FRI_{res}) for each flame type of Table 4.2 versus mesh size Δ. Horizontal red line: $FRI_t = 5$.

4.3 Minimum resolution in flame-vortex interaction

Section 4.2 suggests that a proper limit for resolution of a steady strained H2 - air diffusion flames is $FRI_{res} = 5$. This criterion is tested now in an unsteady curved diffusion flame with an interaction with a vortex pair (FVI).

The FVI configuration (Table 4.5) is symmetrical and only the right side is displayed. A pair of counter rotating vortices interacts with a 2D non-premixed strained flame. The vortices are generated on the air side. The system is 2 by 10 cm, at 1 bar. Initial flames are generated using a CANTERA solution at low strain and the *CAN2AV* tool of AVBP . The characteristics of each flame are summarized in the table 4.6. The pair of vortices pushes the flame, strains and curves it (Fig. 4.15). The analysis focuses on the central symmetric axis since this is where the flame is submitted to the maximum strain and has the smallest flame thickness [102] (then the smallest FRI_{res}). Due to significant strain increase imposed by the interaction with vortices, only initial flames at low strain have been tested. The goal here is not to extinct locally the flame but to observe and analyze the impact of low resolution on results. DNS are repeated for all conditions of Table 4.7 that summarizes evaluations of initial resolution of the flame before any interaction with vortices.

Figure 4.15 displays instantaneous solutions of high (on the left) and low (on the right) resolution simulations for case F of Table 4.6 at $t^+ = 0.5$, $t^+ = 20.5$ and $t^+ = 40.5$. Here, time is normalized using the vortex time: $t^+ = u_{max}t/R$ where R is the vortex radius and u_{max} its maximum velocity within the vortex. On each instantaneous solution, two

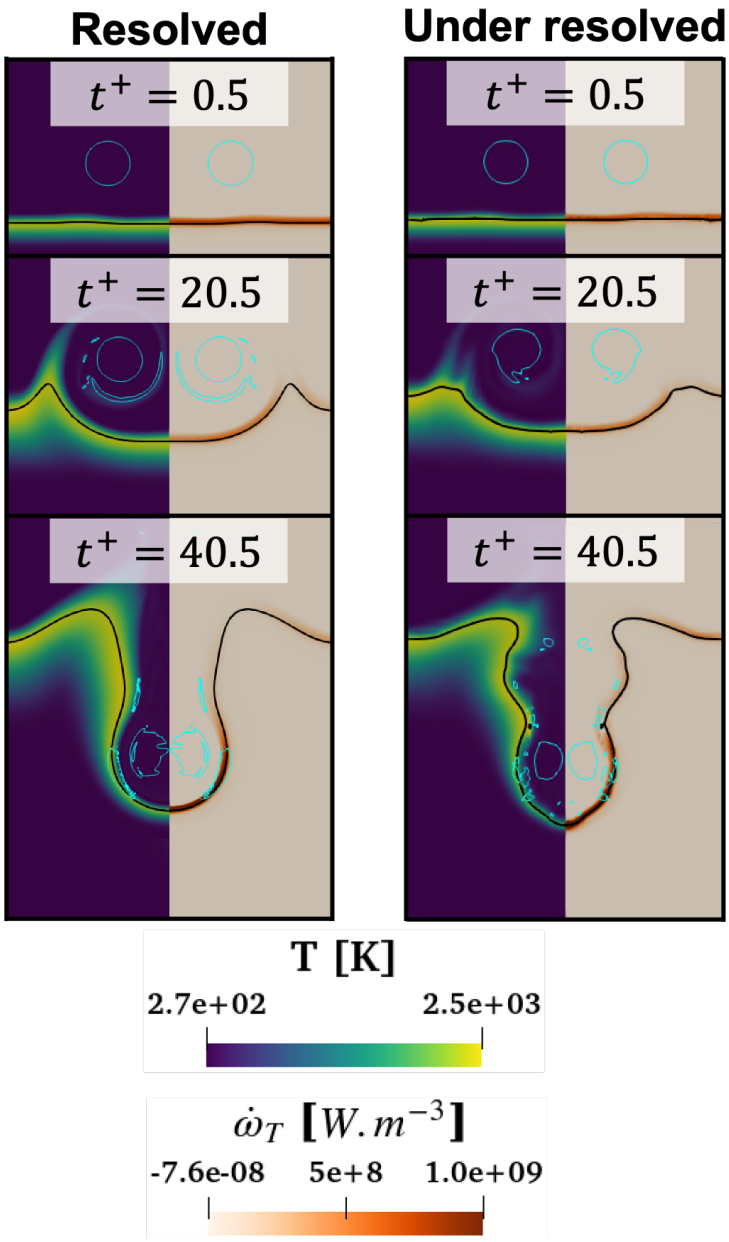
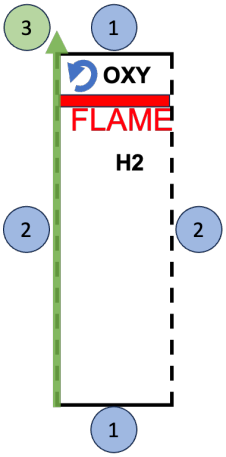


Figure 4.15: Interaction between diffusion flame and a vortex pair for a spatially resolved case (left) and an unresolved case (right). Snapshots of temperature and heat release $\dot{\omega}_T$ at three instants. The black line is the stoichiometric line.

Table 4.5: Boundary conditions of FVI (Flame/Vortex Interaction)

Scheme	Reference	Boundary Conditions
	1	Outlet
	2	Symmetry
	3	Study axis
		Vortex

fields are shown, on the left the temperature T and on the right the heat release $\dot{\omega}_T$. Looking first into the most resolved one, along the study axis (Tab. 4.5). At $t^+ = 0.5$, the flame does not present influence from the couple of vortices. After a certain time, the flame starts to being push by vortices, thus strain increases and the flame became thinner. Moreover, since no additional velocity field is included during the simulation, the flame shows a downward vertical displacement. The simulation is always terminated when the flame touch the bottom boundary frontier of the configuration (which happens approximately at $t^+ \approx 55$). On the opposite, the low resolution on the left seems to not present strong variation before real influence of vortices on the flame at $t^+ = 0.5$. After a certain time and far from the study axis, at $t^+ = 20.5$, differences starts to appear where interactions with vortices seems less strong (peaks along the stoichiometric line are lower), yet on the study axis (middle) the results show coherent behavior (small dissipation of heat release and temperature). Then after a long time, vortices fully interact with the flame and under resolved simulation behaviors show up. Uneven stoichiometric isoline is estimated, noisy field of $\dot{\omega}_T$ is presented and a higher concentration of $\dot{\omega}_T$ seems to appear on the study axis compared to the resolved simulation.

Table 4.6: Flame Vortex interaction point used to determine minimum FRI_t values for stable and consistent solutions.

Case Name	Strain $[s^{-1}]$	Thickness $[\mu m]$	Maximum Temperature $[K]$
F	602	796	2339
G	1169	573	2242
H	1729	472	2176
I	2285	411	2125
J	2837	370	2083

Table 4.7: FRI_t values reached in the meshes used for Flame Vortex Interaction.

Case Name	Mesh 3 ($40\mu m$)	Mesh 4 ($80\mu m$)	Mesh 5 ($160\mu m$)	Mesh 6 ($320\mu m$)
F	19.91	9.95	4.98	2.49
G	14.34	7.17	3.59	1.80
H	11.81	5.91	2.96	1.48
I	10.28	5.14	2.57	1.29
J	9.23	4.62	2.31	1.15

Simultaneously, all characteristics of the flame evolve during the simulation and especially the resolution because strain varies strongly on the flame. Figures 4.16 to 4.20 show temporal evaluations of FRI_{res}^{min} and integral of heat release $\dot{\Omega}_T$ along the study axis (Tab. 4.5) for, respectively, case F to J (Tab. 4.6). First, we focus on case F (Fig. 4.16). The results obtained on mesh 3 of Table 4.7 ($\Delta = 40\mu m$) are used as the right reference because it is highly resolved ($FRI_{res} \approx 20$) During the first stage of the interaction, from $t^+ = 0$ to 25, FRI_{res} grows because the flame is not yet strained and becomes thicker [87] and easier to resolve. Then the vortex enters the flame, the strain seen by the flame tip increases and FRI_t decreases, making the simulation less resolved. As long as FRI_{res} is larger than 5 (represented by the solid horizontal red line), the total heat release, $\dot{\Omega}_T$, is correctly captured for all meshes 3 to 6 of Table 4.7. Mesh 3 is actually always well resolved and used as a reference. Meshes 4 to 6 all give reasonable values for $\dot{\Omega}_T$ until strain continues to diminish FRI_{res}. When FRI_{res} becomes too small, the impact on the total heat release, $\dot{\Omega}_T$, is instantaneous: as soon as FRI_{res} at the flame tip goes under 5, errors on $\dot{\Omega}_T$ appear for each mesh. The rest of the cases substantiate this conclusion. Thus, during an active and unsteady configuration, simulations became under resolved when locally $FRI_{res} < 5$, independently of resolution or of initial strain. Additionally, chaotic behaviors (appear under unresolved configurations). Thus, compared to steady cases from the previous section, it seems hard to estimate a priori a typical behavior of an under resolved simulation of an interaction between a diffusion flame and a simple turbulence. Furthermore, if we look into the rest of cases, from G to J the initial strain increases (Tab. 4.6), then the initial resolution decreases (Tab. 4.7). Thus, at the beginning, inconsistent evaluation of $\dot{\Omega}_T$ shows up for mesh 6. However, the increase of FRI_{res}^{min} above 5 helps those simulations to almost recover the right value of $\dot{\Omega}_T$ until FRI_{res}^{min} returns under 5. This observation suggests that even if a simulation starts under resolved, if we succeed (with the help of a model for example) to increase the resolution it may be able to recover correct evaluations of key variables such as $\dot{\Omega}_T$.

These results show that for this strained, curved, unsteady flame (typical of flamelets encountered in turbulent flames), like for the steady strained flames of Section 4.2, resolution is an issue which can be characterized to first order by FRI_{res}: monitoring FRI_{res}, independently of how subgrid scale terms are handled, allows to detect when the diffusion flame becomes under resolved. When this happens, a model must be triggered when FRI_{res} is less than 5. The model will increase the resolution until $FRI_{res} > 5$ and then creates a simulation that does not suffer from under resolved resolution and can be the socle for flame-turbulence models.

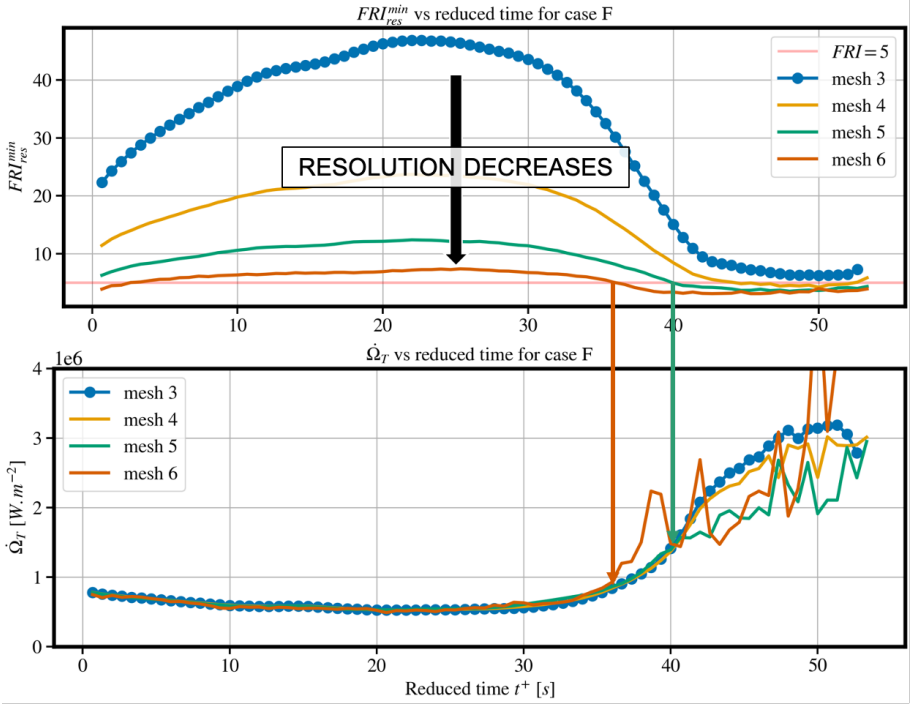


Figure 4.16: Diagnostics on symmetry axis for flame - vortex interaction for case F (Tab. 4.6). Circles: fully resolved case (mesh 3). Top: evolution of FRI_t versus reduced time. Bottom: Evolution of $\dot{\Omega}_T$ versus reduced time. When $FRI_{res} < 5$, the integrated reaction rate $\dot{\Omega}_T$ starts to deviate from the resolved case (circles). Horizontal red line: $FRI_{res} = 5$.

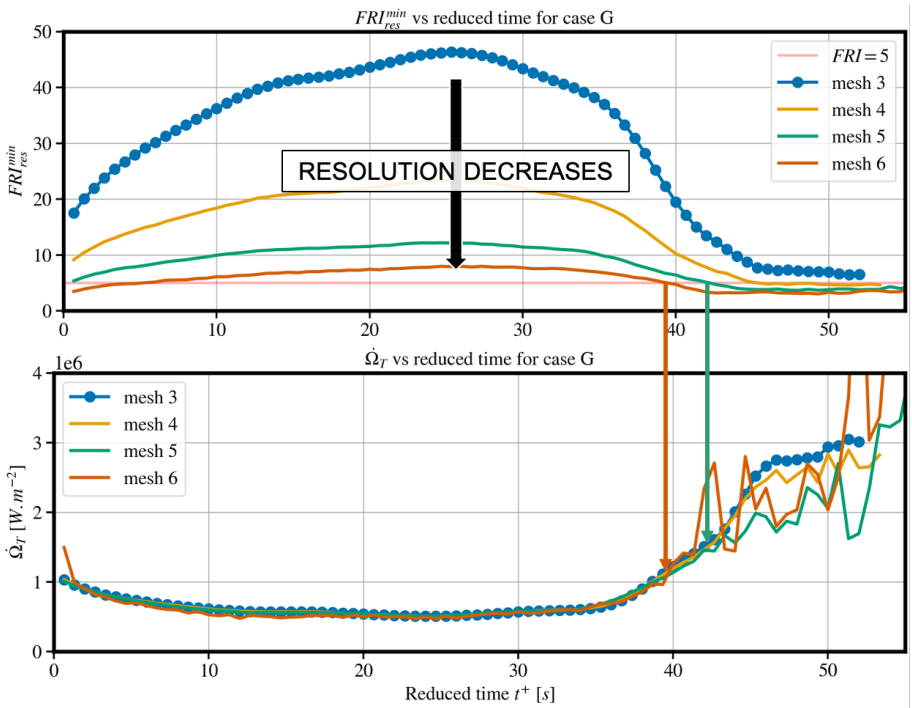


Figure 4.17: Diagnostics on symmetry axis for flame - vortex interaction for case G (Tab. 4.6). Circles: fully resolved case (mesh 3). Top: evolution of FRI_t versus reduced time. Bottom: Evolution of $\dot{\Omega}_T$ versus reduced time. When $FRI_{res} < 5$, the integrated reaction rate $\dot{\Omega}_T$ starts to deviate from the resolved case (circles). Horizontal red line: $FRI_{res} = 5$.

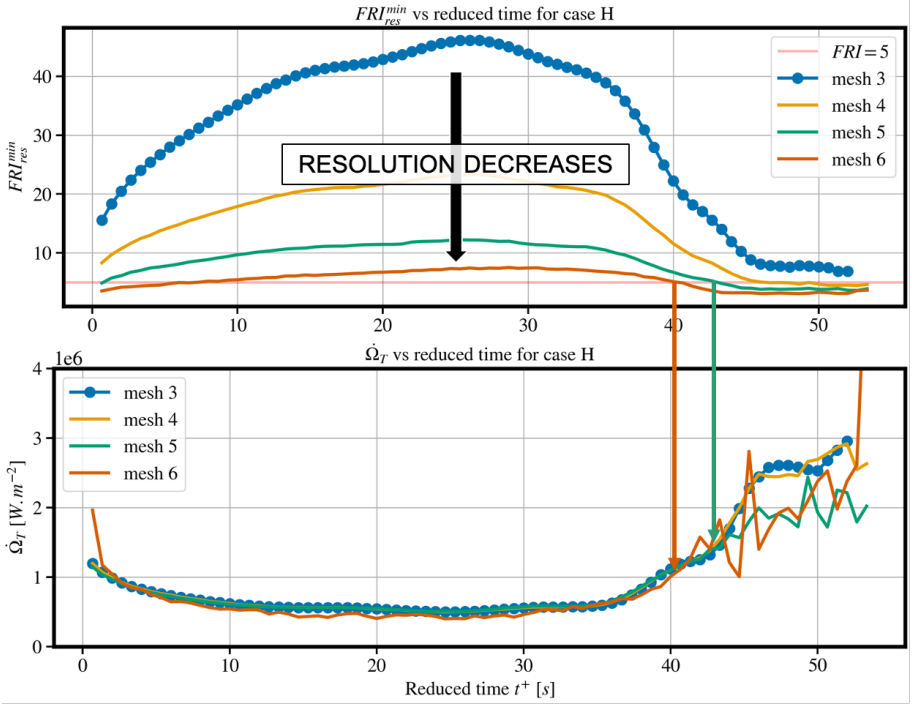


Figure 4.18: Diagnostics on symmetry axis for flame - vortex interaction for case H (Tab. 4.6). Circles: fully resolved case (mesh 3). Top: evolution of FRI_t versus reduced time. Bottom: Evolution of $\dot{\Omega}_T$ versus reduced time. When $FRI_{res} < 5$, the integrated reaction rate $\dot{\Omega}_T$ starts to deviate from the resolved case (circles). Horizontal red line: $FRI_{res} = 5$.

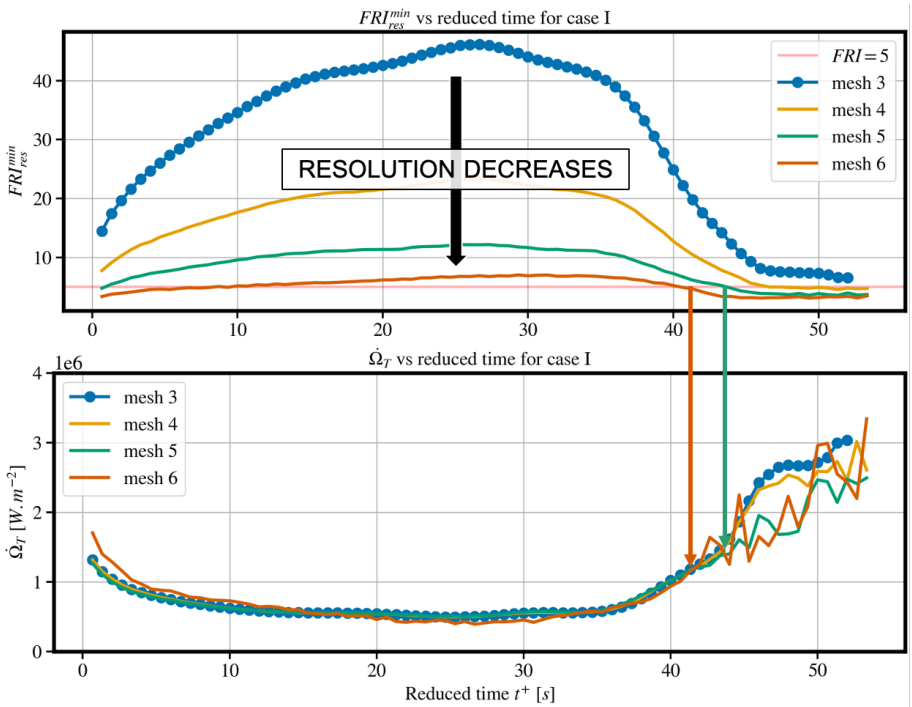


Figure 4.19: Diagnostics on symmetry axis for flame - vortex interaction for case I (Tab. 4.6). Circles: fully resolved case (mesh 3). Top: evolution of FRI_t versus reduced time. Bottom: Evolution of $\dot{\Omega}_T$ versus reduced time. When $FRI_{res} < 5$, the integrated reaction rate $\dot{\Omega}_T$ starts to deviate from the resolved case (circles). Horizontal red line: $FRI_{res} = 5$.

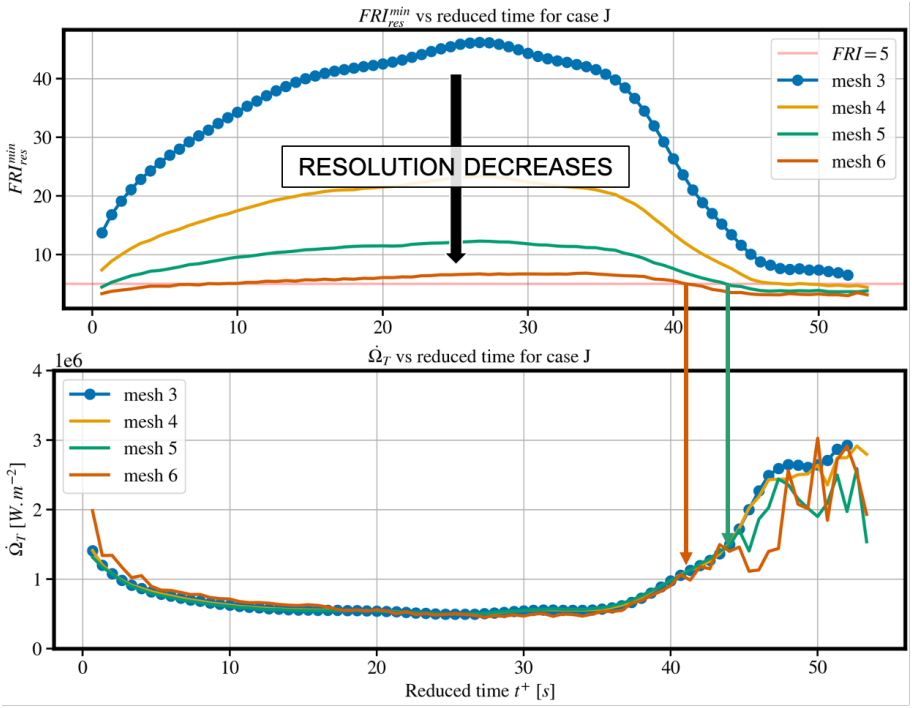


Figure 4.20: Diagnostics on symmetry axis for flame - vortex interaction for case J (Tab. 4.6). Circles: fully resolved case (mesh 3). Top: evolution of FRI_t versus reduced time. Bottom: Evolution of $\dot{\Omega}_T$ versus reduced time. When $FRI_{res} < 5$, the integrated reaction rate $\dot{\Omega}_T$ starts to deviate from the resolved case (circles). Horizontal red line: $FRI_{res} = 5$.

This is the objective of the model described in the next chapter called Diffusion Front Thickness Control (DFTC). Note that, from now on, FRI_t will not be accessible in the LES since only FRI_{res} can be measured on the LES mesh. We will assume that a threshold of 5 for FRI_{res} is the right criterion and verify it.

4.4 Towards a model based on resolution

This chapter has presented a simple methodology to characterize the resolution of a diffusion flame during a simulation. The expression of FRI, which is the ratio between the thickness of the flame δ_f and the local mesh size Δ, allows to follow local resolution during the simulation. Steady (Strained stagnation Point Laminar Flame - SPLF) and unsteady (Flame/Vortex interaction - FVI) cases have been studied to prove that FRI is the adequate metric for resolution. Similar conclusions, from both cases, depict a barrier between under resolved and resolved simulation at $FRI = 5$. Yet, this value should be taken into account carefully due to the distinction between the real FRI_t and the one that we can measure during the simulation FRI_{res}. T This chapter demonstrates that, in theory FRI_t captures perfectly the resolution, however inside a simulation, the thickness of the flame (directly link to the gradient of mixture fraction Δz) starts to be dependent of the quality of the mesh (ie the cell size) when the simulation is under resolved. Therefore in under resolved configurations, FRI_{res} is lower-bounded by 2 or 3 (Fig 4.14). At this point, what we can say with certainty is that $FRI_{res} < 5$ implies an under resolved context.

When FRI_{res} goes under five, the criterion starts to be dependent of the mesh and then it is no longer possible to express precisely how far we are from a resolved environment. Nevertheless, something must be done to retrieve a resolved configuration in order to capture realistic physics before any consideration of flame/turbulence modeling.
The next chapter presents a model called Diffusion Front Thickness Control (DFTC) which thicken diffusion fronts so that $FRI_{res} > 5$, its applications on test cases, the list of the goals that it ideally should respect and its reliability. Note that using artificial viscosity is another way used today by many codes to reach the same objectives: we will show that even though DFTC and artificial viscosity struggle to completely solve the under resolution issues, yet DFTC offers more advantages and control over the situation.

Chapter 5

DFTC : a model for diffusion flame based on resolution monitoring

The previous chapter allowed us to understand resolution issues and build a criterion that separates unresolved from resolved flame surface area during a simulation. This chapter focuses on building a dynamic model dedicated to diffusion flame resolution in LES, called Diffusion Front Thickness Control (DFTC). Starting from the philosophy of the thickened flame LES model (TFLES), section 5.1 explains how DFTC modifies the governing equations, the conditions that it must respect and more importantly, how thickening is dynamically applied to keep a sufficient resolution over the whole flame front. After the theory, steady and unsteady test cases are used to validate DFTC:

- Strained stagnation Point Laminar Flame (SPLF) (section 5.2).
- Flame/Turbulence Interaction (FTI) (section 5.3).

The main goal for each test case is to ensure proper resolution by thickening the flame front via the laminar diffusivity of the mixture. Obviously, such a correction changes the physics of the flame (increase of the heat release integral $\dot{\Omega}_T$ for example). In order to counteract that, a few attempts are done in section 5.4 by modifying the turbulent viscosity or the reaction terms. This section shows that this technical hurdle is more challenging for non-premixed flames than it is for premixed flames due to their fundamentally different natures. In fact, section 5.4 demonstrates that diffusion flames are mainly controlled by the diffusivity which strongly limits our scope of actions. However this conclusion must be balanced by the fact that thickening also has drawbacks for premixed flames (TFLES). Since DFTC and TFLES are essentially based on the same concept (modify the diffusivity by multiplying the mixture diffusivity by a factor F) it is relevant to compare the two models. Section 5.5 lists the pros and cons of both DFTC and TFLES. Finally, before concluding and discussing perspectives around DFTC in section 5.7, section 5.6 presents several subtleties pertaining to DFTC in order to discuss crucial choices that have been made and why.

5.1 DFTC: a model to thicken diffusion fronts

Among the multiple models for flame/turbulence interaction, the idea of a 'mapping' proposed initially in the thickened flame model of O'Rourke and Bracco [78] has been

extended through different variations of this thickening approach [26, 57, 89]. These models directly address the issue of spatial resolution by building a master equation for the filtered mass fraction $\widetilde{Y}_k$ of species k, starting from the conservation equation:

$$\overline{\rho}\frac{D\widetilde{Y}_k}{Dt} = \nabla \cdot [\overline{\rho}(D_k + D_t)\nabla \widetilde{X}_k] + \overline{\dot{\omega}}_k \tag{5.1}$$

where D_k and D_t are respectively the laminar and turbulent diffusivity, $\widetilde{X}_k$ the filtered molar fraction and $\overline{\dot{\omega}}_k$ the filtered reaction rate.

The 'mapping' operation consists in modifying the diffusion and reaction terms as follows:

$$\overline{\rho}\frac{D\widetilde{Y}_k}{Dt} = \nabla \cdot [\overline{\rho}(\boldsymbol{F}D_k + \boldsymbol{\alpha}D_t)\nabla \widetilde{X}_k] + \boldsymbol{F}_r\overline{\dot{\omega}}_k \tag{5.2}$$

Note that the 'mapping' operation is not based on a rigorous derivation of the conservation equations as done in pdf models [87] or in filtered models [27, 28]. However, it has various advantages: it is simple to implement, degenerates naturally to DNS when the mesh is refined (F, α and F_r go to unity) and allows to control spatial resolution: this explains its choice for the present work.

The 'mapping' introduces three terms F, α and F_r which are added to Eq. (5.1):

1. modification of the local laminar diffusivity D_k by a factor F (analog to the thickening factor of the TFLES model for premixed flames). Models that modify this term are labeled 'LAM';

2. modification of the local turbulent diffusivity D_t by a factor α. Models that modify this term are labeled 'TUR';

3. modification of the reaction term $\dot{\omega}_k$ by a factor F_r (in the TFLES model for premixed flames, $F_r = 1/F$). Models that modify this term are labeled 'REAC'.

A single model can combine any of these three terms[1].

Before discussing what F, α and F_r should be, objectives must be defined (Tab. 5.1): RES, LQ, LTMAX, LC and FTIC are criteria which are deemed essential for LES simulations even if it is not clear that they can all be satisfied with Eq. (5.2). The choice of the criteria of Tab. 5.1 is guided by flamelet ideas: the model should work first for a typical flamelet, ie for a stagnation point, laminar, steady diffusion flame and then be applied to a turbulent flame, viewed as a collection of flamelets. In many approaches the FTIC condition is often cited as the only important one: conserve the reaction rate per unit area. We will show that:

1. FTIC is very difficult to reach with any model;

2. the other criteria RES, LQ, LTMAX are as important as LC and can be satisfied using DFTC.

[1]For example, the classical TFLES model for premixed flames is a "LAM REAC" model with $F_r = 1/F$.

Table 5.1: Objectives for LES modeling (SPLF = Strained stagnation Point Laminar Flame, FTI= Flame / turbulence Interaction).

Objectives	Measure	Test case
RES: Resolution The diffusion front must be resolved so that results do not depend on mesh or artificial viscosity	FRI_m minimum value	SPLF FVI FTI
LQ: Laminar Quenching In laminar resolved diffusion fronts, quenching must occur at the right strain	Quenching strain a_Q	SPLF
LTMAX: Maximum temperature In laminar resolved diffusion fronts, the maximum temperature evolution vs strain must be conserved	Maximum temperature vs strain a	SPLF
LC: Laminar Consumption In laminar resolved diffusion fronts, the total heat release per unit flame area vs strain must be predicted correctly	Integrated heat release $\dot{\Omega}_T$ vs strain a	SPLF
FTIC: Flame / Turbulence Interaction Consumption In DNS of diffusion flames interaction with turbulence, the total reaction rate must be captured	Integrated heat release $\dot{\Omega}_T$ vs strain a	FTI

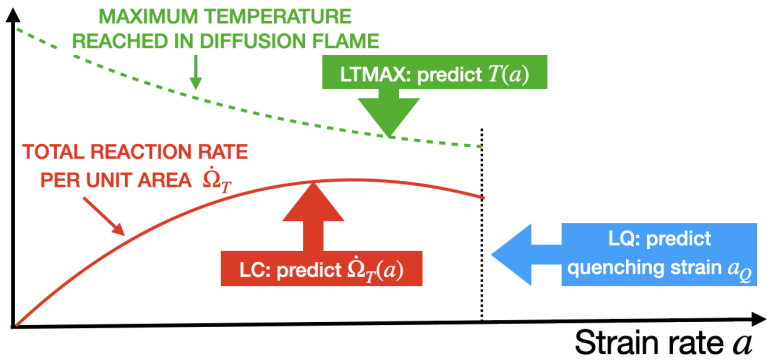


Figure 5.1: Total reaction rate per unit area vs strain or scalar dissipation in a laminar, strained, diffusion flame [87].

Fig. 5.1 summarizes the main arguments used to guide the choices of F, α and F_r obtained from classical laminar strained diffusion flames analysis. The first point is that, unlike premixed flames, diffusion flames are controlled by strain: their total reaction rate increases with strain until quenching. The second point is that, at low strain values, the infinitely fast chemistry assumptions holds, so that for these low strain flames, we should not expect F_r to affect the results.

Table 5.1 includes five objectives for the model:

- RES: spatial resolution is mandatory. If the modeled front leads to a discontinuous z field where z jumps from 0 to 1 in one cell, no FTI model will work because the under resolution will prevent any precise resolution.
- LQ: a key issue in many diffusion fronts, stabilized for example on injector lips [1, 4, 66] is to know if the diffusion front is quenched or not. In a flamelet framework, this can be translated into a question of strain value at quenching, called a_Q, for a stagnation point flame. The resolved flame in the LES should not quench at the wrong stretch value.
- LTMAX: since the prediction of NO is a key issue, predicting the correct maximum temperature reached in a strained diffusion flame would be useful because it controls NO_x emissions [40].
- LC: the model should predict the overall consumption rate of a laminar stagnation diffusion flame over the whole range of strains, up to quenching.
- FTIC: the ultimate goal of all LES models is to capture the mean reaction rate during the interaction of a flame with turbulence. This question has generated many DNS over the last decade for hydrogen flames [71, 74, 102] but only few real LES models.

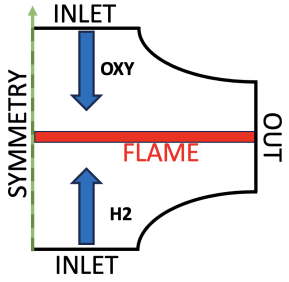
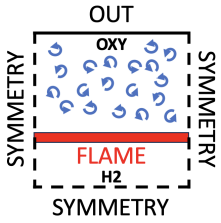
There is obviously some arbitrariness in the choice of criteria RES, LQ, LC, LTMAX and FTIC. For certain cases, such as flame quenching on a lip, LQ for example is important since it determine whether the flame will quench or not near the lip [12, 105]. For NO_x, LTMAX is also certainly critical [112]. We will limit this work to these five criteria,

insisting that RES is mandatory. The wish list of Tab. 5.1 is probably impossible to fully satisfy. We show below that the DFTC model satisfies RES, LQ, LTMAX very well but not LC and FTIC.

Two pure diffusion flame configurations will be used to test DFTC (Tab. 5.2):

- SPLF: strained stagnation point laminar flames using either CANTERA or AVBP (section 5.2).
- FTI: diffusion flame interacting with decaying two-dimensional turbulence using AVBP (section 5.3).

Table 5.2: Schemes of test cases for DFTC model

Case Acronym	Scheme
SPLF	
FTI	

Chapter 4 has shown that RES would be satisfied for hydrogen-air flames when $FRI_{res} > FRI_{target}$ with $FRI_{target} = 5$ where FRI is obtained at each point by $FRI = 1/(\Delta\nabla z)$ (eq. (4.3)). Interestingly, this value of 5 for $FRI_{target} = 5$ is close to the value used in premixed fronts for TFLES. Note that this is a remarkably small number of points to resolve a complex chemistry front.

The simplest way to ensure the $FRI_{res} > FRI_{target}$ condition is to use a LAM model and increase the laminar diffusivities by a factor F. In LES, turbulent diffusivities (D_t in Eq. 5.2) would also make the flame thicker but they are not designed to control the resolved flame thickness. Moreover, D_t can go to zero in many quasi-laminar zones of the flow, leading to under resolution. Therefore, controlling the resolution through a change of laminar diffusivities is a first logical choice.

Theory shows that at low local strains, the diffusion flame front thickness δ_L (and therefore FRI) increases like $\sqrt{F}$. Unlike in premixed flames, the thickening factor F to apply cannot be determined a priori since the local diffusion flame thickness depends on local strain and is unknown. Therefore DFTC uses an active control loop at every mesh point where F is changed at each time step until a satisfactory value is reached for

FRI_{res}. The local value of F is set to[2]:

$$F = 1 + (F_{max} - 1)f(z) \tag{5.3}$$

where the maximum thickening factor F_{max} is obtained by a time-dependent process:

$$\frac{dF_{max}}{dt} = \frac{(FRI_{target} - FRI_{res}^{min})}{\tau} \tag{5.4}$$

where FRI_{res}^{min} is the minimum value reached by FRI_{res} in the domain (usually along the $z = 0.5$ line) and FRI_{target} is the target value needed for resolution (5 as shown in chapter 4). The $f(z)$ function has a parabolic shape and provides a maximum thickening factor at $z = 0.5$ where the gradient of z is maximum (Fig. 5.2):

$$f(z) = \exp\left\{-\left[\text{erf}^{-1}(1 - 2z)\right]^2\right\} \tag{5.5}$$

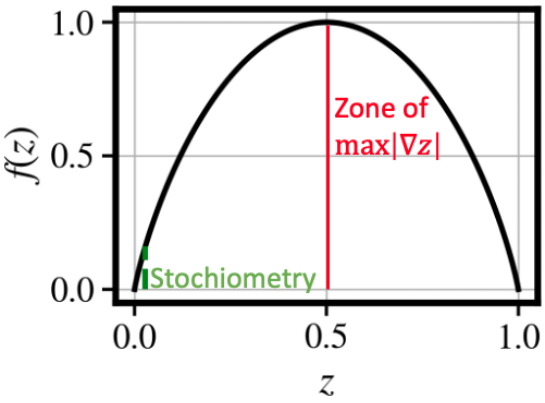


Figure 5.2: The $f(z)$ function controlling the distribution of thickening in z space. Red vertical line: maximum z gradient ($z = 0.5$). Green dashed line: $z = z_{st}$.

No thickening is applied in the pure streams ($z = 0$ and $z = 1$). Choices that have been made for f are developed in appendix B.1.
The time τ is a relaxation time, controlling the speed at which the controller follows local strain and FRI_{res} variations. A convenient method to evaluate τ is to scale it by the time step Δt of the simulation:

$$\tau = K\Delta t \tag{5.6}$$

where K is a free model parameter. Small K values will lead to fast adjustment of F and vice-versa. The choice of the value for K is discussed in subsection 5.6.1. In a time-dependent code, Eq. 5.4 becomes:

$$F_{max}^{n+1} = F_{max}^{n} + (FRI_{target} - FRI_{res}^{min})/K \tag{5.7}$$

where n is the iteration index, leading to a simple implementation.

Table 5.3: Skeletal algorithm of DFTC for RES criterion

Steps	Equations	Objectives
Step 1	$FRI_{res}(z) = \frac{\delta_f(z)}{\Delta(z)} = \frac{1}{\Delta \nabla z}$	Estimation of the local resolution field.
Step 2	$FRI_{res}^{min} = \min_z (FRI_{res}(z))$	Estimation of the minimum resolution over the whole domain.
Step 3	$F_{n+1}^{max} = F_n^{max} + \frac{FRI_{target} - FRI_{res}^{min}}{K}$	Evaluation of F^{max} at $n+1$ time iteration following the difference between target value and minimum resolution.
Step 4	$F(z) = 1 + (F^{max} - 1) f(z)$	Apply the thickening factor F over the whole domain following the distribution function f (eq 5.5).

Criteria LQ, LTMAX, LC and FTIC are all desirable once RES is ensured. Table 5.3 summarizes and describes the algorithm of DFTC implemented in AVBP for the RES criterion in diffusion flame cases.

The following sections show how DFTC satisfies these criteria in stagnation point flames (Section 5.2) and in turbulent flames (Section 5.3) before discussing possible improvements using LAM TUR REAC models in Section 5.4.

5.2 Tests of DFTC in stagnation point laminar flames (SPLF)

The SPLF case has already been studied in section 4.2 for resolution. This section focusing of DFTC addition on it in a three-part discussion. First CANTERA is used to study the effect of thickening when the resolution is DNS like (more than 20 points in the front flame). Second, on AVBP, the same set of flame for the SPLF is used (Tab. 4.2) to test DFTC for under resolved cases (Tab. 4.4) following the algorithm described in table 5.3 . Additionally, on AVBP, a focus is done on the effect of artificial viscosity when DFTC is activated or not to highlight the similarities and differences between using DFTC and applying artificial viscosity.

Fig. 5.3 shows changes of $\dot{\Omega}_T$ and T_{max} versus strain a in stagnation point flames computed with CANTERA for increasing imposed values of F_{max} (Here F_{max} is imposed and steps 1 to 3 of table 5.3 are by passed): as expected, when thickening, measured by

[2]An alternative and more complex solution would be to write a conservation equation for F with convection and diffusion terms. This was not done here because DFTC of Eq. 5.3 seems sufficient

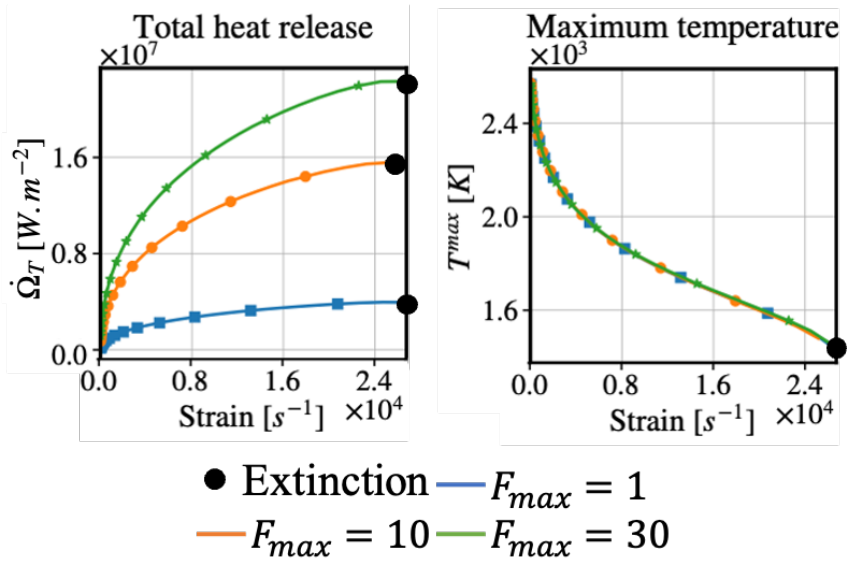


Figure 5.3: Steady strained planar flames (SPLF) using CANTERA for increasing thickening factors F_{max} from 1 to 30. Left: total reaction rate per unit area $\dot{\Omega}_T$ vs strain. Right: maximum temperature T_{max}. Black solid circles: quenching points.

F_{max}, increases, $\dot{\Omega}_T$ increases (like $\sqrt{F}$) so that criterion LC is not satisfied. However, LQ is satisfied: for all thickening factors, quenching (visualized by black circles) occurs at the same strain value (Fig. 5.3 left). Criterion LTMAX (Fig. 5.3 right) is also satisfied: the maximum temperature reached in the diffusion front is the same for all F_{max} values. The results of Fig. 5.3 are all consistent with diffusion flame theory: the total reaction rate per unit flame area $\dot{\Omega}_T$ increases by $\sqrt{F}$ while the quenching strain rate a_Q and the maximum temperature T_{max} are independent of diffusivities and therefore of F as expected from theory [64]. We will discuss the interest of satisfying LC in stagnation point flames in Section 5.4 after discussing the turbulent flame case in Section 5.3.

Fig. 5.4 displays the variation of the integral of heat release $\dot{\Omega}_T$ (black solid line) along the study axis for each combination of flame (Tab. 4.2) and mesh (Tab. 4.4) coupled with F^{max} (dotted line and right axis) and a "normalized" $\dot{\Omega}_T\sqrt{F^{max}}$ to check if we observe the same behavior as with CANTERA. The full algorithm of DFTC is activated (Tab 5.3) with $K = 2000$ and $FRI_{target} = 6$ to ensure a correct resolution.
Looking first specifically into case A several features can be analyzed. First, delving into F^{max} behavior, it is clear that when mesh is coarse (Δ is large), F^{max} increases. This is expected since a poor mesh quality will require more thickening to reach the proper resolution, yet it shows that DFTC can adapt in time. In response to a greater thickening factor F, $\dot{\Omega}_T$ obviously increases too. However, it increases following a predictable pattern which is $\sqrt{F^{max}}$. In fact, after dividing by $\sqrt{F^{max}}$, the result of the simulation is almost constant (Fig. 5.4). Actually, the relation between $\dot{\Omega}_T$ thickened or not is strictly equal to $\sqrt{F^{max}}$ when F^{max} is applied everywhere from $z = 0$ to $z = 1$, which is not the case here. Now, looking into the other cases, high strains (from flame A to flame E) lead to high thickening factors up to $F^{max} \approx 20$ in the most critical mesh with flame A to over 150 with flame E. This demonstrates the robustness of DFTC against any combination of strain/mesh. Concerning RES criterion, figure 5.5 displays the minium

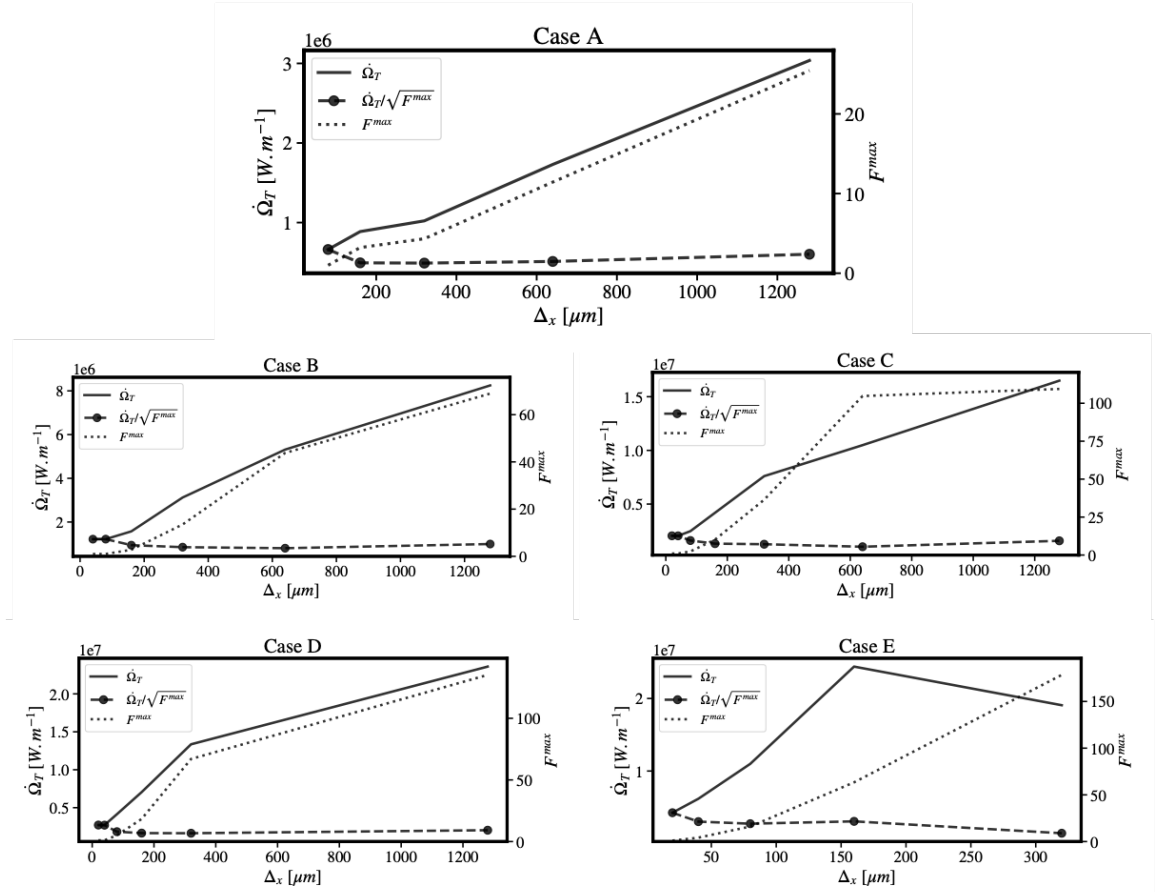


Figure 5.4: Strained stagnation Point Laminar Flame (SPLF): on the right axis, integral of heat release $\dot{\Omega}_T$ and $\dot{\Omega}_T/\sqrt{F^{max}}$ and on the left axis the maximum thickening factor F^{max} as function of the cell size Δ_x for each flame (Tab. 4.2)

resolution measured FRI_{res}^{min} in the simulation for flames A to D in critical meshes (where $FRI_{res}^{min} < 5$ without DFTC) as function of the cell size Δ_x. For each case, DFTC succeeds to validate RES. We can consider that DFTC is validated for steady cases concerning the RES criterion (and LQ and TMAX). The next section will challenge DFTC in a turbulence/flame interaction. Yet, before moving on, one last advantage of DFTC must be discussed.

Fig. 5.6 illustrates this key point of DFTC: instead of thickening the flame using DFTC, an alternative path is to use artificial viscosity. AVBP has centered schemes with second-order artificial viscosity which was used to compute the stagnation point flame of the present section. Fig. 5.6 compares, for the flame C (Tab. 4.2) with mesh 6 (Tab. 4.4), the effects of this artificial viscosity (quantified here by the ratio of artificial hydrogen diffusivity $D_{H_2}^{artif}$ to true laminar diffusivity $D_{H_2}^{lam}$) with and without DFTC. On the left, without DFTC (Case NO MODEL), the code crashes (as in Fig. 4.10) if $D_{H_2}^{artif}/D_{H_2}^{lam}$ is less than two. And when $D_{H_2}^{artif}/D_{H_2}^{lam}$ is larger than two, the code does not crash but results depend strongly on the artificial viscosity $D_{H_2}^{artif}$, making the simulation useless. With DFTC (Fig. 5.6 right), any value of $D_{H_2}^{artif}/D_{H_2}^{lam}$ can be used and results do not

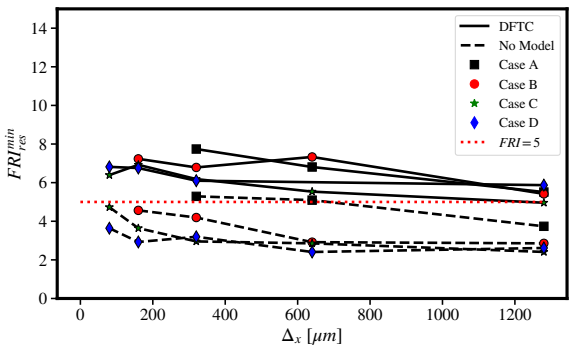


Figure 5.5: Strained stagnation Point Laminar Flame (SPLF): Impact of DFTC on the resolution FRI_{res}^{min} when it is activated (solid line) or not (dashed line) as function of cell size Δ_x for flames A to D (Tab. 4.2). Horizontal red dotted line: $FRI_{res} = 5$.

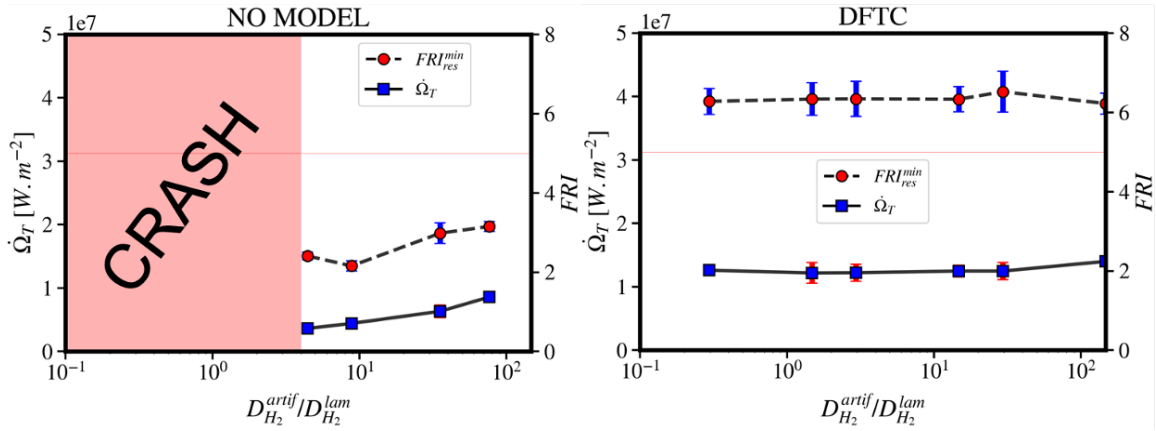


Figure 5.6: Stagnation point diffusion flame: effects of artificial viscosity measured by the ratio $D_{H_2}^{artif}/D_{H_2}^{lam}$. FRI_{res}^{min} and $\dot{\Omega}_T$ are plotted vs $D_{H_2}^{artif}/D_{H_2}^{lam}$ without DFTC (left figure) and with DFTC (right figure). Horizontal red line: $FRI_{res} = 5$.

depend on it: DFTC leads to a simulation which is independent of numerical dissipation.

5.3 Tests of DFTC in a turbulent flame

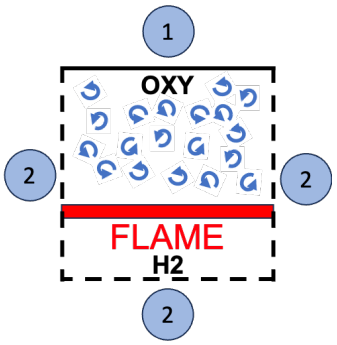
DFTC is tested now on a DNS of the interaction between turbulence and a non-premixed flame in two dimensions, using the AVBP solver [94], the San Diego mechanism [100] and the turbulence generation model of Guezennec et al [45]. Four cases have been tested and summarized in table 5.4

Table 5.4: Characteristics of the turbulence in flame/turbulence interaction cases

Case Name	Integral lengthscale l_t $[mm]$	turbulent velocity fluctuation u' $[m.s^{-1}]$
Alpha α	1	10.
Beta β	2	8.66
Gamma γ	1	15.8
Epsilon ε	1	18.7

The box size is 80 by 80 mm. The initial flame front corresponds to a strained diffusion laminar H2-air flame where both streams are at 300 K. Turbulence is added at $t = 0$ in the air stream only using a superposition of vortices to mimic a Passot Pouquet spectrum [81] with imposed integral scale l_t and RMS velocity u' (Tab. 5.4). Table 5.5 displays the scheme and the boundary conditions of FTI configuration.

Table 5.5: Boundary conditions of FTI (Flame/Turbulence Interaction)

Scheme	Reference	Boundary Conditions
	  	Outlet Symmetry 2D pseudo-HIT

DNS (with 2000 by 2000 points) is used as a reference and compared to two LES using 250 by 250 points without or with the DFTC model. The full algorithm of DFTC is activated (Tab 5.3) with $K = 100$ to ensure a rapid reaction under the effect of turbulence and $FRI_{target} = 6$ to ensure a correct resolution. For DNS, the resolution is high and the minimum FRI_{res} is always greater than 20. For the main part of the section, we will focus on the case gamma (Tab. 5.4). Fig. 5.7 compares a DNS field (left) with an LES

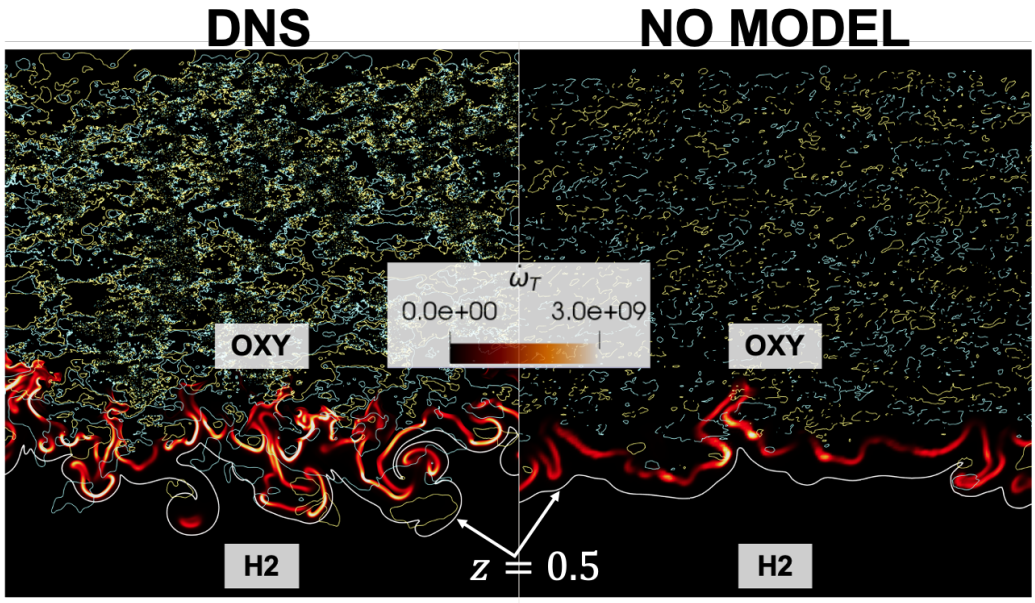


Figure 5.7: Comparison of DNS (left) and LES (right) of heat release fields at $t = 1ms$. White line: $z = 0.5$. Blue line $u = 10\ m.s^{-1}$. Yellow line $u = -10\ m.s^{-1}$ (u is the horizontal velocity).

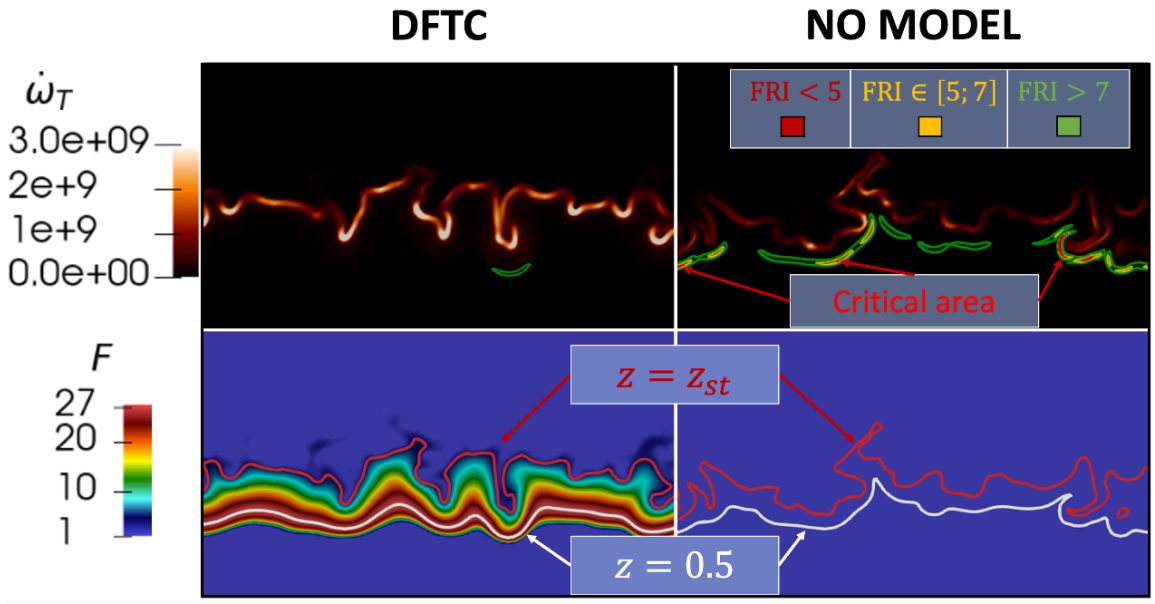


Figure 5.8: FRI_{res} fields for LES with DFTC ($K = 100$) and LES without DFTC at $t = 0.75ms$. The thickening F field is displayed only for DFTC (F is one for the NO MODEL case). White line: $z = 0.5$. Red line: $z = z_{st} = 0.028$.

field (right). Obviously, the first effect of LES is that the coarse mesh strongly limits the wrinkling which the flame can produce.

Fig. 5.8 compares LES with DFTC and without DFTC. The white $z = z_{st}$ stoichiometric line visualizes the instantaneous flame front. Initially and until $t = 0.2$ ms, the flame is well resolved in both LES. The RES criterion is satisfied everywhere: $FRI_{res} > FRI_{target} = 5$. After $t = 0.2$ ms, turbulence wrinkles the flame and strains it,

reducing its thickness and diminishing FRI_{res} at a few points below FRI_{target} in the NO MODEL case where no thickening is applied (Red zones in the field of FRI in 5.8 right). With DFTC, local thickening is introduced when FRI_{res}^{min} becomes close to FRI_{target}: as a result, FRI_{res}^{min} is never less than FRI_{target}, ensuring that the solution is well resolved at all times ($FRI_{res} > 5$ is everywhere in Fig. 5.8 left).

Fig. 5.8 captures a typical zoomed snapshot for both LES after 0.75 ms, showing fields of heat release $\dot{\Omega}_T$ and of FRI. The thickening factor F is added for the DFTC case (F is unity without DFTC). Two isolines of z of interest are also displayed: the stoichiometric line $z = z_{st}$ and the line where $z = 0.5$ where the maximum gradients of z are found. The FRI_{res} field is colored using the following code:

- green: $FRI_{res} > 7$, fronts are well resolved, no thickening is required;
- yellow: $FRI_{res} \in [5, 7]$, fronts are close to being under resolved;
- red: $FRI_{res} < 5$, fronts are under resolved and DFTC must be activated.

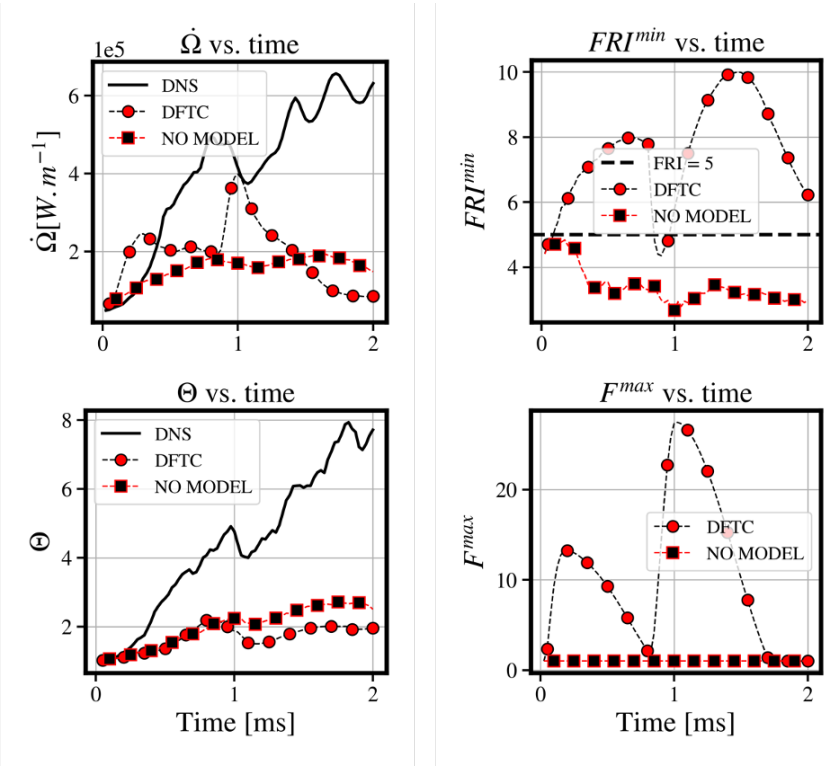


Figure 5.9: From left to right, first row, Time variations of total reaction rate $\dot{\Omega}$, minimum FRI value and second row, wrinkling Θ and maximum thickening factor F. Comparison of DNS and LES without and with DFTC model. Fig. 5.8 is extracted from those simulations

DFTC avoids under resolution zones. No red zone is observed (Fig. 5.8 top) when DFTC is on: criterion RES is satisfied everywhere. In all regions where FRI_{res} decreases towards 5, DFTC increases F and prevents under resolution. Since the cost function used by Eq. 5.4 uses the minimum value of FRI_{res} (FRI_{res}^{min}) to decide whether thickening is needed or not, the model is not local in two ways: (1) along the flame normal, FRI_{res}^{min} is not reached at stoichiometry ($z = z_{st}$) but closer to $z = 0.5$. Still, the value of

$FRI_{res}(z = 0.5)$ is the one which decides the DFTC activation. This is not a major issue: $FRI_{res}(z = 0.5)$ is proportional to $FRI_{res}(z = z_{st})$ [87] and is clearly a good sensor to decide whether the entire local flamelet is strained or not ; (2) along the flame surface, when DFTC detects that thickening is needed at one point, it thickens the whole flame. This limitation is not an issue for the homogeneous flame of Fig. 5.7 but extensions of the model will be required to relax this simplification in real chambers.

Fig. 5.9 shows how DFTC adjusts to the changes in the flame which experiences strain fields changing over time. At time $t = 0.8$ ms, the flame has no strongly strained point any more: DFTC senses this because FRI_{res}^{min} is large enough so that thickening is not needed: F_{max} goes back to unity (right image). Slightly later, at $t = 1$ ms, straining starts again and DFTC increases again thickening to avoid small FRI_{res}^{min} values. The speed at which DFTC reacts is controlled by the relaxation time τ in Eq. 5.6.

Ensuring that the flame is resolved as shown in Fig. 5.8 does not ensure that it is predicted accurately. Tests in laminar flames of Section 5.2 have shown that the quenching strain criterion LQ and the maximum temperature criterion LTMAX are probably satisfied in the turbulent flame too. However, the FTIC criterion, focusing on the total consumption rate, is not satisfied (Fig. 5.9) as shown by the evolution of the total reaction rate $\dot{\Omega}$ over the whole domain, defined as $\dot{\Omega} = \int \dot{\omega}_t dS$. In the flamelet limit, $\dot{\Omega}$ can be written as $\dot{\Omega} = \dot{\Omega}_T * L_f$ where $\dot{\Omega}_T$ is the reaction rate per unit length, studied in the laminar flame cases and L_f is the flame length in this two-dimensional case. This length is controlled by the wrinkling factor $\Theta = L_f/L_y$ in Fig. 5.9 where L_y is the length of the DNS domain.

Fig. 5.9 shows that both LES underpredict the total reaction rate $\dot{\Omega}$ measured in the DNS: this effect is opposite to the one observed in laminar flames where DFTC increased $\dot{\Omega}_T$ too much. In turbulent cases, the decrease of $\dot{\Omega}_T$ is mainly due to the fact that LES underestimates flame wrinkling Θ. While the local reaction rate per unit area $\dot{\Omega}_T$ increases when DFTC is used, as shown for SPLF flames (Fig. 5.3), wrinkling decreases more and the overall effect is that $\dot{\Omega}_T$ is too small in the two LES, even with DFTC. Before discussing this issue in Section 5.4 the other cases can be analyzed.
Fig. 5.10 displays the same information as Fig. 5.9 for cases α, β and ε (Tab. 5.4). First as for the gamma case, DFTC maintains a proper resolution over time for all cases. For each case $K = 100$ and $FRI_{target} = 6$

5.4 Effects of turbulent viscosity and reaction terms: tests of LAM TUR REAC models

Previous sections have shown that DFTC allows to predict quenching by strain at the right value (criterion LQ in Table 5.1), maximum flame temperature (criterion LTMAX) while ensuring that the flame was properly resolved (criterion RES) and independent of numerical dissipation. When it comes to the mean heat release, however, DFTC overpredicts the exact results in laminar stagnation point flames (Criterion LC, Section 5.2) because molecular diffusivities are increased by F but underpredicts them in turbulent cases (Criterion FTIC, Section 5.3) because wrinkling is reduced when the mesh is decreased from 2000^2 to 250^2.

A logical next step is to wonder whether criterion FTIC which is an important objective

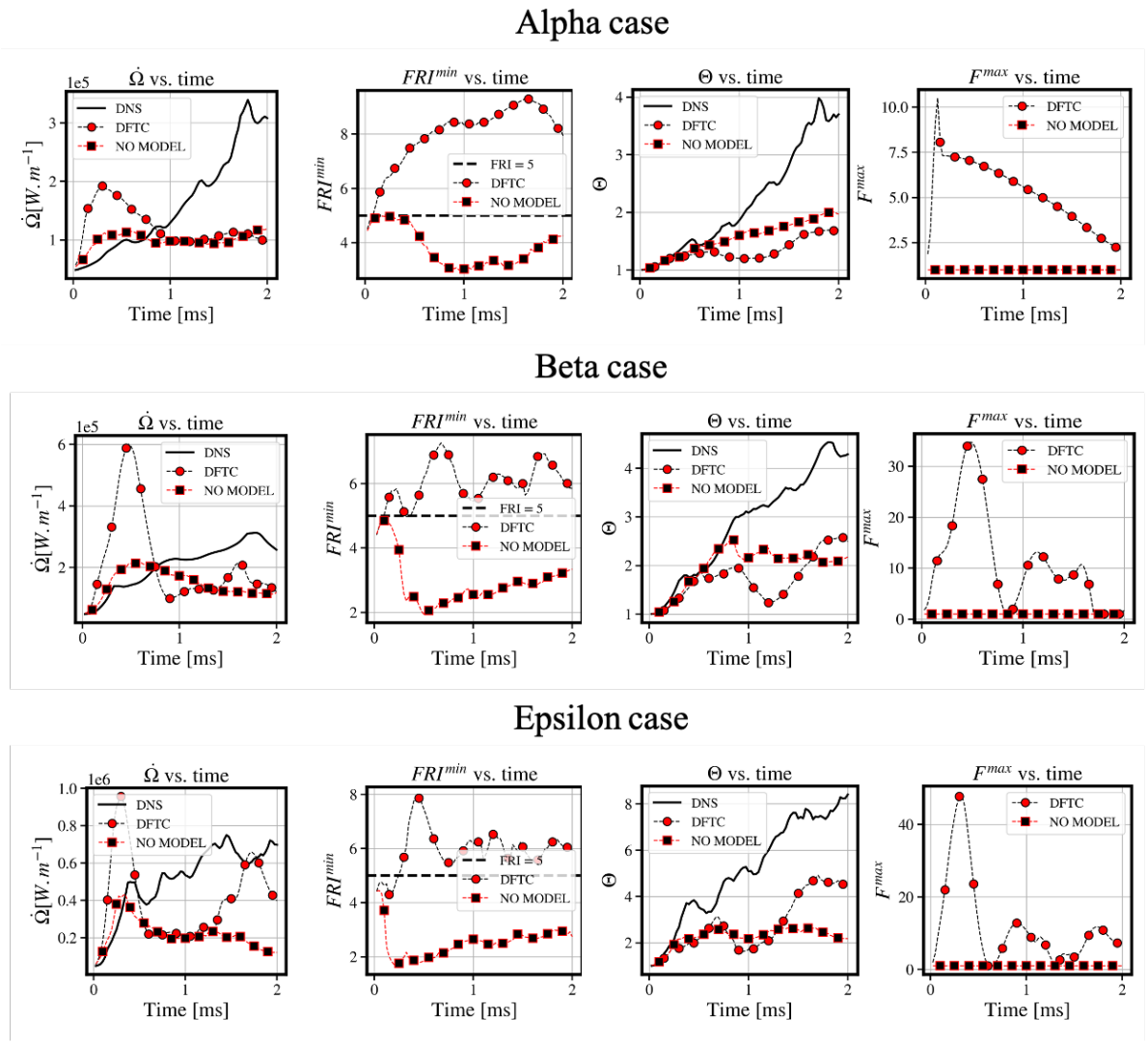


Figure 5.10: From left to right: Time variations of total reaction rate $\dot{\Omega}$, minimum FRI value, wrinkling Θ and maximum thickening factor F. Comparison of DNS and LES without and with DFTC model for cases α β and ε of table 5.4.

of a proper LES subgrid model can be satisfied by introducing modifications of α (the turbulent diffusion term) or F_r (the chemical term) which means going to LAM TUR REAC formulations in Eq. 5.2 instead of a LAM model which is the case of DFTC.

A first test is to explore LAM TUR models and add the 'TUR' option to the existing DFTC. This is done by modifying the turbulent diffusivity term in Eq. 5.2 and multiplying it by α. Fig. 5.11 shows that increasing turbulent viscosity by $\alpha = 10$ (left) or even 20 (right) does not allow to increase Ω (Fig. 5.11 top). Increasing the turbulent viscosity actually decreases the flame wrinkling Θ (Fig. 5.11 middle) so that Ω is smaller when $\alpha > 1$ and not larger as expected initially. The bottom images of Fig. 5.11 show the ratio of the turbulent diffusivity to the laminar diffusivity $r_{eff} = (\alpha D_{H_2}^{turb})/(F D_{H_2}^{lam})$ averaged along the $z = z_{st}$ line. It shows that increasing α indeed increases the effects of the turbulent diffusivity on the flame front but does not allow to increase Ω.

To increase the reaction rate $\dot{\Omega}$ in Fig. 5.9, another natural approach following what is done when using efficiency functions for premixed flames is to increase the reaction term by increasing F_r in Eq. 5.2 and test LAM REAC models. Fig. 5.12 shows however that this idea does not work well for the diffusion hydrogen flame studied here. With $F_r = 10$, even though chemistry is ten times faster, the overall reaction rate $\dot{\Omega}$ does not increase. This is clearly due to the fact that in this hydrogen flame, chemical reactions are very fast compared to diffusion in all cases with $F_r = 1$ or $F_r = 10$. Thus, diffusion controls the overall reaction rate for low strains: increasing F_r makes the reaction zone thinner without changing the overall reaction rate which is controlled by diffusion only.
This result gives rise to two important observations:

- The first remark is that, from this perspective, diffusion flames exhibit a significantly higher level of complexity than premixed flames. In the latter case, the global heat release can always be modulated by modifying the reaction terms (this being the role of the efficiency function in the TFLES framework). Such flexibility is not available for diffusion flames.

- The second remark is that the choice of hydrogen flames places the present study in a particularly stringent framework. Owing to the very fast chemical kinetics in hydrogen–air flames, combustion often occurs on timescales shorter than those of molecular diffusion. As a result, adjusting the reaction-rate scaling factor F_r does not allow to modify the global heat release significantly. By contrast, for fuels characterized by slower chemistry in air such as hydrocarbons or ammonia, it may be possible, at least partially, to restore the overall heat release through an appropriate tuning of F_r.

From a practical point of view, these results suggest that, for hydrogen flames, the standard LAM model used in the previous sections is probably the best compromise which can be achieved in the LAM TUR REAC class of models and that changing α or F_r brings limited improvement for hydrogen-air flames. When it comes to LES, Fig. 5.7 illustrates the main problem: the loss of flame wrinkling Θ which is a direct consequence of the coarser meshes used in the LES. At this point mesh refinement is certainly a useful approach (such as Automatic Mesh Refinement - AMR) even if it will not fully solve the problem for highly strained diffusion flamelets [24].

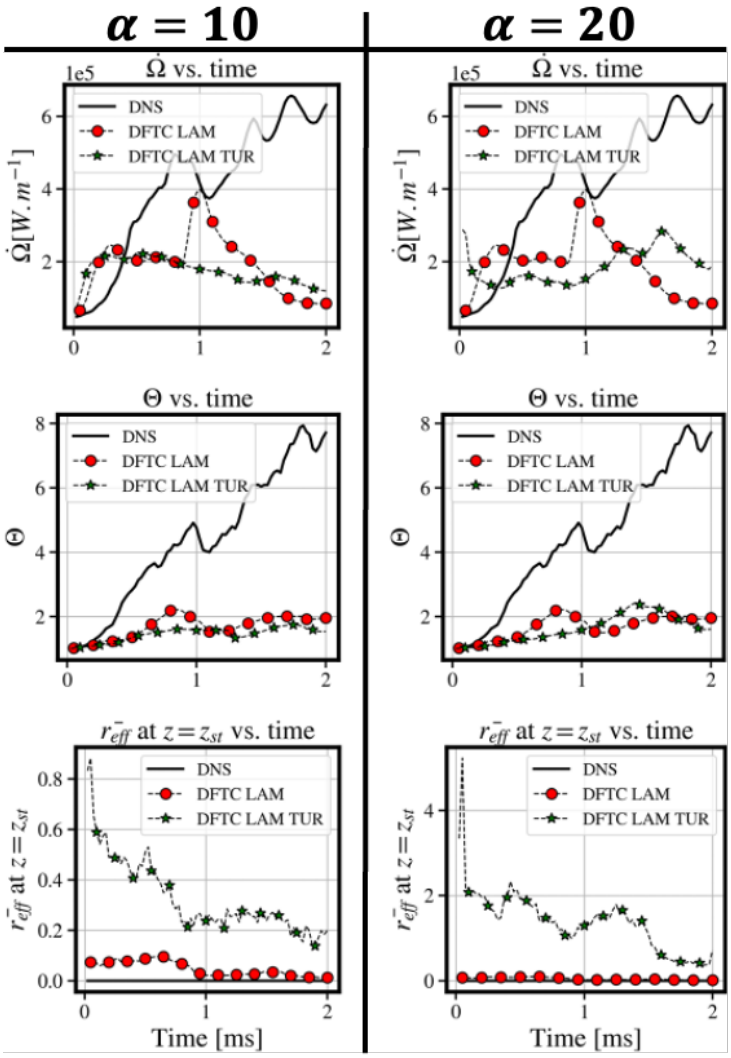


Figure 5.11: Effects of α: time variations of total reaction rate $\dot{\Omega}$ (top), wrinkling factor Θ (middle) and average ratio $r_{eff} = (\alpha D_{H_2}^{turb})/(F D_{H_2}^{lam})$ along the $z = z_{st}$ line (bottom). Comparison of DNS and LES without and with DFTC model when α is changed. Left: $\alpha = 10$. Right: $\alpha = 20$.

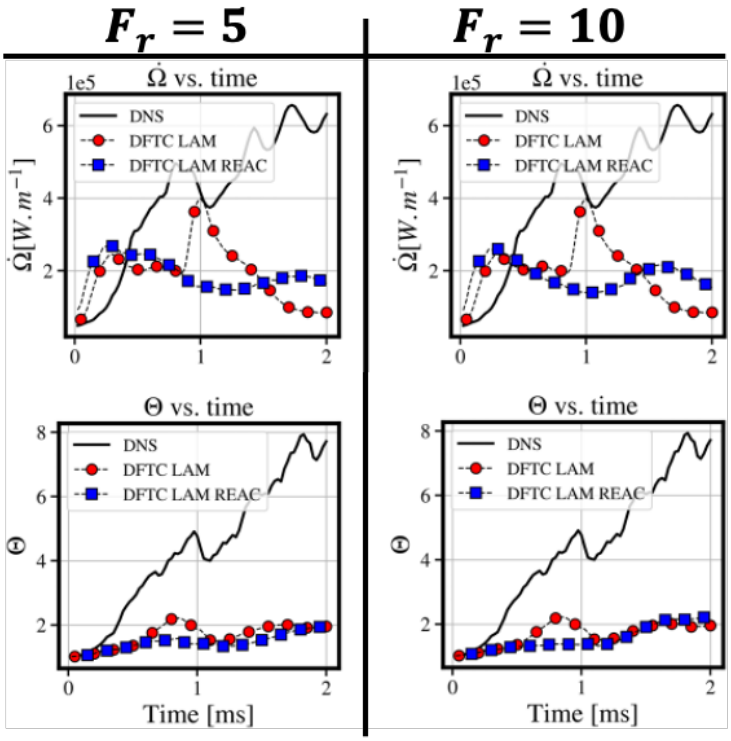


Figure 5.12: Effects of F_r: time variations of total reaction rate $\dot{\Omega}$ (top) and wrinkling factor Θ (middle). Comparison of DNS and LES without and with DFTC model when F_r is changed from 1 to 10.

5.5 Comparison of thickening models for premixed and diffusion flames

Since DFTC proposes to thicken diffusion fronts, it is interesting to compare it to the well established TFLES model for premixed fronts. The initial papers of O'Rourke and Bracco [78] and Colin et al [17] are often used to justify TFLES by the fact that thickening conserves the speed of a laminar unstrained premixed flame (first line in Table 4.2). This is right but with one important caveat: it works only if the flame is not strained and not curved, something which is never true in turbulent flames. Recent results show that TFLES does not conserve maximum temperature, flame speed or quenching strain rate of strained flames as soon as they are thickened [26, 57]. In that sense, DFTC does better than TFLES as shown by Table 5.6, especially for quenching limits: a diffusion flame thickened by DFTC will quench at the right strain while a premixed flame thickened by TFLES with quench at a strain rate which diminishes like $1/F$, leading to large errors. Overall, thickening fronts may be as reasonable for diffusion flames as it is for premixed flames.

Table 5.6: Comparison of thickening effects on laminar diffusion and premixed flames

PROPERTY TO CONSERVE	DFTC (Diffusion flames)	TFLES (Premixed flames)
1 Laminar unstrained flame speed	N/A	YES
2 Quenching strain rate	YES	NO
3 Maximum temperature vs strain	YES	NO
4 Heat release vs strain	NO	NO

5.6 Status and guide for current DFTC

5.6.1 Effect of the K relaxation parameter

The goal of DFTC is to detect unresolved areas within an LES simulation and dynamically thicken it in order to increase local resolution without (in the current status) modifying the mesh. Facing an unresolved steady solution, SPLF for example, this must lead to an increase of thickening factor until stabilization is attained (when the criterion is reached). However, facing an unresolved unsteady simulation, where turbulence is present for example, the model will, on one hand, control and terminate initial unresolved areas and, on the other hand, perceive new areas created under flame/turbulence interaction and then apply sufficient thickening factor to ensure proper resolution (following once again the FRI criterion). The second objective can be quite challenging because a balance must be considered: the model should adapt directly (in the course of a few iterations) to kill the unresolved area but it is numerically impossible. Tremendous changes during a small amount of iterations lead to numerical instabilities causing crashes or non-physical behaviors of the flame. On the opposite side, too slow variations of F can lead to an absence of impact from the model to prevent under resolution. Therefore, the choice of K (and of z) is important. Several tests have been performed, on steady and unsteady

configurations, to determine K. The conclusion of those tests leads to a value of 2000 for steady cases and 100 for unsteady ones and a weak dependence on K. The idea to link K to physical time scales was also considered by looking into characteristic physical times depending on the configuration type following 5.6.

- Steady cases (SPLF for exemple): a reasonable τ time value in a steady flame is the inverse of strain. For an hydrogen diffusion flame, as shown in chapter 2, extinction is reached at approximately 27 000 s^{-1}. Then $\tau_K = 1/a \approx 40\mu s$ (since K must be the smallest). For the present case, $\Delta_t \approx 10ns$ of (classical for LES), so that K is estimated at 4000 which is coherent with the value that we find empirically.

- Unsteady cases (FTI for exemple): it is also possible to use turbulent timescales such as $\tau_t = l_t/u'$ (turbulent timescale τ_t as function of integral lengthscale l_t and root mean square of fluctuation speed u') or Kolmogorov timescale $\tau_\eta = (\nu/\varepsilon)^{1/2}$ (Kolmogorov timescale τ_η as function of viscosity ν and the turbulent dissipation rate ε). Since K must be small as possible (to capture the unsteady behavior of the system) and $\tau_\eta < \tau_t$ always, then an estimation of τ_η can give an estimation of the proper K. Assuming a 3D context with similar turbulence as in our configuration of section 5.3 and that $\varepsilon \approx (u'^3/l_t) \approx 41.10^6$. Then (if the turbulence is only in the air stream) $\tau_\eta \approx (1.7.10^{-5}/41.10^6)^{1/2} \approx 2\mu s$ which imply $K \approx 200$. This estimation is also coherent with the one used in the present LES.

To conclude, choosing τ (or equivalent K) can be done by scaling τ by flow and flame times. Since this is the only free parameter of the model, this will not be a major limitation for the practical use of DFTC.

5.7 State of DFTC and perspectives

Many hydrogen flames of practical interest do not operate in fully premixed modes for safety reasons. Therefore they exhibit premixed but also diffusion fronts and these diffusion fronts raise a daunting challenge for simulations especially at high pressures. The scaling laws given by simple asymptotic theory are used here to show that strained diffusion flamelets (extinguished or not) are the thinnest flame elements in turbulent flames (Fig. 4.1) and that they become systematically under resolved at high pressures, leading to considerable risks when codes validated in lab-scale experiments at 1 bar are ported to high pressures. A thickening method (called DFTC for Diffusion Front Thickness Control) to prevent this systemic under-resolution in LES is proposed by defining a resolution index called FRI_{res} (FRI_{res} indicates how many mesh points are used inside the resolved flame front) and implementing a closed-loop thickening algorithm (Eq.(5.2) which guarantees that these zones are spatially resolved and have large enough FRI_{res} values. DFTC automatically deactivates when strain is reduced and diffusion flamelets are resolved on the LES grid. DFTC introduces additional viscosity like an artificial viscosity does but in a controlled fashion which does not depend on the grid or on the numerical scheme.

Tests on stagnation point laminar flames show that $FRI_t > 5$ is mandatory to ensure minimum resolution. DFTC is then developed using an on-line, active control method to thicken the front where the flame is too thin, ensuring values of FRI_{res} larger than 5. Results show that multiplying all laminar diffusivities by the thickening factor F provided

by DFTC allows to maintain FRI_{res} to values larger than 5 so that the simulation is no longer influenced by the mesh or the numerical method. The total reaction rate $\dot{\Omega}_T$ per unit area is overestimated in laminar flames but the quenching strain rate a_Q as well as the maximum temperature T_{max} reached in the diffusion flame are conserved. Results are not as good in turbulent flames where the loss of flame wrinkling due to low resolution becomes the first issue for LES. This could be compensated only partially by smart meshing strategies such as AMR because at high pressures, wrinkling scales will be too small to be meshed anyway even by using mesh refinement. Combining DFTC and AMR methods is probably a good approach for high pressure diffusion flames: DFTC will ensure that criteria RES, LQ and LTMAX (Table 5.1) will be satisfied while AMR will be needed to satisfy FTIC.

In conclusion, it is quite clear that even if the current status of DFTC is not perfect it has multiple advantages:

- it is as interesting as TFLES for premixed flames which does not satisfy any of the criterion LQ, LTMAX, LC or FTIC.
- it is very simple to implement.
- it is compatible with full chemical schemes like TFLES.
- it satisfies RES, providing a numerical solution which is the same and unaffected by numerics.
- it does not require any tabulation.

In addition, extensions of DFTC to premixed flames or to mixing regions have already been written (PhD of A. Schioppa and H. Tchetche) so that a single DFTC model may have the possibility to handle all flame regime.

Chapter 6

Conclusion

The transition toward low-carbon energy systems has renewed interest in hydrogen as a fuel for future gas turbines, aero-engines, and power-generation systems. While hydrogen offers significant environmental advantages, its combustion introduces major scientific and numerical challenges. Among them, the prevalence of highly strained non-premixed flames, the strong density variations inherent to hydrogen-air mixtures, and the extremely small reactive scales encountered at elevated pressures all complicate the predictive simulation of practical combustion systems. Within this context, the objective of this dissertation was to improve the understanding and modeling of hydrogen diffusion flames and to develop numerical tools suitable for Large Eddy Simulation (LES) of turbulent hydrogen combustion.

The work was initiated around theoretical analysis of laminar diffusion flames and finished with the development of a dedicated LES model. Four major scientific contributions emerge from this research.

The first contribution concerns the definition and characterization of flame strain in diffusion flames. Although strain is universally recognized as the dominant parameter controlling the structure, thickness, extinction behavior, and overall dynamics of non-premixed flames, its practical evaluation remains surprisingly ambiguous in the literature. Multiple definitions coexist, often leading to inconsistent interpretations when comparing laminar flamelets, numerical simulations, and combustion models. This thesis revisited the notion of strain from first principles and established exact definitions applicable to stagnation-point diffusion flames which are essential information for any model using flamelet assumption. These definitions were systematically compared with commonly used approximations over a broad range of operating conditions, including variations in pressure, fuel temperature, oxidizer temperature, and fuel type. The analysis demonstrated the equivalence of a few formulations called "exact" while highlighting the limitations of simplified strain definitions, particularly in hydrogen flames where density variations are significant. The results provide a coherent framework for evaluating strain and establish a robust foundation for future flamelet-based modeling approaches.

The second major contribution is the extension of classical diffusion-flame theory to variable-density hydrogen flames. Most analytical models available in the literature were originally derived under the assumption of constant density, an approximation that may be acceptable for methane-air flames but becomes questionable for hydrogen-air mixtures. In hydrogen combustion, density can vary by more than an order of magnitude across the

mixing layer, even in the absence of chemical reactions. This thesis demonstrates that such variations have a non-negligible influence on diffusion H_2-air flame structure. To address this limitation, a new analytical framework was developed in which density, molecular weight, temperature, and diffusivity variations are explicitly incorporated. The resulting Variable Density (VD) model was compared against both the classical Constant Density (CD) theory and detailed one-dimensional simulations performed with CANTERA. Over a wide range of operating conditions, the VD model consistently improved the prediction of key quantities such as mixture fraction distributions, scalar dissipation rate, flame thickness, maximum temperature, and integrated fuel consumption. Beyond its immediate applications, this development contributes to bridging the gap between idealized analytical theories and the physical reality of hydrogen combustion, where density effects cannot be ignored.

The third contribution of this dissertation concerns the resolution of diffusion flames in LES and the development of a new model dedicated to under-resolved hydrogen flames. Through an extensive series of laminar and turbulent simulations, it was shown that flame resolution constitutes a fundamental prerequisite for any successful turbulence-chemistry interaction model. In practical high-pressure configurations, diffusion flame thicknesses may reach only a few micrometers, making their direct resolution impossible in industrial LES where the mesh reaches only around one hundred micrometers with most existing solvers today. The thesis demonstrated that unresolved diffusion fronts lead to significant errors in flame structure, heat release, and extinction behavior, regardless of the turbulence-combustion closure employed. To quantify these effects, a resolution metric based on the Flame Resolution Index (FRI) was introduced and systematically investigated. A critical threshold for FRI separating resolved and under-resolved configurations was identified, providing a practical criterion for monitoring simulation quality.

The last and most significant contribution is a new model called Diffusion Flame Thickness Control (DFTC). Inspired by the philosophy of Thickened Flame LES methods for premixed combustion, DFTC dynamically modifies the local transport properties to ensure adequate resolution of diffusion fronts while preserving the essential physical behavior of the flame. The model was validated in both steady strained diffusion flames and unsteady flame/turbulence interaction. The results demonstrated that DFTC significantly improves numerical robustness, maintains sufficient flame resolution, and recovers key global quantities such as maximum temperature or quenching value. Unlike approaches relying primarily on artificial viscosity, the proposed methodology provides direct control over flame resolution and offers a physically motivated framework for future LES developments. This represents one of the first attempts to construct a dedicated resolution-control model specifically designed for non-premixed hydrogen flames.

Beyond these four principal achievements, the dissertation also generated a comprehensive database describing the evolution of diffusion flame properties under variations of pressure, strain, and boundary temperatures. This database provides valuable physical insight into the behavior of hydrogen diffusion flames and may serve as a reference for future model development and validation activities. Furthermore, the work established a clear methodological link between laminar flamelet theory, DNS-level understanding, and practical LES modeling, thereby strengthening the consistency of the overall modeling strategy.

Although significant progress has been achieved, several perspectives remain open.

The current version of DFTC can be supported by adding adaptive mesh refinement strategies and improved coupling with turbulence-chemistry interaction models once the resolved flame front is achieved. The extension of the methodology to multi-regime combustion, where premixed and non-premixed flame structures coexist, represents another important challenge. Additional developments may also include applications to realistic industrial burners operating under high pressure. Finally, the theoretical developments proposed for variable-density diffusion flames could be extended to account for finite-rate chemistry effects and differential diffusion.

In conclusion, this dissertation contributes to the understanding and modeling of hydrogen diffusion flames at multiple levels, from fundamental flame theory to practical LES applications. By clarifying the definition of strain, extending classical diffusion-flame theory to variable-density conditions, identifying the critical role of flame resolution, and introducing the DFTC model, this work provides new tools and concepts for the simulation of turbulent hydrogen combustion. These developments constitute a step toward more reliable predictive simulations of future hydrogen-fueled combustion systems and contribute to the broader effort of enabling low-carbon energy technologies.

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Appendix A

Chapter 2

A.1 Database of key variables (total consumption rate) at a stagnation point laminar diffusion flames under strain, temperature and pressure variation

The goal of this part is to create a database of axisymmetric strained H_2-Air at a stagnation point flame that can be used as a reference and also to keep in mind how key variables react against different configuration points.

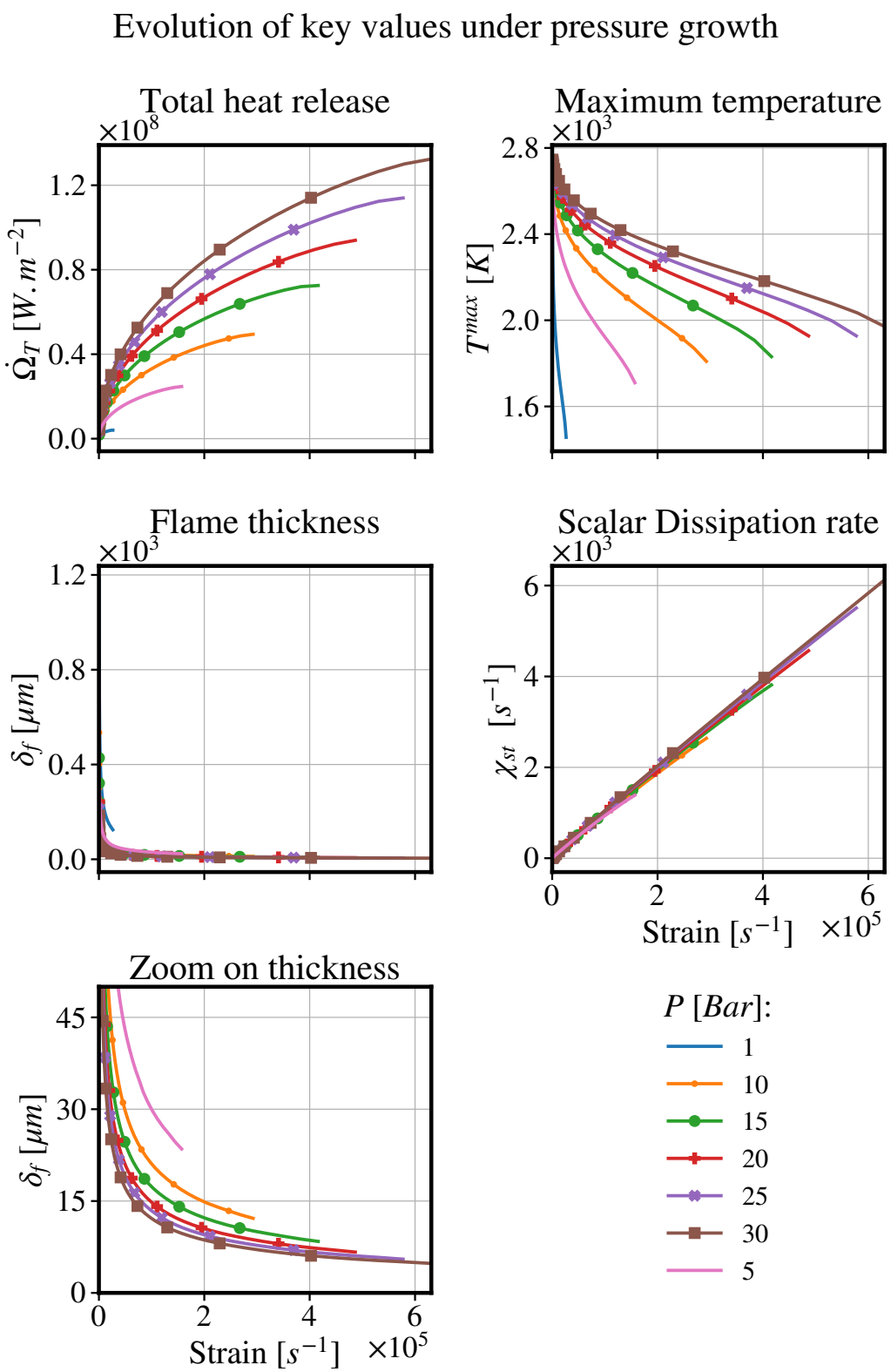


Figure A.1: Evolution of key variables as function of pressure increases, from 1 *bar* to 30 *bar*. On the first column, from top to bottom, the integral of heat release $\dot{\Omega}_T$, the flame thickness δ_f and zoom on small value of δ_f. On the second column, the maximum temperature T^{max} and the scalar dissipation rate evaluated at the stoichiometry χ_{st}

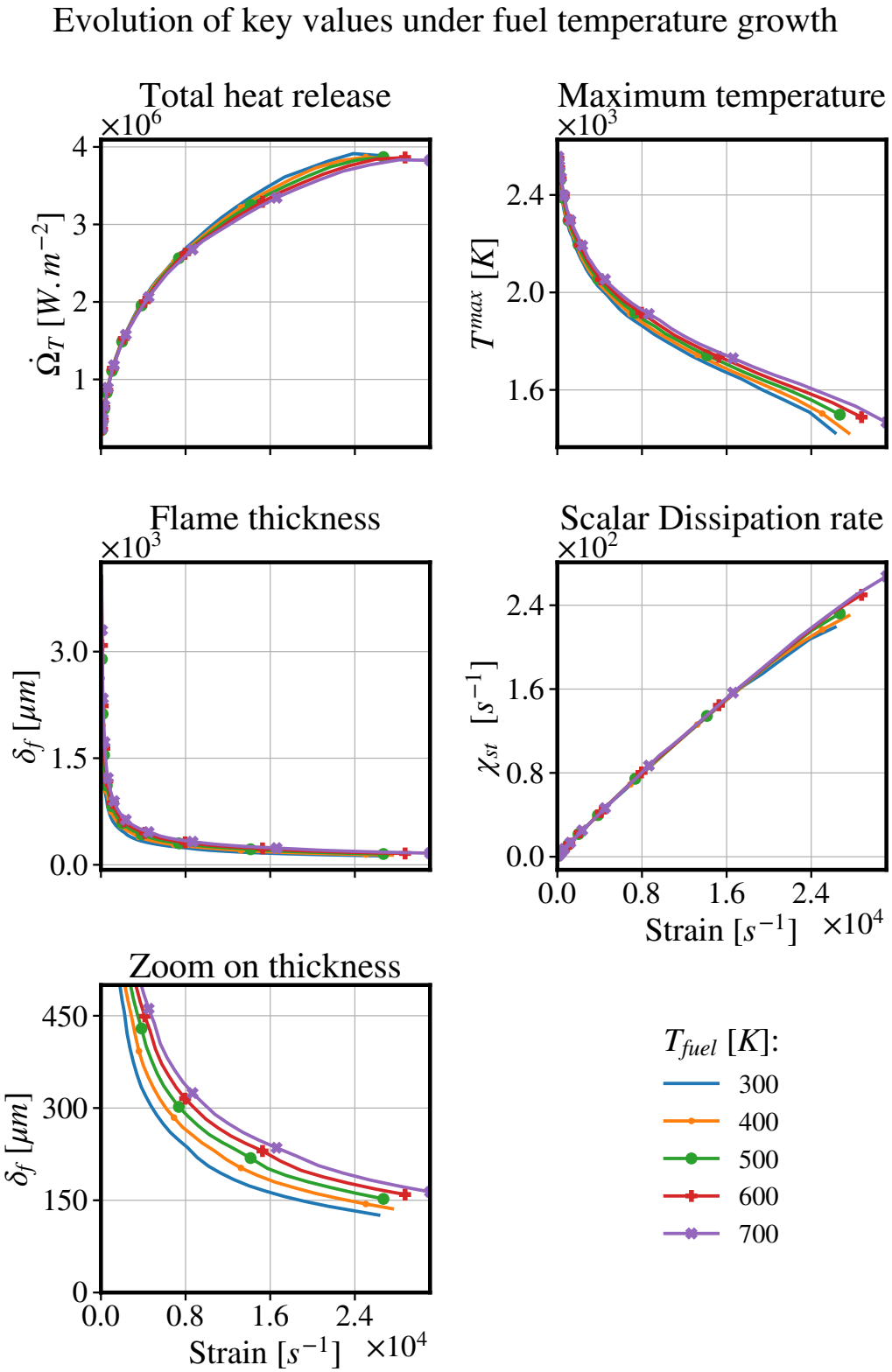


Figure A.2: Evolution of key variables as function of fuel temperature increases, from 1 *bar* to 30 *bar*. On the first column, from top to bottom, the integral of heat release $\dot{\Omega}_T$, the flame thickness δ_f and zoom on small value of δ_f. On the second column, the maximum temperature T^{max} and the scalar dissipation rate evaluated at the stoichiometry χ_{st}.

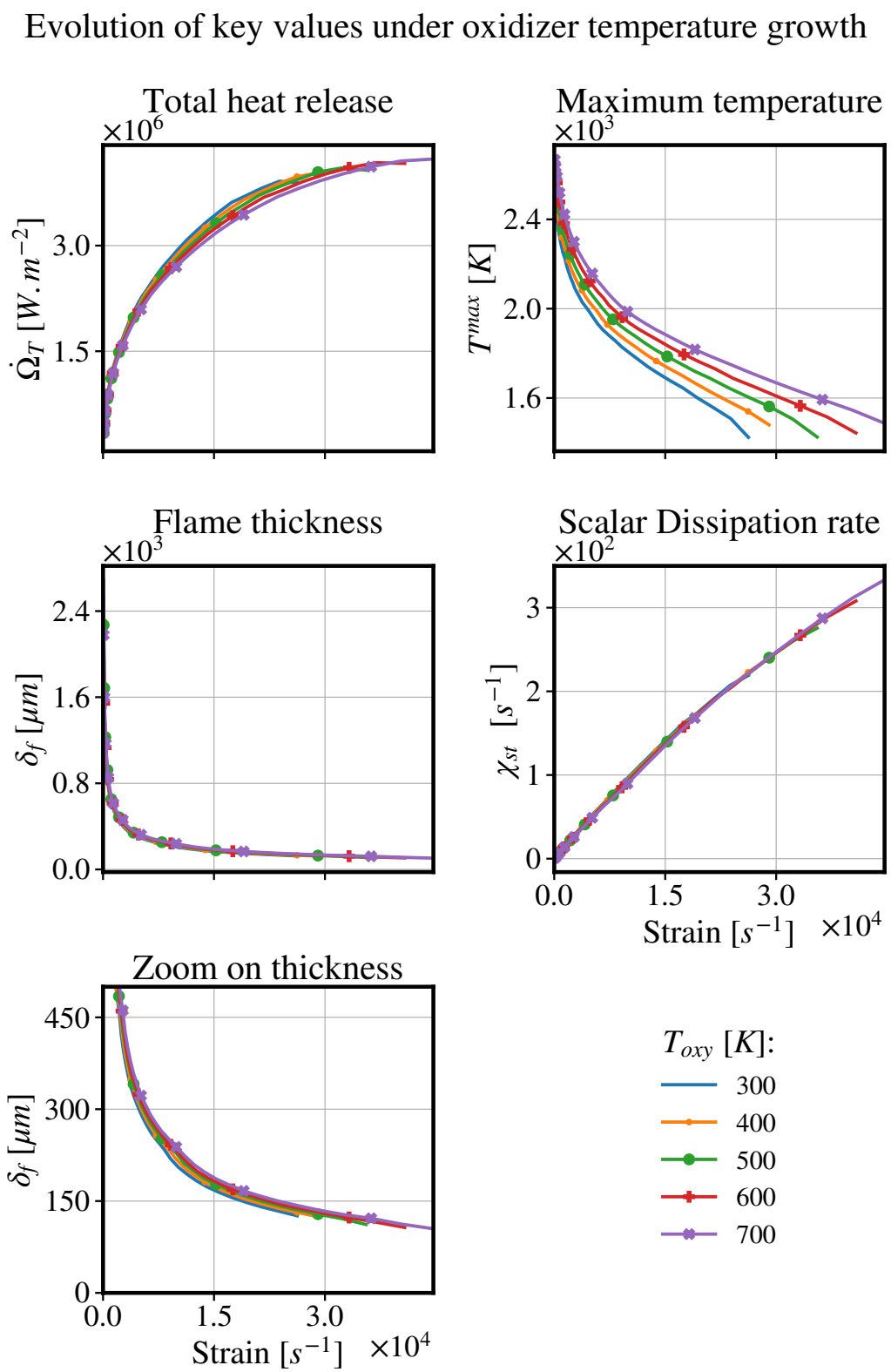


Figure A.3: Evolution of key variables as function of oxidizer temperature increases, from 1 *bar* to 30 *bar*. On the first column, from top to bottom, the integral of heat release $\dot{\Omega}_T$, the flame thickness δ_f and zoom on small value of δ_f. On the second column, the maximum temperature T^{max} and the scalar dissipation rate evaluated at the stoichiometry χ_{st}.

Appendix B

Chapter 2

B.1 Effect of f, the thickening factor distribution function

Another key point of the model is the "shape" of f (Fig. 5.2). This function controls the amount of thickening factor that will be applied as a function of the local mixture fraction z (after the estimation of F^{max}). There are several important features:

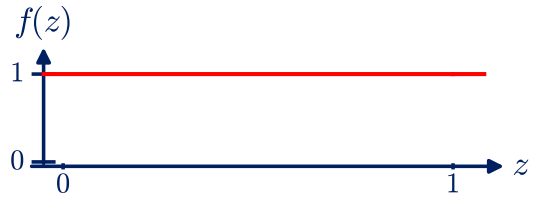
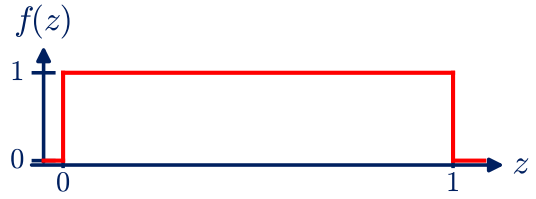
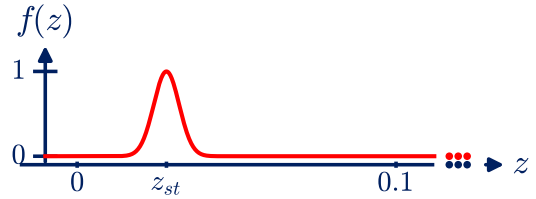
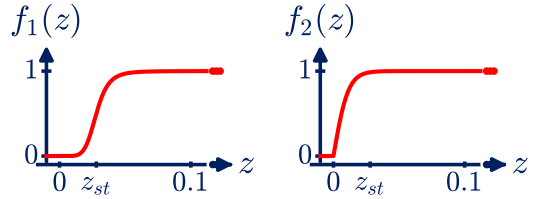
1. obviously, there is no need to impose thickening factor outside of the flame. Therefore, $f(z=0) = f(z=1) = 0$ is a must.

2. Following the last point, f should not exhibit strong variations around particular values, such as the stoichiometric mixture fraction z_{st} or close to the boundaries.

3. For hydrogen, as well as hydrocarbon fuels (methane or kerosene for example), the stoichiometric mixture fraction value is very low ($z_{st}^{H_2} = 0.028$, $z_{st}^{CH_4} = 0.055$ and $z_{st}^{kero} \approx 0.07$). Therefore, it is important that f rapidly takes high values close to $f(z=0)$ in order to have thickening at the stoichiometry.

This metric imposes a limited amount of choices for f and f profile displayed in figure 5.2 is a good balance between all points. Table B.1 shows unsatisfactory f functions that would imply wrong thickening factors: these functions were tested and table B.1 last column says why they were rejected.

Table B.1: Effect of the thickening factor distribution function and their consequences.

Conditions	f Profile as function of z	Unwanted consequences
Condition 1		Thickening presents in the air and fuel streams results
Condition 2		Numerical instabilities appears around the interface between the flame front and the rest of the system
Condition 2		$F(z) = \begin{cases} F^{\max} & \text{if } z \approx z_{st} \\ 0 & \text{otherwise} \end{cases}$ where F is the thickening factor. Numerical instabilities appears around z_{st} and most of the flame is not thickened.
Condition 3		$F_1(z_{st})$ is too low, no thickening factor will be applied over the stoichiometric surface. $F_2(z_{st})$ is higher and presents the right balance between high value and a smooth transition from the front flame to the rest.

Appendix C

Combustion and Flame Article

Spatial under-resolution and thickening models for LES of turbulent diffusion flames

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Abstract

This paper presents a model able to handle spatial under resolution in simulations of diffusion flames where the combination of high strain and high pressure creates very thin flames. These thin diffusion fronts are resolved on a few points only, leading to numerical solutions which are mainly controlled by the mesh and artificial viscosity. The paper proposes a thickening strategy for these diffusion flames, similar to the TFLES machinery used in premixed flames: a flame resolution index (FRI) is first introduced to measure the thickness of diffusion fronts using the gradient of mixture fraction (FRI indicates how many points are used in the flame thickness, measured on the mixture fraction profile). Then, an active control technique is used to thicken the flame locally and ensure that FRI is larger than a threshold for numerical accuracy which is found to be 5 in hydrogen air laminar flames. The model is tested in laminar stagnation point flames, flame-vortex interaction configurations and turbulent diffusion flames and an evaluation of the its accuracy is provided using theory.

Keywords: Non-premixed, Numerical simulation, Resolution, Modeling, Hydrogen, Flame-turbulence interaction

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Novelty and significance statement

Models for turbulent diffusion flames are a weak point of numerical simulation of combustion with unsteady approaches like LES. In these flames, the first difficulty is linked to spatial resolution: strained diffusion flames can become extremely thin and lead to under resolved fronts and ill-posed numerical solutions especially at high pressures, when all flame thicknesses decrease drastically. The present paper demonstrates that codes, validated against experimental data at 1 bar on lab-scale diffusion flames, might fail at high pressures because of a systemic spatial under-resolution of diffusion fronts and shows how this problem can be fixed by using a new model, similar to the TFLES (Thickened Flame model for LES) for premixed flames. Like TFLES, this model can be easily implemented in most codes. It provides a new framework for LES of turbulent high-pressure diffusion flames and should attract interest from many groups.

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1. Introduction

In many practical combustors, premixed and diffusion flames coexist within the same reactive flow field [3, 4]. In engine-relevant configurations, the combination of elevated pressures (ranging from several tens of bar in automotive and aero-engines to several hundreds of bar in rocket propulsion systems) and intense strain rates leads to extremely thin instantaneous flame thicknesses. Although the exact flame thickness depends on the local thermochemical state and transport properties, its proper numerical resolution would typically require grid spacings of the order of a few micrometers.

Thus, while many studies focus on flame-turbulence interaction (FTI) models, an issue which is often ignored is the spatial resolution of the flame. A simple metric for spatial resolution is a quantity called FRI (Flame Resolution Index), which measures the number of points used to resolve the laminar flame thickness δ_L:

$$FRI = \delta_L/\Delta \tag{1}$$

where Δ is the local grid size. Definition 1 applies to both premixed and diffusion flames, using an appropriate definition of flame thickness δ_L in each case. FRI indicates therefore how many mesh points are used to solve the conservation equations describing

Nomenclature	
α: modification of turbulent term in Eq.3	$\dot{\omega}$: reaction rate $[W.m^{-3}]$
a: strain $[s^{-1}]$	$\dot{\omega}_T$: heat release $[W.m^{-3}]$
δ_L: thickness of the flame $[m]$	$\dot{\Omega}$: total reaction rate $[W.m^{-1}]$
δt: simulation step time $[s]$	$\dot{\Omega}_T$: integrated heat release along flame normal $[W.m^{-2}]$
Δ: local mesh size $[m]$	ρ: density $[kg.m^{-3}]$
D: species diffusivity $[m^2.s^{-1}]$	r: ratio
f: function used in DFTC	R: radius
F: modification of laminar term in Eq.3	S: Surface $[m^2]$
F_r: modification of reaction term in Eq.3	τ: relaxation time $[s]$
FRI: number of points inside flame front	Θ: wrinkling of the flame
K: free DFTC model parameter	t: time $[s]$
l_t: integral length scale $[m]$	t^+: reduced time
L: length $[m]$	T: Temperature $[K]$
∇: gradient operator	u': turbulent velocity fluctuation $[m.s^{-1}]$
n: simulation iteration index	Y_k: Mass fraction of species k
	z: mixture fraction

Subscripts	
artif: artificial	min: minimum
b: burned	Q: quenching
eff: effective	st: stochiometry
f: flame	res: resolved
k: species k	t: true
lam: laminar	turb: turbulent
max: maximum	u: unburned
	y: along the y axis

Abbreviations	
AMR: adaptive mesh refinement	LAM: modification of Laminar term
AVBP: solver [1]	LC: Laminar Consumption
CANTERA: solver [2]	LES: Large Eddy Simulation
DFTC: Diffusion Front Thickness Control	LQ: Laminar Quenching
DNS: Direct Numerical Simulation	LTMAX: Laminar Maximal Temperature
F-TACLES: Filtered TAbulated Chemistry for LES	NO MODEL: absence of combustion model
FRI: Flame Resolution Index = δ_F/Δ	REAC: modification of Reaction term
FTI: Flame Turbulence Interaction	RES: Resolution
FTIC: Flame Turbulence Interaction Consumption	SPLF: Strained stagnation Point Laminar Flame
FVI: Flame Vortex Interaction	TFLES: Thickened Flame Large Eddy Simulation
HYLON: experimental setup [3]	TNF: Turbulent Non-premixed Flames workshop
	TUR: modification of turbulent term

chemical reactions and diffusion processes within the flame. Most DNS and one-dimensional flame solvers like CANTERA [2] use high FRI values (typically 20 for DNS and more than 100 using mesh adaptation in 1D solvers for laminar flames).

For premixed configurations, flamelet-based approaches such as TFLES [5, 6, 7] and F-TACLES [8, 9] explicitly address the need to control the FRI of premixed fronts, either through artificial thickening or by filtering the flame structure. In this framework, the underlying mechanisms are well established. Within TFLES, a laminar premixed flame can be thickened by enhancing molecular diffusivities, while theoretical arguments provide a straightforward scaling to preserve the laminar flame speed, which is a key intrinsic property [10]. Moreover, a correct turbulent flame speed can, at least in principle, be recovered by introducing an efficiency function that compensates for the reduced wrinkling of the artificially thickened flame.

For diffusion flames, the situation is fundamentally different. The present work therefore addresses the evaluation of FRI and the development of flamelet-based models that ensure adequate resolution within LES frameworks applied to diffusion flames. This issue is generic and does not depend on the specific fuel considered. Hydrogen flames are adopted here as a representative test case, as they allow to avoid two-phase flow effects and rely on comparatively simple chemical kinetics relative to hydrocarbons. In addition, hydrogen–air combustion naturally promotes diffusion-type flame structures and provides a particularly stringent framework for model development and assessment.

Flame resolution for diffusion flames is rarely discussed: LES users assume that the numerical method somehow will self-adjust if the spatial resolution is insufficient, i.e. if FRI is small. When FRI becomes less than 3, the flame is resolved on two or three mesh points: the numerical method can not handle such

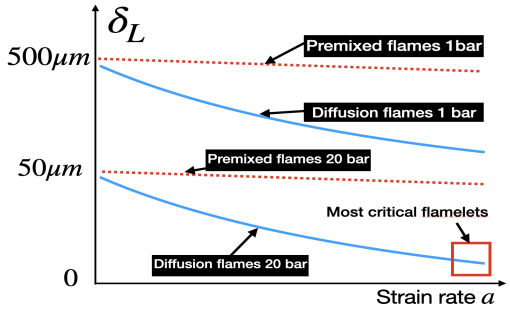


Fig. 1: Flame thicknesses as a function of local strain rate a and pressure for premixed and diffusion flamelets.

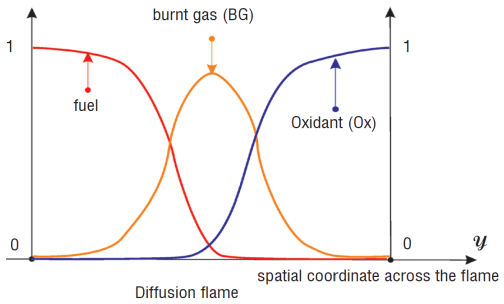


Fig. 2: Schematic representation of profiles (fuel mass fractions in red, oxidizer mass fractions in blue, heat release in orange) in a diffusion flame.

steep fronts accurately. The usual solution is to add numerical viscosity [11] or to use upwind schemes for spatial differentiation, in order to avoid code crashes, hoping that this numerical adjustment does not interfere with the solution.

The fact that most studies linked to resolution focus on premixed flames and ignore diffusion fronts is due to a crucial difference between premixed and diffusion fronts when it comes to evaluate $FRI = \delta_L/\Delta$. For premixed flames, the thickness δ_L is easy to estimate using the usual definition $\delta_L = (T_b - T_u)/\max(\nabla T)$ [10]. Moreover, δ_L is only weakly sensitive to strain so that this estimate can be used before the simulation to determine resolution needs. For non-premixed flames, however, the situation is more complicated. A simple method to measure δ_L in a diffusion flame is $\delta_L = 1/\max(\nabla z)$ in flames where the mixture fraction z varies from 0 in the oxidizer stream to 1 in the fuel stream. In the flamelet regime, the thickness of the diffusion flames varies like $1/\sqrt{a}$ where a is the local strain [12]. Since a can change over a wide range of values in a turbulent diffusion flame, and is unknown a priori, estimating resolution requirements for a diffusion flame is difficult and depends strongly on the flow. However, simple estimates show that the thickness of diffusion fronts can become very small at high strains and high pressures (Fig. 1) so that highly strained diffusion fronts are the zones controlling resolution in most high pressure simulations. Note that non-reacting fronts (such as quenched or non ignited flames) raise the same problem.

A difficulty for diffusion flames is that two values of FRI must be distinguished: (1) the true value of FRI called FRI_t is δ_L^t/Δ where δ_L^t is the exact flame thickness which is not known a priori; (2) the resolved value of FRI, called FRI_{res} is δ_L^{res}/Δ where δ_L^{res} is the flame thickness seen by the LES on its under refined grid. Even if the true thickness δ_L^t is very small, leading to very small FRI_t values, δ_L^{res} can not be less that approximately two times the mesh size Δ when the flame is resolved on two mesh points so that FRI_{res} is always larger than two. FRI_{res} and FRI_t are large and equal only when the flame is well resolved. When the flame becomes under resolved, FRI_{res} never goes below two while FRI_t can become arbitrarily small.

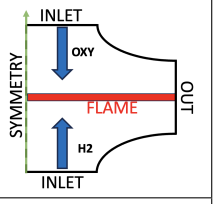
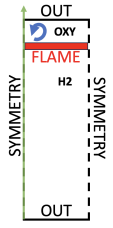
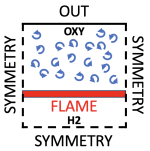
The present paper focuses on two issues. First, the resolution needs in terms of FRI are discussed using laminar stagnation point flames (Section 2) and flame / vortex interaction (Section 3) studies. In the flamelet regime limit, these two laminar flames are proper cases to study the effects of the resolution measured by FRI. Figure 2 schematically represents the temperature, species and heat release profiles typically found in a canonical diffusion flame. The heat release (like the mass source terms for most species) follows a bell curve with three inflection points. A simple piecewise linear representation that preserves the extrema of the curve and inflection points requires at least 5 points: it would be difficult to use less than 5 points to discretize this curve ($FRI_t = 5$). However, the primary question that naturally arises concerns the minimum number of points required to preserve the essential features of the flame like quenching strain rates a_Q or maximum temperature T_{max}. A second - and more fundamental - question is to what extent the minimum resolution criterion established for the physical flame, formulated in terms of FRI_t, can be transposed to FRI_{res}, based on the resolved flame thickness.

This paper shows that, for hydrogen-air flames solved using the San Diego chemical scheme [13], $FRI_{res} = 5$ is indeed the right threshold and that it is not necessary to go beyond, confirming results of Cuenot et al [14]. More generally, a direct examination of many recent LES papers at atmospheric pressure, including during the recent HYLON campaign of the TNF workshop (tnfworkshop.org/hylon2/), shows that this condition is usually reached in most codes: at 1 bar and typical flow rates, these flames have thicknesses of the order of 500 microns. Since many solvers today have meshes of the order of 100 microns, FRI_{res} is of the order of 5. The situation changes drastically at high pressures where flame thicknesses decrease to a few microns leading to a systemic under resolution and a significant risk

for the community: using codes correctly validated at 1 bar, at high pressures where they become systematically under resolved.

In a second step, the present work proposes a model to address the under resolved issue. Since evaluating resolution requirements is difficult a priori, a dynamic procedure allowing to thicken diffusion fronts during the LES until they reach $FRI_{res} = 5$ is proposed in Section 4. The model is called DFTC (Diffusion Front Thickness Control) and it allows to locally modify the flame to obtain $FRI_{res} > 5$ even if FRI_t is very small. DFTC ensures that the thickened flame is well resolved but it does not guarantee that the thickened flame is similar to the true flame for all quantities of interest: even if DFTC provides an adequate spatial flame resolution, it has consequences on the solution (as TFLES does for premixed flames [5, 15]). These effects are evaluated on the three cases of Table 1: laminar stagnation point flames (SPLF), flame-vortex interaction (FVI) and turbulent flames (FTI) using DNS for validation. Results show that DFTC makes the solution sufficiently spatially resolved, conserves the quenching strain rates a_Q and the maximum temperatures T_{max} while overestimating the heat release per unit flame area.

Table 1: Schemes of test cases

Case Acronym	Scheme
SPLF	
FVI	
FTI	

Under resolved context interferes with Flame Turbulence Interaction (FTI) models: no proper FTI model can be built on an under resolved solution so that the this issue must be solved before introducing any FTI model. For example, many FTI models for diffusion flames are based on scalar dissipation χ i.e. on ∇z [16, 17, 10]. If the z front is resolved on only three points separated by a distance equal to the mesh size Δ, ∇z will be of the order of $(1-0)/\Delta$: it will be imposed by the mesh size Δ and have no link with the real gradients of mixture fraction so that the FTI model may work but will lead to wrong subgrid effects and non-physical responses.

Two test cases extracted from Table 1 are then used to test the DFTC model:

- Strained stagnation Point Laminar Flame (SPLF): the model is first tested on steady, laminar stagnation point flames (Section 5).
- Turbulent flame (FTI): finally, the DFTC model is used in LES mode and compared to a highly resolved DNS of a diffusion flame interacting with synthetic turbulence (Section 6).

Limits and possible extensions of DFTC are presented in Section 7 and a comparison of thickening in premixed and diffusion flames is proposed in Section 8.

2. Minimum resolution in a strained stagnation point laminar flame

For hydrogen-air flames, a parametric study was first performed for H_2-air, atmospheric stagnation point flames where both streams are injected at 300 K (Table 2). These flames were computed on different grids (Table 3) ensuring values of FRI_t from 50 (DNS-like resolution) to 0.1 (coarse grid). They are computed either using CANTERA [2] in a one-dimensional formulation on the symmetry axis or AVBP [1] in a two-dimensional stagnation point flow. Fig. 3 shows the difference between a flame with $FRI_t = 7.2$ (left) and with $FRI_t = 0.22$ (right): the $FRI_t = 7.2$ flame is steady as it should and exhibits no instability while the $FRI_t = 0.2$ flame is clearly unstable, submitted to nonphysical oscillations and dispersive effects. The integrated heat release on the symmetry axis is constant and correct for $FRI_t = 7.2$ and oscillates strongly far from the right value for $FRI_t = 0.22$.

Table 2: Strained stagnation point flames used to determine minimum FRI_t values for stable and consistent solutions.

Case Name	Strain $[s^{-1}]$	Thickness $[\mu m]$	Maximum Temperature $[K]$
A	398	1045	2410
B	1603	526	2201
C	5279	289	1962
D	10165	205	1802
E	23868	133	1503

Results using a variety of strains and resolutions (Table 3) confirm that the limit between stable flames

Table 3: FRI_t values reached in the meshes used for stagnation point flames.

Case Name	Mesh 2 ($20\mu m$)	Mesh 3 ($40\mu m$)	Mesh 6 ($320\mu m$)	Mesh 8 ($1280\mu m$)
A	52.25	26.13	3.27	0.82
B	26.30	13.15	1.64	0.41
C	14.45	7.23	0.91	0.27
D	10.25	5.13	0.64	0.16
E	6.95	3.48	0.43	0.11

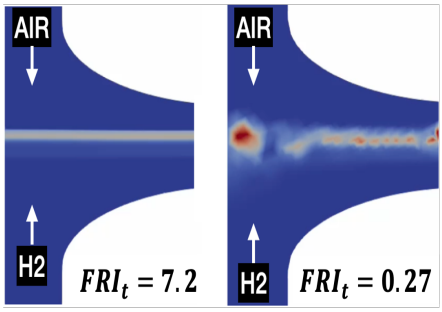


Fig. 3: Comparison of heat release fields for flame C (Table 2). Left: high resolution (mesh 3 in Table 3) - $FRI_t = 7.2$. Right: low resolution (mesh 8 in Table 3) - $FRI_t = 0.27$.

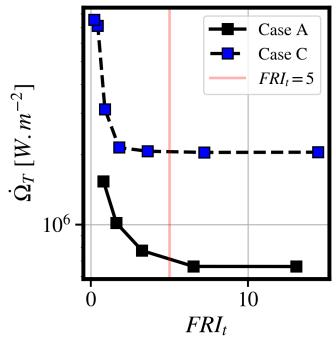


Fig. 4: Variations of $\dot{\Omega}_T$ for flame type A and C (Table 2) versus FRI_t. Horizontal red line: $FRI_t = 5$. See supp. material for cases B,D and E.

like Fig. 3 left and under resolved flames (Fig. 3 right) is $FRI_t = 5$. They also show that this condition $FRI_t > 5$ provides a correct value for the integrated heat release per unit length along the flame $\dot{\Omega}_T = \int_S \dot{\omega}_T dn$ where n is the flame normal: Fig. 4 shows that $\dot{\Omega}_T$ does not vary as soon as $FRI_t > 5$ for flames A and C of Table 3.

A key issue in the control of resolution using FRI is that its resolved value, FRI_{res} can be measured during an LES but its true value FRI_t cannot because it would require to know the true flame thickness δ_L^t. Fortunately, Fig. 5 shows that both definitions pro-

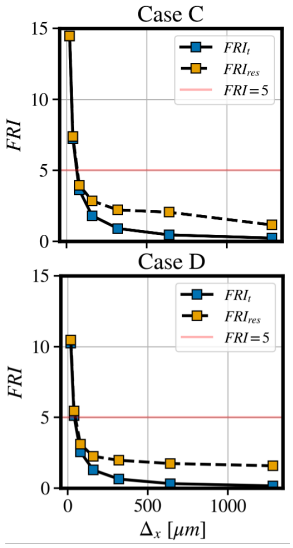


Fig. 5: Comparison between true FRI (FRI_t) and resolved FRI (FRI_{res}) for flames type C (top) and D (bottom) of Table 2 versus mesh size Δ. Horizontal red line: $FRI_t = 5$. See supp. material for cases A,B and E.

vide the same value when the resolution is sufficient: $FRI_t = FRI_{res}$ as soon as $FRI_t > 5$. Therefore in the model which will be presented in Section 4, the resolution criterion which will be used to determine whether a simulation is well resolved or not will be $FRI_{res} > 5$. Note that when FRI_{res} is less than 5, it is not possible to evaluate the exact value of FRI_t; however, in this situation, it is possible to state that the simulation is under resolved and that $FRI_t < 5$.

3. Minimum resolution in flame-vortex interaction

Section 2 suggests that a proper limit for resolution of a steady strained H2 - air diffusion flames is $FRI = 5$. This criterion is tested now in a more complicated, unsteady curved flame: the interaction of a diffusion flame with a vortex pair (FVI). The FVI configuration (Table 1) is symmetrical and only the right side is displayed. A pair of counter rotating vortices interacts with a 2D non-premixed strained flame. The vortices are generated on the air side. The system is 2 by 10 cm, at 1 bar. The initial flame is generated using a CANTERA solution at a strain $602 s^{-1}$ (flame F of table 4). The pair of vortices pushes the flame, strains and curves it (Fig. 6). The analysis focuses on the central symmetry axis because this is where the flame is submitted to the maximum strain and has the smallest flame thickness [18]. DNS are repeated for all conditions of Table 5.

Figure 7 shows the time evolution of FRI_t on the symmetry axis (the flame tip). Time is normalized using the vortex time: $t^+ = u_{max} t/R$ where R is the

vortex radius and u_{max} its maximum velocity within the vortex.

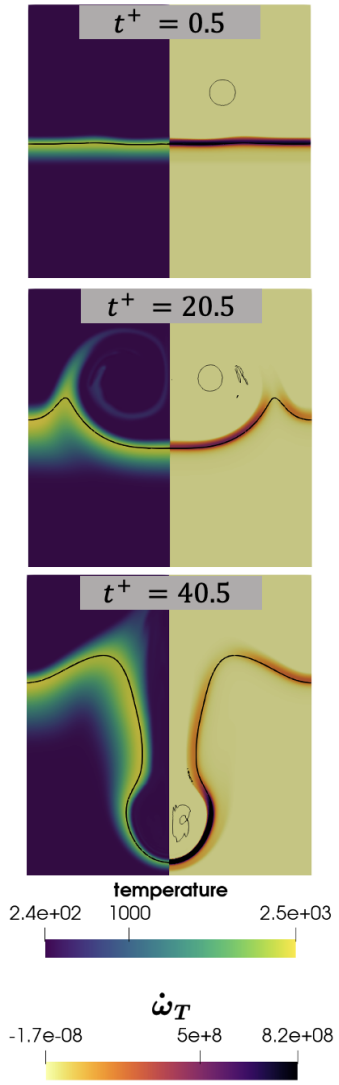


Fig. 6: Interaction between diffusion flame and a vortex pair. Snapshots of temperature (left, in K) and heat release $\dot{\omega}_T$ (right, in $W.m^{-2}$) at three instants. The black line is the stoichiometric line.

Table 4: Flame Vortex interaction point used to determine minimum FRI_t values for stable and consistent solutions.

Case Name	Strain $[s^{-1}]$	Thickness $[\mu m]$	Maximum Temperature $[K]$
F	602	845	2345

Table 5: FRI_t values reached in the meshes used for Flame Vortex Interaction.

Case Name	Mesh 3 ($40\mu m$)	Mesh 4 ($80\mu m$)	Mesh 5 ($160\mu m$)	Mesh 6 ($320\mu m$)
F	21.125	10.56	5.28	2.64

During the first stage of the interaction from $t^+ = 0$ to 25, FRI_t grows because the flame is not yet strained and becomes thicker [10] and easier to resolve. Then the vortex enters the flame, the strain seen by the flame tip increases and FRI_t decreases, making the simulation less resolved. As long as FRI_t is larger than 5 (represented by the solid horizontal black line), the total heat release, $\dot{\Omega}_T$, is correctly captured for all meshes 3 to 6 of Table 5. Mesh 3 is actually always well resolved and used as a reference. Meshes 4 to 6 all give reasonable values for $\dot{\Omega}_T$ until strain continues to diminish FRI_t. When FRI_t becomes too small, the impact on the total heat release, $\dot{\Omega}_T$, is instantaneous: as soon as FRI_t at the flame tip goes under 5, errors on $\dot{\Omega}_T$ appear for each test case, independently of resolution or of initial strain.

These results show that for this strained, curved, unsteady flame (typical of flamelets encountered in turbulent flames), like for the steady strained flames of Section 2, resolution is an issue which can be characterized to first order by FRI_t: monitoring FRI_t, independently of how subgrid scale terms are handled, allows to detect when the diffusion flame becomes under resolved. When this happens, a model must be triggered when FRI_t is less than 5. This is the objective of the DFTC model described in the next section. Note that, from now on, FRI_t will not be accessible in the LES since only FRI_{res} can be measured on the LES mesh. We will assume that a threshold of 5 for FRI_{res} is the right criterion and verify it.

4. DFTC: a model to thicken diffusion fronts

Among the multiple models for flame / turbulence interaction, the idea of a 'mapping' proposed initially in the thickened flame model of O'Rourke and Bracco [19] has been extended through different variations of this thickening approach [6, 20, 7]. These models directly address the issue of spatial resolution by building a master equation for the filtered mass fraction $\widetilde{Y}_k$ of species k, starting from the conservation equation:

$$\overline{\rho}\frac{D\widetilde{Y}_k}{Dt} = \nabla\cdot[\overline{\rho}(D_k + D_t)\nabla\widetilde{X}_k] + \overline{\dot{\omega}}_k \qquad (2)$$

where D_k and D_t are respectively the laminar and turbulent diffusivity, $\widetilde{X}_k$ the filtered molar fraction and $\overline{\dot{\omega}}_k$ the filtered reaction rate.

The 'mapping' operation consists in modifying the diffusion and reaction terms as follows:

$$\overline{\rho}\frac{D\widetilde{Y}_k}{Dt} = \nabla\cdot[\overline{\rho}(\boldsymbol{F}D_k + \boldsymbol{\alpha}D_t)\nabla\widetilde{X}_k] + \boldsymbol{F}_r\overline{\dot{\omega}}_k \qquad (3)$$

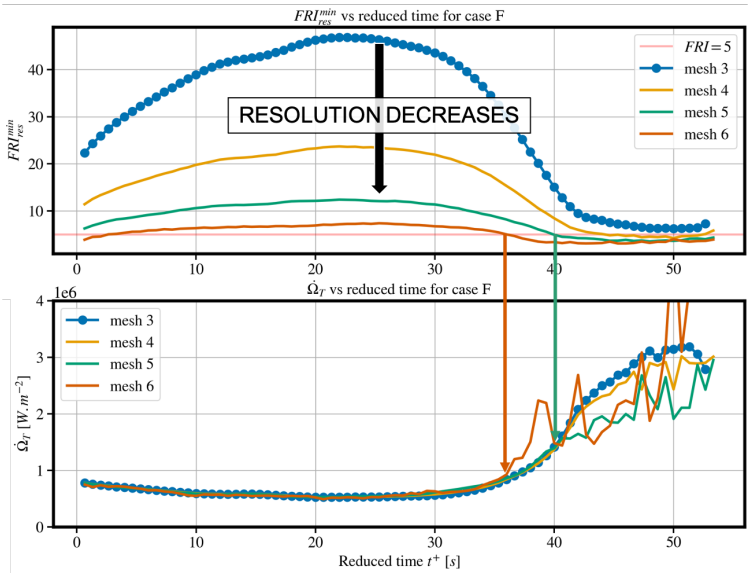


Fig. 7: Diagnostics on symmetry axis for flame - vortex interaction. Circles: fully resolved case (mesh 3). Top: evolution of FRI_t versus reduced time. Bottom: Evolution of $\dot{\Omega}_T$ versus reduced time. When $FRI_t = 5$ becomes less than 5, the integrated reaction rate $\dot{\Omega}_T$ starts to deviate from the resolved case (circles). Horizontal red line: $FRI_{res} = 5$.

Note that the 'mapping' operation is not based on a rigorous derivation of the conservation equations as done in pdf models [10] or in filtered models [21, 9]. However, it has various advantages: it is simple to implement, degenerates naturally to DNS when the mesh is refined and allows to control spatial resolution: this explains its choice for the present work.

The 'mapping' introduces three terms F, α and F_r which are added to Eq. (2): (1) modification of the local laminar diffusivity D_k by a factor F (analog to the thickening factor of the TFLES model for premixed flames). Models that modify this term are labeled 'LAM' (2) modification of the local turbulent diffusivity D_t by a factor α. Models that modify this term are labeled 'TUR' and (3) modification of the reaction term $\dot{\omega}_k$ by a factor F_r (in the TFLES model for premixed flames, $F_r = 1/F$). Models that modify this term are labeled 'REAC'. A single model can combine any of these three terms[1]

Before discussing what F, α and F_r should be, objectives must be defined (Tab. 6): RES, LQ, LTMAX, LC and FTIC are criteria which are deemed essential for LES simulations even if it is not clear that they can all be satisfied with Eq. (3). The choice of the criteria of Tab. 6 is guided by from flamelet ideas: the model should work first for a typical flamelet, ie for a stagnation point, laminar, steady diffusion flame and then be applied to a turbulent flame, viewed as a collection of flamelets.

Fig. 8 summarizes the main arguments used to guide the choices of F, α and F_r obtained from classical laminar strained diffusion flames analysis. The first point is that, unlike premixed flames, diffusion flames are controlled by strain: their total reaction rate increases with strain until quenching. The second point is that, at low strain values, the infinitely fast chemistry assumptions holds, so that for these low strain flames, we should not expect F_r to affect the results.

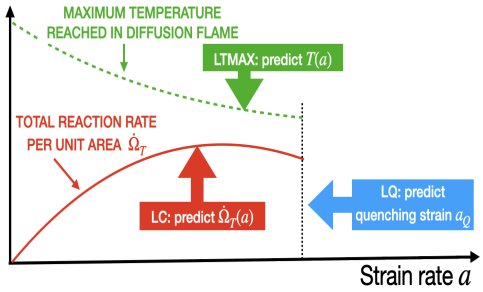


Fig. 8: Total reaction rate per unit area vs strain or scalar dissipation in a laminar, strained, diffusion flame [10].

Table 6 includes five objectives for the model:

- RES: spatial resolution is mandatory. If the modeled front leads to a discontinuous z field

[1]The classical TFLES model for premixed flames is a LAM REAC model with $F_r = 1/F$.

Table 6: Objectives for LES modeling (SPLF = Strained stagnation Point Laminar Flame, FTI= Flame / turbulence Interaction).

Objectives	Measure	Model	Test case
RES: Resolution The diffusion front must be resolved so that results do not depend on mesh or artificial viscosity	FRI_m minimum value	LAM or TUR	SPLF FVI FTI
LQ: Laminar Quenching In laminar resolved diffusion fronts, quenching must occurs at the right strain	Quenching strain a_Q	LAM or TUR	SPLF
LTMAX: Maximum temperature In laminar resolved diffusion fronts, the maximum temperature evolution vs strain must be conserved	Quenching strain a_Q	LAM or TUR	SPLF
LC: Laminar Consumption In laminar resolved diffusion fronts, the total heat release per unit flame area vs strain must be predicted correctly	Integrated heat release $\dot{\Omega}_T$ vs strain a	LAM, TUR or REAC	SPLF
FTIC: Flame / Turbulence Interaction Consumption In DNS of diffusion flames interaction with turbulence, the total reaction rate must be captured	Integrated heat release $\dot{\Omega}_T$ vs strain a	LAM, TUR or REAC	FTI

where z jumps from 0 to 1 in one cell, no FTI model will work.

- LQ: a key issue in many diffusion fronts, stabilized for example on injector lips [22, 3, 23] is to know if the diffusion front is quenched or not. In a flamelet framework, this can be translated into a question of strain value at quenching, called a_Q, for a stagnation point flame. The resolved flame in the LES should not quench at the wrong stretch value.

- LTMAX: since the prediction of NO is a key issue, predicting the correct maximum temperature reached in a strained diffusion flame would be useful [24].

- LC: the model should predict the overall consumption rate of a laminar stagnation diffusion flame over the whole range of strains, up to quenching.

- FTIC: the ultimate goal of all LES models is to capture the mean reaction rate during the interaction of a flame with turbulence. This question has generated many DNS over the last decade for hydrogen flames [18, 25, 26] but only few real LES models.

The wish list of Tab. 6 is probably impossible to fully satisfy. We show below that the DFTC model allows to satisfy RES, LQ, LTMAX very well but not LC and FTIC.

Two pure diffusion flame configurations were used to test DFTC:

- SPLF: strained stagnation point laminar flames using either CANTERA or AVBP.

- FTI: diffusion flame interacting with decaying two-dimensional turbulence using AVBP.

Sections 2 and 3 have shown that RES would be satisfied for hydrogen-air flames when $FRI_{res} > FRI_{target}$ with $FRI_{target} = 5$. Interestingly, this value of 5 for $FRI_{target} = 5$ is close to the value used in premixed fronts for TFLES. Note that this is a remarkably small number of points to resolve a complex chemistry front. The simplest way to ensure the $FRI_{res} > FRI_{target}$ condition is to use a LAM model and increase the laminar diffusivities by a factor F. In a LES, turbulent diffusivities (D_t in Eq. 3) would also make the flame thicker but they are not designed to control the resolved flame thickness. Moreover, D_t can go to zero in many quasi-laminar zones of the flow, leading to under resolution. Therefore, controlling the resolution through a change of laminar diffusivites is a logical choice.

Theory shows that at low local strains, the flame front thickness δ_L (and therefore FRI) increases like $\sqrt{F}$. Unlike in premixed flames, the thickening factor F to apply cannot be determined a priori since the local diffusion flame thickness depends on local strain and is unknown. Therefore DFTC uses an active control loop at every mesh point where F is changed at each time step until a satisfactory value is reached for FRI_{res}. The local value of F is set to:

$$F = 1 + (F_{max} - 1) f(z) \tag{4}$$

where the maximum thickening factor F_{max} is obtained by a time-dependent process:

$$dF_{max}/dt = (FRI_{target} - FRI_{res}^{min})/\tau \tag{5}$$

where FRI_{res}^{min} is the minimum value reached by FRI_{res} in the domain (usually along the $z = 0.5$ line) and FRI_{target} is the target value needed for resolution (5 as shown in Section 2 and 3). The $f(z)$ function has a

parabolic shape and provides a maximum thickening factor at $z = 0.5$ where the gradient of z is maximum (Fig. 9):

$$f(z) = \exp\left\{-\left[\mathrm{erf}^{-1}(1-2z)\right]^2\right\} \quad (6)$$

No thickening is applied in the pure streams ($z = 0$ and $z = 1$). The time τ is a relaxation time, controlling the speed at which the controller follows local strain and FRI_m variations. A convenient method to evaluate τ is to scale it by the time step Δt of the simulation: $\tau = K\Delta t$ where K is a free model parameter. Small K values will lead to fast adjustment of F and vice-versa. In a time-dependent code, Eq. 5 becomes $F^{n+1}_{max} = F^{n}_{max} + (FRI_{target} - FRI^{min}_{res})/K$ where n is the iteration index, leading to a simple implementation.

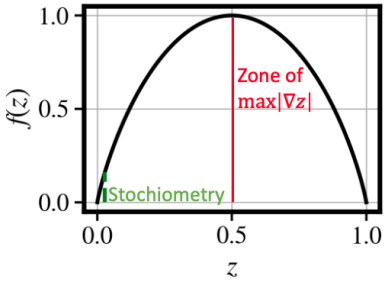


Fig. 9: The $f(z)$ function controlling the distribution of thickening in z space. Red vertical line: maximum z gradient ($z = 0.5$). Green dashed line: $z = z_{st}$.

Criteria LQ, LTMAX, LC and FTIC are all desirable once RES is ensured. The following sections show how DFTC satisfies these criteria in stagnation point flames (Section 5) and in turbulent flames (Section 6) before discussing possible improvements using LAM TUR REAC models in Section 7.

5. Tests of DFTC in stagnation point flames

Fig. 10 shows changes of $\dot{\Omega}_T$ and T_{max} versus strain a in stagnation point flames computed with CANTERA for increasing imposed values of F_{max}: as expected, when thickening, measured by F_{max}, increases, $\dot{\Omega}_T$ increases (like $\sqrt{F}$) so that criterion LC is not satisfied. However, LQ is satisfied: for all thickening factors, quenching (visualized by black circles) occurs at the same strain value (Fig. 10 left).

Criterion LTMAX (Fig. 10 right) is also satisfied: the maximum temperature reached in the diffusion front is the same for all F_{max} values.

The results of Fig. 10 are all consistent with diffusion flame theory: the total reaction rate per unit flame area $\dot{\Omega}_T$ increases by $\sqrt{F}$ while the quenching strain rate a_Q and the maximum temperature T_{max} are independent of F as expected from theory [27]. We will discuss the interest of satisfying LC in stagnation

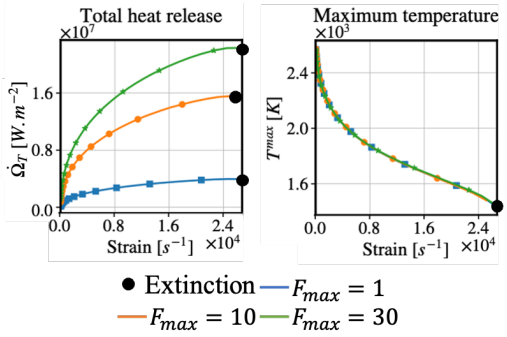


Fig. 10: Steady strained planar flames (SPLF) using DFTC for increasing thickening factors F_{max} from 1 to 30. Left: total reaction rate per unit area $\dot{\Omega}_T$ vs strain. Right: maximum temperature T_{max}. Black solid circles: quenching points.

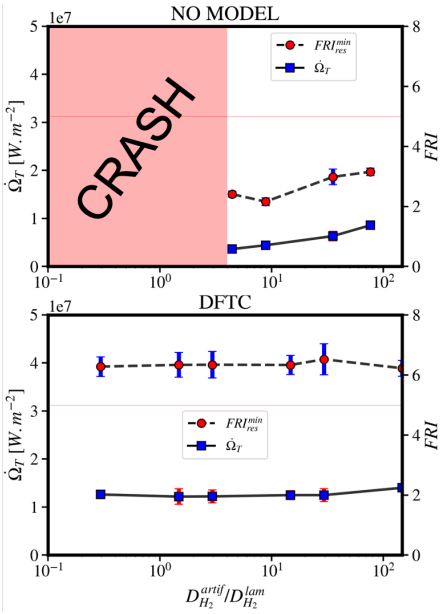


Fig. 11: Stagnation point diffusion flame: effects of artificial viscosity measured by the ratio $D^{artif}_{H_2}/D^{lam}_{H_2}$. FRI^{min}_{res} and $\dot{\Omega}_T$ are plotted vs $D^{artif}_{H_2}/D^{lam}_{H_2}$ without DFTC (Top figure) and with DFTC (bottom figure). Horizontal red line: $FRI_{res} = 5$.

point flames in Section 7 after discussing the turbulent flame case in Section 6.

Finally Fig. 11 illustrates a key point of DFTC: instead of thickening the flame using DFTC, an alternative path is to use artificial viscosity. AVBP has centered schemes with second-order artificial viscosity which was used to compute the stagnation point flame of the present section. Fig. 11 compares the effects

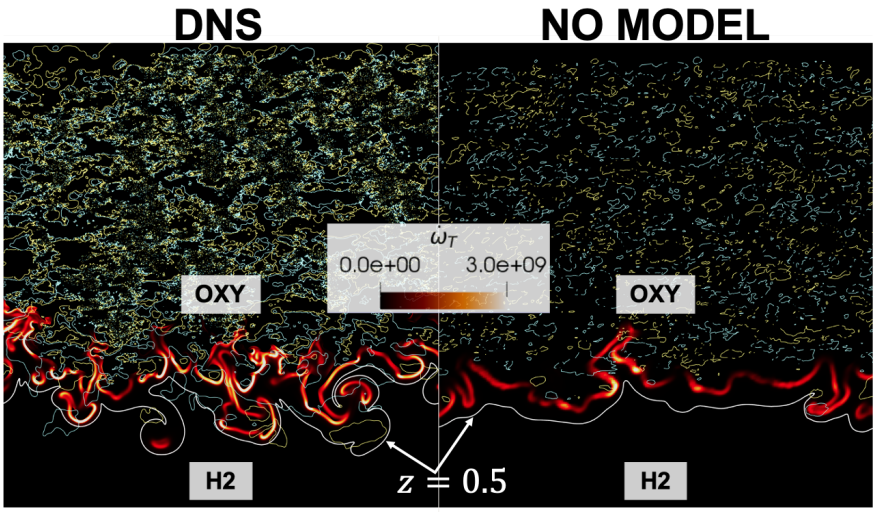


Fig. 12: Comparison of DNS (left) and LES (right) of heat release fields at $t = 1ms$. White line: $z = 0.5$. Blue line $u = 10\ m.s^{-1}$. Yellow line $u = -10\ m.s^{-1}$ (u is the horizontal velocity).

of this artificial viscosity (quantified here by the ratio of artificial hydrogen diffusivity $D^{artif}_{H_2}$ to true laminar diffusivity $D^{lam}_{H_2}$) with and without DFTC. On top, without DFTC (Case NO MODEL), the code crashes (as in Fig. 3) if $D^{artif}_{H_2}/D^{lam}_{H_2}$ is less than two. And when $D^{artif}_{H_2}/D^{lam}_{H_2}$ is larger than 2, the code does not crash but results depend strongly on the artificial viscosity, making the simulation useless. With DFTC (Fig. 11 bottom), any value of $D^{artif}_{H_2}/D^{lam}_{H_2}$ can be used and results do not depend on it: DFTC leads to a simulation which is independent of numerical dissipation.

6. Tests of DFTC in a turbulent flame

DFTC is tested now on a DNS of the interaction between turbulence and a non-premixed flame in two dimensions, using the AVBP solver [1], the San diego mechanism [13] and the turbulence generation model of Guezennec et al [28], where the integral length scale $l_t = 1\ mm$ and the turbulent velocity fluctuation $u' = 15.8\ m.s^{-1}$. The box size is 80 by 80 mm. The initial flame front corresponds to a strained diffusion laminar H2-air flame where both streams are at 300 K. Turbulence is added at $t = 0$ in the air stream only. DNS (with 2000 by 2000 points) is used as a reference and compared to two LES using 250 by 250 points without or with the DFTC model. For DNS, the resolution is high and the minimum FRI_{res} is always greater than 20.

Fig. 12 compares a DNS field (left) with an LES field (right). Obviously, the first effect of LES is that the coarse mesh strongly limits the wrinkling which the flame can produce.

Fig. 13 compares LES with DFTC and without DFTC. The white $z = z_{st}$ stoechiometric line visualizes the instantaneous flame front. Initially and until $t = 0.2$ ms, the flame is well resolved in both LES. The RES criterion is satisfied everywhere: $FRI_{res} > FRI_{target} = 5$. After $t = 0.2$ ms, turbulence wrinkles the flame and strains it, reducing its thickness and diminishing FRI_{res} at a few points below FRI_{target} in the NO MODEL case where no thickening is applied. With DFTC, local thickening is introduced when FRI^{min}_{res} becomes close to FRI_{target}: as a result, FRI^{min}_{res} is never less than FRI_{target}, ensuring that the solution is well resolved at all times.

Fig. 13 captures a typical zoomed snapshot for both LES after 0.75 ms, showing fields of heat release $\dot{\Omega}_T$ and of FRI. The thickening factor F is added for the DFTC case (F is unity without DFTC). Two isolines of z of interest are also displayed: the stoechiometric line $z = z_{st}$ and the line where $z = 0.5$ where the maximum gradients of z are found. The FRI_{res} field is colored using the following code:

- green: $FRI_{res} > 7$, fronts are well resolved, no thickening is required,
- yellow: $FRI_{res} \in [5, 7]$, fronts are close to under resolution,
- red: $FRI_{res} < 5$, fronts are under resolved and DFTC must be activated.

DFTC avoids under resolution zones. No red zone is observed (Fig. 13 top) when DFTC is on: criterion RES is satisfied everywhere. In all regions where FRI_{res} decreases towards 5, the DFTC model

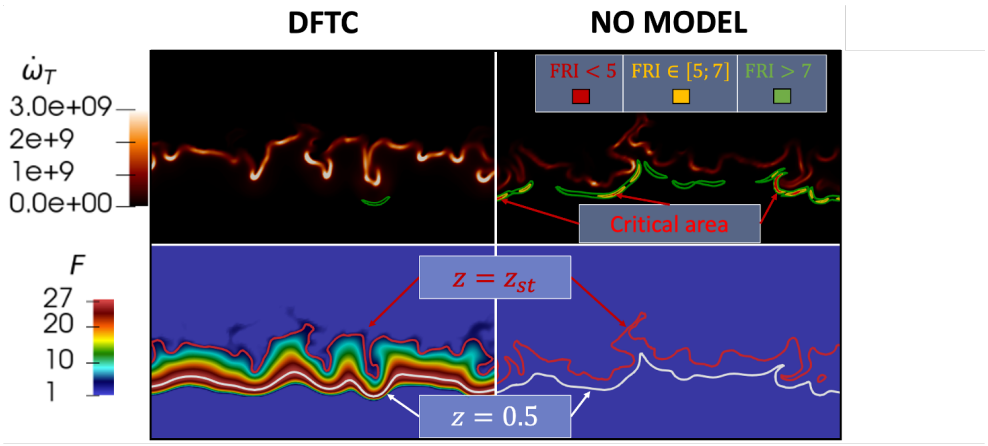


Fig. 13: FRI_{res} fields for LES with DFTC ($K = 100$) and LES without DFTC. The thickening F field is displayed only for DFTC (F is one for the NO MODEL case). White line: $z = 0.5$. Red line: $z = z_{st} = 0.028$.

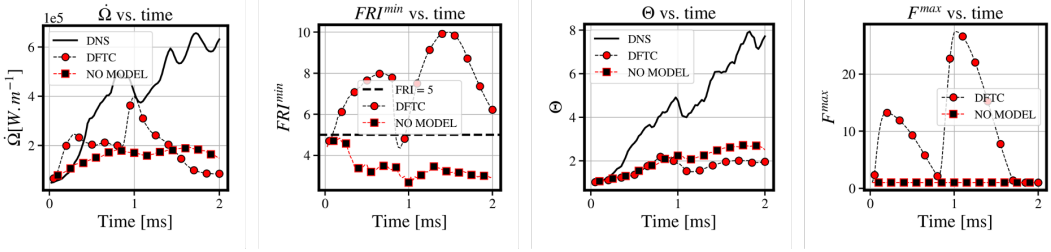


Fig. 14: From left to right, Time variations of total reaction rate $\dot{\Omega}$, maximum thickening factor F, minimum FRI value and wrinkling Θ. Comparison of DNS and LES without and with DFTC model. Fig. 13 is extracted form those simulations

increases F and prevents under resolution. Since the cost function used by Eq. 5 uses the minimum value of FRI_{res} (FRI_{res}^{min}) to decide whether thickening is needed or not, the model is not local in two ways: (1) along the flame normal, FRI_{res}^{min} is not reached at stoechiometry ($z = z_{st}$) but closer to $z = 0.5$. Still, the value of $FRI_{res}(z = 0.5)$ is the one which decides the DFTC activation. This is not a major issue: $FRI_{res}(z = 0.5)$ is proportional to $FRI_{res}(z = z_{st})$ [10] and is clearly a good sensor to decide whether the entire local flamelet is strained or not ; (2) along the flame surface, when DFTC detects that thickening is needed at one point, it thickens the whole flame. This limitation is not an issue for the homogeneous flame of Fig. 12 but extensions of the model will be required to relax this simplification in real chambers.

Fig. 14 shows how DFTC adjusts to the changes in the flame. At time $t = 0.8$ ms, the flame has no strongly strained point any more: DFTC senses this because FRI_{res}^{min} is large enough so that thickening is not needed: F_{max} goes back to unity (right image). Slightly later, at $t = 1$ ms, straining starts again and DFTC increases thickening to avoid small FRI_{res}^{min} values. The speed at which DFTC reacts is controlled by the relaxation time τ in Eq. 3.

Ensuring that the flame is resolved as shown in Fig. 13 does not ensure that it is predicted accurately. Tests in laminar flames of Section 5 have shown that the quenching strain criterion LQ and the maximum temperature criterion LTMAX are probably satisfied in the turbulent flame too. However, the FTIC criterion, focusing on the total consumption rate, is not satisfied (Fig. 14) as shown by the evolution of the total reaction rate $\dot{\Omega}$ over the whole domain, defined as $\dot{\Omega} = \int \dot{\omega}_t dS$. In the flamelet limit, $\dot{\Omega}$ can be written as $\dot{\Omega} = \dot{\Omega}_T * L_f$ where $\dot{\Omega}_T$ is the reaction rate per unit length, studied in the laminar flame cases and L_f is the flame length in this two-dimensional case. This length is controlled by the wrinkling factor $\Theta = L_f/L_y$ in Fig. 14 where L_y is the length of the DNS domain.

Fig. 14 shows that both LES underpredict the total reaction rate $\dot{\Omega}$ measured in the DNS: this is mainly due to the fact that they both underestimate flame wrinkling Θ. While the local reaction rate per unit area $\dot{\Omega}_T$ increases, wrinkling decreases more and the overall effect is that $\dot{\Omega}$ is too small in the two LES, even with DFTC. This issue is discussed in Section 7.

7. Effects of turbulent viscosity and reaction terms: tests of LAM TUR REAC models

Previous sections have shown that DFTC allows to predict quenching by strain at the right value (criterion LQ in Table 6), maximum flame temperature (criterion LTMAX) while ensuring that the flame was properly resolved (criterion RES) and independent of numerical dissipation. When it comes to the mean heat release, however, DFTC overpredicts the exact results in laminar stagnation point flames (Criterion LC, Section 5) because molecular diffusivities are increased by F but underpredicts it in turbulent cases (Criterion FTIC, Section 6) because wrinkling is reduced when the mesh is decreased from 2000^2 to 250^2.

A logical next step is to wonder whether criterion FTIC which is the final objective of a proper LES subgrid model can be satisfied by introducing modifications of α (the turbulent diffusion term) or F_r (the chemical term) which means going to LAM TUR REAC formulations in Eq. 3.

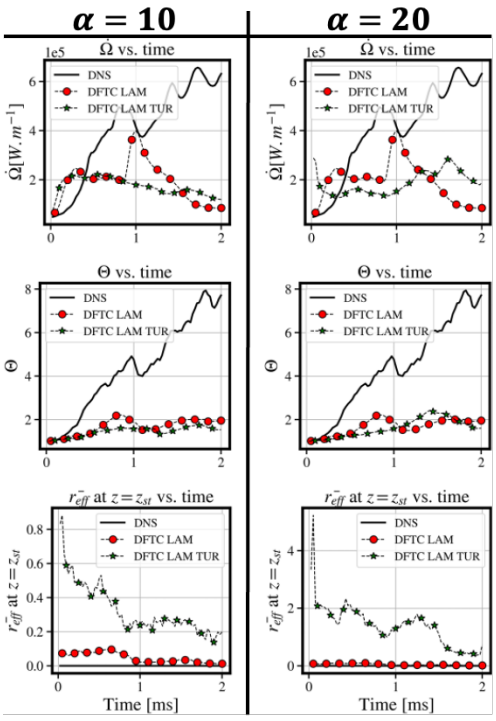


Fig. 15: Effects of α: time variations of total reaction rate $\dot{\Omega}$ (top), wrinkling factor Θ (middle) and average ratio $r_{eff} = (\alpha D^{turb}_{H_2})/(F D^{lam}_{H_2})$ along the $z = z_{st}$ line (bottom). Comparison of DNS and LES without and with DFTC model when α is changed. Left: $\alpha = 10$. Right: $\alpha = 20$.

A first test is to explore LAM TUR models and add the 'TUR' option to the existing DFTC. This is done by modifying the turbulent diffusivity term in Eq. 3 and multiplying it by α. Fig. 15 shows that increasing turbulent viscosity by $\alpha = 10$ (left) or even 20 (right) does not allow to increase Ω (Fig. 15 top). Increasing the turbulent viscosity actually decreases the flame wrinkling Θ (Fig. 15 middle) so that Ω is smaller when $\alpha > 1$ and not larger as expected initially. The bottom images of Fig. 15 shows the ratio of the turbulent diffusivity to the laminar diffusivity $r_{eff} = (\alpha D^{turb}_{H_2})/(F D^{lam}_{H_2})$ averaged along the $z = z_{st}$ line. It shows that increasing α indeed increases the effects of the turbulent diffusivity on the flame front but does not allow to increase Ω.

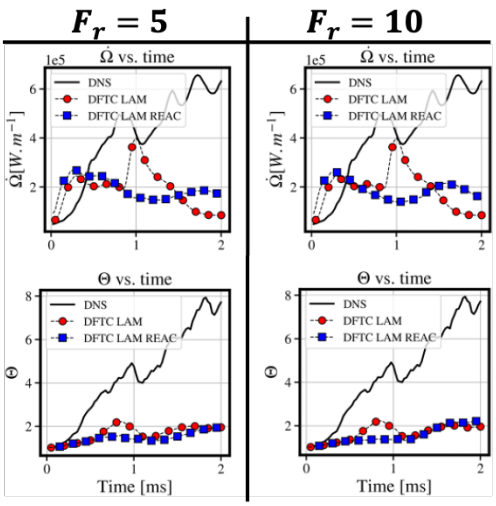


Fig. 16: Effects of F_r: time variations of total reaction rate $\dot{\Omega}$ (top) and wrinkling factor Θ (middle). Comparison of DNS and LES without and with DFTC model when F_r is changed from 1 to 10.

To increase the reaction rate $\dot{\Omega}$ in Fig. 14, another natural approach following what is done when using efficiency functions for premixed flames is to increase the reaction term by increasing F_r in Eq. 3 and test LAM REAC models. Fig. 16 shows however that this idea does not work well for the hydrogen flame studied here. With $F_r = 10$, even though chemistry is ten times faster, the overall reaction rate $\dot{\Omega}$ does not increase. This is clearly due to the fact that in this hydrogen flame, chemical reactions are very fast compared to the diffusion phenomena. Thus, diffusion controls the overall reaction rate for low strains: increasing F_r makes the reaction zone thinner without changing the overall reaction rate. This result gives rise to two important observations. The first remark is that, from this perspective, diffusion flames exhibit a significantly higher level of complexity than premixed flames. In the latter case, the global heat release can always be modulated by modifying the reaction terms

(this being the role of the efficiency function in the TFLES framework) . Such flexibility is not available for diffusion flames. The second remark is that the choice of hydrogen flames places the present study in a particularly stringent framework. Owing to the very fast chemical kinetics in hydrogen–air flames, combustion often occurs on timescales shorter than those of molecular diffusion. As a result, adjusting the reaction-rate scaling factor F_r does not allow to modify the global heat release significantly. By contrast, for fuels characterized by slower chemistry in air such as hydrocarbons or ammonia, it may be possible, at least partially, to restore the overall heat release through an appropriate tuning of F_r.

From a practical point of view, these results suggest that, for hydrogen flames, the standard LAM model used in the previous sections is probably the best compromise which can be achieved in the LAM TUR REAC class of models and that changing α or F_r brings limited improvement for hydrogen-air flames. When it comes to LES, Fig. 12 illustrates the main problem: the loss of flame wrinkling Θ which is a direct consequence of the coarser meshes used in the LES. At this point mesh refinement is certainly a useful approach (such as Automatic Mesh Refinement - AMR) even if it will not fully solve the problem for highly strained diffusion flamelets.

8. Comparison of thickening models for premixed and diffusion flames

Since DFTC proposes to thicken diffusion fronts, it is interesting to compare it to the well established TFLES model for premixed fronts. The initial papers of O'Rourke and Bracco [19] and Colin et al [5] are often used to justify TFLES by the fact that thickening conserves the speed of a laminar unstrained premixed flame (first line in Table 2). This is right but with one important caveat: it works only if the flame is not strained and not curved, something which is never true in turbulent flames. Recent results show that TFLES does not conserve maximum temperature, flame speed or quenching strain rate of strained flames as soon as they are thickened [6, 20]. In that sense, DFTC does better than TFLES as shown by Table 7, especially for quenching limits: a diffusion flame thickened by DFTC will quench at the right strain while a premixed flame thickened by TFLES with quench at a strain rate which diminishes like $1/F$, leading to large errors. Overall, thickening fronts is as reasonable for diffusion flames as it is for premixed flames.

Table 7: Comparison of thickening effects on laminar diffusion and premixed flames

PROPERTY TO CONSERVE	DFTC Diffusion flames	TFLES Premixed flames
1 Laminar unstrained flame speed	N/A	YES
2 Quenching strain rate	YES	NO
3 Maximum temperature vs strain	YES	NO
4 Heat release vs strain	NO	NO

9. Conclusions

Most hydrogen flames of practical interest do not operate in fully premixed modes for safety reasons. Therefore they exhibit premixed but also diffusion fronts and these diffusion fronts raise a daunting challenge for simulations especially at high pressures. Simple asymptotic theory is used here to show that strained diffusion flamelets (extinguished or not) are the thinnest flame elements in turbulent flames (Fig. 1) and that they become systematically under resolved at high pressures, leading to considerable risks when codes validated in lab-scale experiments at 1 bar are ported to high pressures. A thickening method (called DFTC for Diffusion Front Thickness Control) to prevent this systemic under-resolution in LES is proposed by defining a resolution index called FRI_{res} (FRI_{res} indicates how many mesh points are used inside the resolved flame front) and implementing a closed-loop thickening algorithm (Eq.(3) which guarantees that these zones are spatially resolved and have large enough FRI_{res} values. DFTC automatically deactivates when strain is reduced and diffusion flamelets are resolved on the LES grid. DFTC introduces additional viscosity like an artificial viscosity does but in a controlled fashion which does not depend on the grid or on the numerical scheme.

Tests on stagnation point laminar flames show that $FRI_t > 5$ is mandatory to ensure minimum resolution. DFTC is then developed using an on-line, active control method to thicken the front where the flame is too thin, ensuring values of FRI_{res} larger than 5. Results show that multiplying all laminar diffusivities by the thickening factor F provided by DFTC allows to maintain FRI_{res} to values larger than 5 so that the simulation is no longer influenced by the mesh or the numerical method. The total reaction rate $\dot{\Omega}_T$ per unit area is overestimated in laminar flames but the quenching strain rate a_Q as well as the maximum temperature T_{max} reached in the diffusion flame are conserved. Results are not as good in turbulent flames where the loss of flame wrinkling due to low resolution becomes the first issue for LES. This could be compensated only partially by smart meshing strategies such as AMR because at high pressures, wrinkling scales will be too small to be meshed anyway even by using mesh refinement. Combining DFTC and AMR methods is probably a good approach for high pressure diffusion flames: DFTC will ensure that criteria RES, LQ and LTMAX (Table 6) will be satisfied while AMR will be needed to satisfy FTIC.

Credit authorship contribution statement

E.M. Berthoumieu: Writing, investigation, methodology. **N. Bertier**: Writing, review and edit-

ing; Validation. **T. Poinsot**: Writing, review and editing; Methodology; Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary information are available for figures 4 & 5

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